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This thesis summarizes the scientific study I carried out as part of the group at the University of Siena and the INFN in Pisa involved in the Hyper-Kamiokande experiment, in collaboration with the software and reconstruction group.

A thesis submitted in partial fulfillment of the requirements for the degree Master of Science

Optimization and performance evaluation of a likelihood-based algorithm for multi-ring event reconstruction in Hyper-Kamiokande

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Introduction

In 1930, the existence of the *neutrino* was hypothesized for the first time by W. Pauli, in a now famous letter [1], as a solution to the continuous energy spectrum of electrons seen in β-decay experiments by Chadwick [2]. Originally, Pauli proposed the name *neutron* for this new particle of spin $\frac{1}{2}$, but thought his proposal was too speculative, and so he did not publish it in a scientific journal until 1934. E. Fermi further developed the theory of β-decay giving the neutrino its modern name and providing a solid theoretical framework for weak interactions. The goal of particle physics is to discover and understand the fundamental laws of nature. Until the discovery of neutrino oscillations, experimental results were consistent with zero neutrino mass. The observation of neutrino oscillations by the Super-Kamiokande (SK) experiment in 1998 provided the first evidence for physics beyond the Standard Model, proving that neutrinos must possess non-zero masses and that the lepton flavors are not conserved during propagation.

From then until now, the determination of the neutrino Mass Ordering (MO) and the search for Charge-Parity Violation (CPV) in the leptonic sector have become central questions in fundamental physics. The Hyper-Kamiokande (HK) experiment aims to discover CPV in the lepton sector, to measure the neutrino oscillation parameters with high precision, and to observe the proton decay. Beyond oscillation physics, HK will function as an astrophysical observatory.

The far detector of the HK experiment is a water-Cherenkov detector. Because neutrinos are electrically neutral and interact only weakly, they do not directly produce Cherenkov light. Instead, when a neutrino interacts with a nucleus in the water, it produces charged particles such as electrons, muons, or pions. If these charged particles travel faster than the speed of light in water, they emit a cone of Cherenkov radiation. This light produces a characteristic ring pattern on the detector wall, where it is collected by PhotoMultiplier Tube (PMT) arrays. These rings are the main signature used to reconstruct the event. The event reconstruction algorithm, `fiTQun`, is designed to identify these rings, count them (*ring counting*), and determine the kinematics (direction, momentum, vertex position) of each particle. It performs a maximum-likelihood fit to the PMTs recorded hit pattern, looking for the particle configuration that best explains the observed time and charge distributions on the PMTs. The `fiTQun` algorithm is being developed to better handle complex events with multiple overlapping rings, the so-called *multi-ring* events, which are the primary focus of this thesis.

This thesis presents a study of the multi-ring reconstruction algorithm `fiTQun`, with the primary objective of optimizing its ring-counting performance at the 8-digit fit level, the first stage where the algorithm decides whether to accept additional rings. The work identifies the limitations of the currently implemented version of the algorithm, which relies on a one-dimensional cut based only on the likelihood ratio comparison. By analyzing two-dimensional distributions, in particular the correlation between the likelihood ratio and the opening angle $\Delta\theta$ between reconstructed directions of the Cherenkov cones, new bi-dimensional selection criteria are proposed. The optimization was performed on a dedicated dataset of about $\sim$50 000 simulated events, using a figure of merit that combines efficiencies and purities for single and two-ring classifications. The new criteria reduce misclassifications, especially for true single-ring events reconstructed as two-ring, without degrading kinematic reconstruction quality.

The thesis is organized as follows:

- chapter 1 introduces the HK experiment, covering the detection principle, the J-PARC neutrino beam, the near detectors (ND280 and IWCD), the water-Cherenkov far detector, the detection principle and event topologies, and, finally, the physics programme;
- chapter 2 describes the reconstruction software stack, including the `WCSim` simulation package and the `fiTQun` maximum-likelihood reconstruction algorithm, with a detailed focus on the multi-ring fit procedure, the standard single-dimensional selection criteria, and the merging procedure used to handle collinear rings;
- chapter 3 presents the simulated dataset, the performance analysis of the standard reconstruction revealing the limitations of the existing cuts, and the main contribution of this work: the development and optimization of new bi-dimensional selection criteria, that include the opening angle $\Delta\theta$ alongside the likelihood ratio. The optimization strategy, the scan methodology, the final performance and the residual distributions improvements are finally presented and discussed.

Chapter 1

The Hyper-Kamiokande experiment

HK is a large infrastructure for particle and astroparticle physics being built in Japan and aiming to start operations in 2028. The main objective is to address the most important questions in science today, for instance how the universe began and evolved. HK aims to answer fundamental questions such as the origin of the baryon asymmetry in the Universe through the discovery of leptonic CPV. Relying both on a Megawatt intense neutrino beam and high-statistics atmospheric neutrino samples, HK will break the degeneracies between the effects of the MO (appendix A.2) and CPV, allowing for their measurement. Thanks to its huge detector mass, HK will search for proton decay, surpassing the existing SK limits after a few years of data taking. Moreover, HK will serve as an astrophysical observatory as well, measuring solar neutrino oscillations and other astrophysics events such as supernova bursts, relic supernova neutrinos and neutrinos in correspondence with gravitational waves and potentially detecting neutrinos from sources such as dark-matter annihilation, gamma-ray burst jets, and pulsar winds, further increasing our understanding of some of the most spectacular, and least understood, phenomena in the Universe. Finally, it will be used to perform searches for proton decay, thanks to the high-statistics due to its size and Particle IDentification (PID) capability.

HK is expected to run at least 20 years from the start of operations. The infrastructure consists of three main facilities:

- section 1.1: the neutrino beam;
- section 1.2: the Near detector (Nd);
- section 1.3: the water-Cherenkov Far detector (Fd).

No one has ever built a particle detector that can detect neutrinos (directly): because neutrinos interact only weakly with matter, their presence must be inferred from the products of their rare interactions. In water Cherenkov detectors such as HK, this is achieved by detecting the Cherenkov radiation emitted by charged secondary particles produced when a neutrino interacts in the detector volume.

1.1 The neutrino beam

The neutrino beam is produced at J-PARC (Japan Proton Accelerator Research Complex). Protons are accelerated up to an energy of 30 GeV and directed onto a graphite target 90 cm length. The interaction of the proton beam with the target produces mainly pions and kaons, whose subsequent decays provide the neutrino flux.

The J-PARC proton accelerator and the neutrino beamline have been recently upgraded [3]. The T2K (Tokai to Kamioka) experiment is currently operating with a beam intensity of 0.8 MW [4], and a further increase to 1.3 MW is planned for 2028. The expected evolution of the main ring fast extraction performance, including beam power, protons per pulse, and repetition rate, is illustrated in figure 1.1.

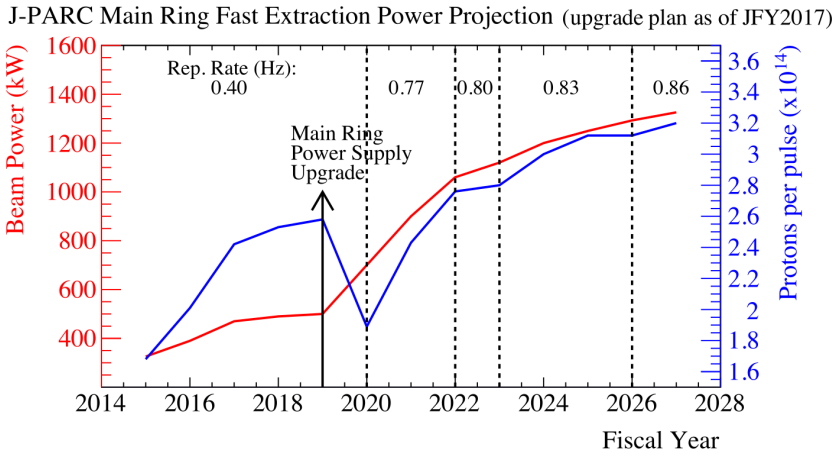


Figure 1.1. The projected main ring fast extraction performance up to 2028, including the beam power, the protons per pulse, and the repetition rate. From [5].

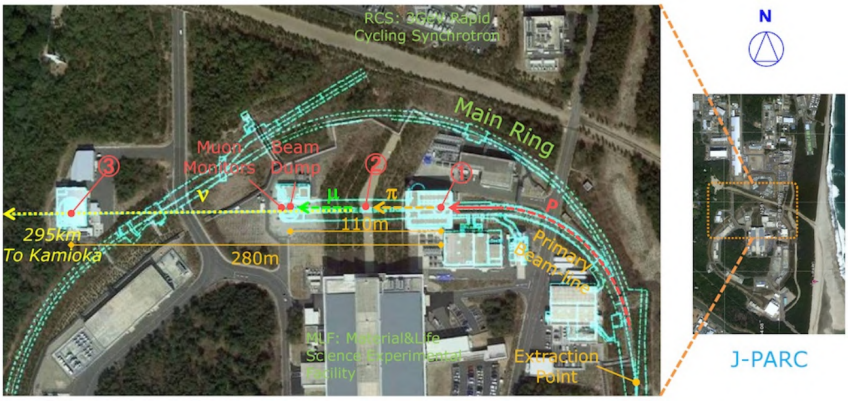


Figure 1.2. The neutrino experimental facility (neutrino beamline) at J-PARC. From [5].

An overview of the neutrino experimental facility [6] is shown in figure 1.2. The extracted proton beam is transported along a primary beamline to a production target/pion-focusing magnetic horn system in a target station. The produced pions and kaons decay during their flight through a 96 m long decay volume into muons and neutrinos. A graphite beam dump is installed at the end of the decay volume, and muon monitors placed downstream of the dump are used to measure the muon profile, which serves as a continuous monitor of the neutrino beam direction and intensity.

The beam is directed at an off-axis of 2.5° relative to the direction from J-PARC to SK [7]. This off-axis configuration allows the neutrino energy spectrum to be tuned, since the energy of neutrinos produced in two-body pion decays depends only weakly on the parent pion momentum at a fixed decay angle. The beamline has the capability to adjust the off-axis angle within a range from 2.0° to 2.5°. The centerline of the beam extends 295 km to the west, passing between the Tochibora and Mozumi sites.

1.2 The near detector

For HK to accomplish a precise determination of the leptonic CPV (δ_{CP}) it is necessary that uncertainties on the neutrino flux and interaction cross sections be controlled at the few % level. In particular, an accurate modeling of the electron neutrino to antineutrino interaction event rate, $\Delta(\sigma_{\nu_e}/\sigma_{\bar{\nu}_e})$, is needed as it could bias the measurement of δ_{CP}.

The current T2K systematic uncertainty for this ratio is 4.9%, while to fully realize HK's physics potential, the collaboration aims to reduce this to 2.7% [8]. This will be achieved through the combined operation of a new Intermediate Water Cherenkov Detector (IWCD) and the upgraded Near Detector 280 (ND280). The main tasks will be: to measure the charge of the final-state lepton, to collect large samples of ν_e and $\bar{\nu}_e$ interactions, to measure the neutrino cross section in water (same medium as the Fd), and to reconstruct the hadronic final-states with sufficient accuracy to support the extrapolation of the $\nu_\mu \rightarrow \nu_e$ cross section to the Fd.

1.2.1 ND280

The ND280 detector, shown in figure 1.3, comprises an on-axis detector, INGRID [9], and an off-axis magnetized detector ND280-OA, at an angle of 2.5° from the beam direction.

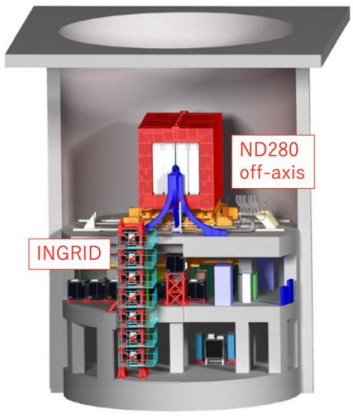


Figure 1.3. 3D model of the ND280 pit. From [10].

INGRID consists of 16 layers of plastic scintillator bars alternating with iron sheets in a cross pattern centered on the neutrino beam axis. Each layer is a sandwich of iron plates and plastic scintillator tracking planes, surrounded by scintillating veto panels that reject interactions originating outside the volume. The tracking planes are built from perpendicular arrays of scintillator bars, with the veto planes segmented along the beam direction. INGRID monitors the neutrino event rate and the beam width by spanning the 0° to 1° off-axis range, allowing it to measure the beam profile and confirm its alignment. A dedicated Proton Module has been added to improve the detection of protons produced via Charged-Current Quasi-Elastic (CC-QE) interactions. The primary purpose of the INGRID detector is to constrain the neutrino beam direction, whilst the off-axis detector is used to characterize the various neutrino interaction rates before oscillations.

ND280-OA is a detector suite situated inside the UA1 magnet, figure 1.4, originally from CERN UA1/NOMAD magnet. The 0.2 T magnetic field provides a horizontal dipole, which allows for momentum measurement and sign selection of charged particles. This capability is essential when operating in antineutrino mode, where the contamination from neutrino background is quite large. The magnet consists of water-cooled aluminum coils, which create the horizontally oriented dipole field, and a flux return yoke. The dimensions of the inner volume of the magnet are 7.0 m × 3.5 m × 3.6 m and the total weight of the yoke is 850 t. The coils are made of aluminum bars with 5.45 cm × 5.45 cm square cross sections, with a central 23 mm diameter bore for water to flow. Then, the inner tracking region is surrounded by Electromagnetic CALorimeters (ECALs) [11] and Side Muon Range Detectors (SMRDs) [12], all of which are held inside the UA1 magnet.

The SMRDs are composed of scintillator modules inserted into the gaps of the magnet yoke, serving a dual role within the ND280-OA complex. On one hand, it is designed to catch muons that leave the inner tracker at steep angles relative to the neutrino beam and to estimate their momentum from the amount of steel they traverse. On the other hand, it acts as a veto for charged particles entering the detector from outside, helping to reject cosmic-ray backgrounds and to flag events that originate not in the fiducial volume, but in the surrounding pit walls or in the magnet material itself. This external tagging capability is important for maintaining the purity of the neutrino interaction sample, especially in antineutrino running where a large fraction of the recorded events are produced by interactions in the upstream detector components.

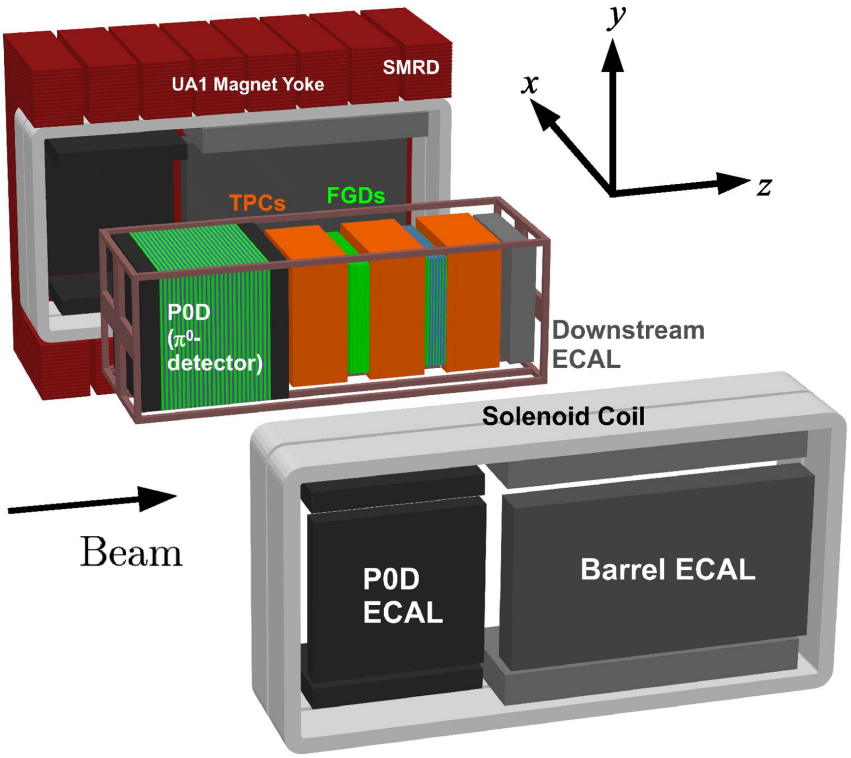


Figure 1.4. 3D view of the ND280-OA reference design. The main basket contains the π^0 detector, three TPCs, two FGDs and the downstream ECAL. The basket is enclosed within the ECALs and the UA1 magnet. The neutrino beam goes from left to right. From [6].

As illustrated in figure 1.5, the internal tracking region has recently undergone a major upgrade to improve high-angle scattering acceptance. The downstream tracking volume retains the original configuration of three vertical Time Projection Chamber (TPC) [13] separated by two Fine-Grained Detector (FGD) [14]. The primary role of the two FGDs is to provide a substantial target mass for neutrino interactions, ensuring a high event rate for cross section measurements. Equally important, they are designed to track charged particles emerging from the interaction vertex with sufficient spatial resolution to allow precise reconstruction of the event topology. Each FGD is constructed from extruded polystyrene scintillator bars with dimensions of 9.61 mm × 9.61 mm × 1864.3 mm. The bars are arranged in alternating layers along the x and y directions, perpendicular to the neutrino beam axis. This orthogonal layering provides two-dimensional spatial information for particles traversing the detector, with the third

dimension reconstructed from the timing of the scintillation signals. The overall dimensions of a single FGD are 2300 mm × 2400 mm × 365 mm, corresponding to approximately 1100 kg of active target material. The scintillator bars are read out individually, allowing for granular tracking of charged particles as they propagate through the detector volume. Each TPC uses an argon-based drift gas and MicroMegas readout planes, achieving a spatial resolution of about 0.7 mm in the readout plane. This resolution is sufficient for PID via energy loss measurements, with a muon-to-electron misidentification probability of 0.2% for tracks below 1 GeV/c.

The upstream tracking volume, formerly occupied by the π^0 detector [15], has been entirely replaced by a new tracking suite. The centrepiece of this new system is the SuperFGD, a highly granular three-dimensional scintillator target made of 2 000 000 optically-isolated 1 cm^3 plastic scintillator cubes [16]. The SuperFGD is sandwiched between two horizontal TPCs [17, 18], which extend the tracking acceptance to high-angle particles. The entire upstream structure is surrounded by a six-plane Time of Flight (TOF) detector [19], which provides precise timing information for PID and background rejection. This upgraded configuration significantly improves the reconstruction of particles emitted at large angles, a crucial capability for constraining the ν_e cross section and reducing the dominant systematic uncertainties.

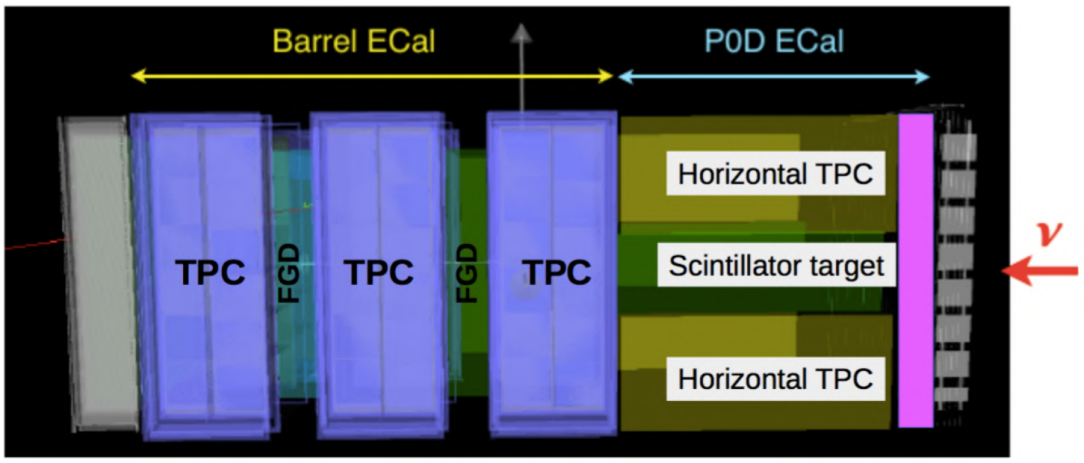


Figure 1.5. Internal configuration of the ND280-OA detector tracker region (TPCs and FGDs). The neutrino beam goes from right to left. From [5].

1.2.2 IWCD

The IWCD features a cylindrical inner detector of 7 m in diameter and 8 m in height, enclosing a total active water mass of 600 t and a fiducial mass of about 150 t. The Cherenkov light is detected by 370 multi-PMT (mPMT), each one containing 19 3-inch diameter PMTs and the integrated readout electronics.

The detector will be suspended in a vertical shaft and moved by adjusting the water level in the excavated pit, allowing the off-axis angle to be varied between 1.5° and 4° [20, 21]. This capability is at the base of the PRISM analysis technique [22]: a mono-energetic ν_μ ($\bar{\nu}_\mu$) beam is synthesised from the combination of measurements of the energy spectrum at different off-axis angles. At a baseline of 832 m, IWCD expects to detect about 17 500 Charged-Current (CC) ν_e

events in the neutrino-dominated beam mode and 19 000 $\bar{\nu}_e$ events in antineutrino-dominated beam mode.

The detector position is controlled by a system of flotation compartments, rails, and free bearings, enabling precise determination of the off-axis angle. Finally, the rejection of the ν_e background in the ν_e cross section measurement relies on the e/μ and e/γ separation, that requires a detailed characterization of the water-Cherenkov detector response. A 3D reconstruction of the IWCD is shown in figure 1.6.

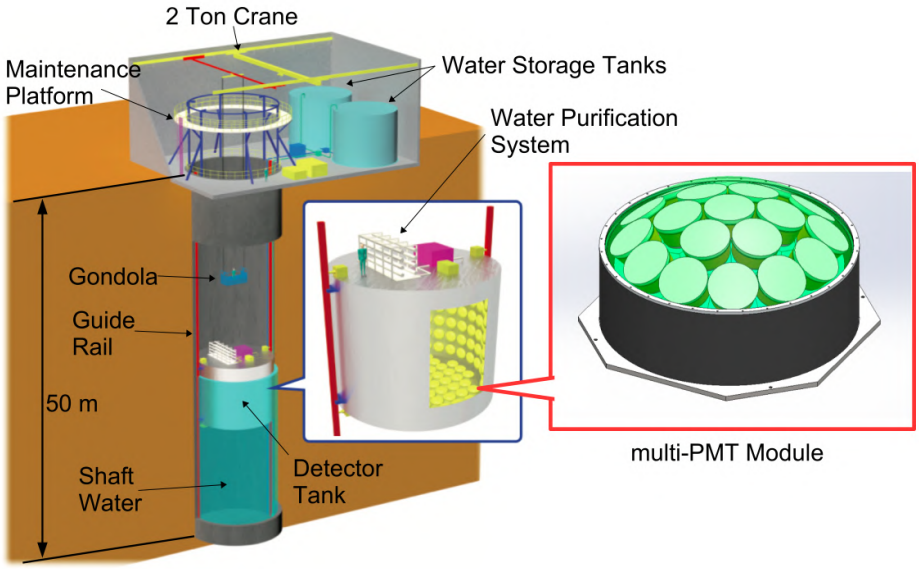


Figure 1.6. The IWCD detector, showing the vertically movable feature and the mPMT module. From [23].

1.3 The water-Cherenkov far detector

HK is the next-generation water Cherenkov neutrino Fd, located in the largest human-made cavern of the Tochibora mine at the Kamioka facility (in Hida, Gifu prefecture, Japan). It employs the ring-imaging water Cherenkov detector technique to detect rare interactions of neutrinos and the possible spontaneous decay of protons and bound neutrons. Detectors of this type are located deep underground in order to be shielded from cosmic rays and their associated secondary particles and thereby to achieve a very low background environment. HK is located $\approx$ 650 m under the peak of Nijuugo mountain. A 3D reconstruction of the entire Fd facility is shown in figure 1.7, while in figure 1.8a shows a picture of the main cavern under construction. The main cavern consists of a cylindrical-shaped tank detector, 69 m in diameter and 71 m in height, filled with 258 kt of ultrapure water. With a fiducial mass of 187 kt, eight times that of SK, it is the largest underground water tank ever built.

A detector mass of at least $O(10^2)$ kt is required in order to accumulate $O(10^3)$ electron neutrino signal events from the J-PARC neutrino beam. This is necessary to measure the CPV effect with a few % accuracy. This mass of water contains $O(10^{35})$ nucleons (protons and

neutrons) which would give an unprecedented sensitivity to nucleon lifetime at the level of 10^{35} years.

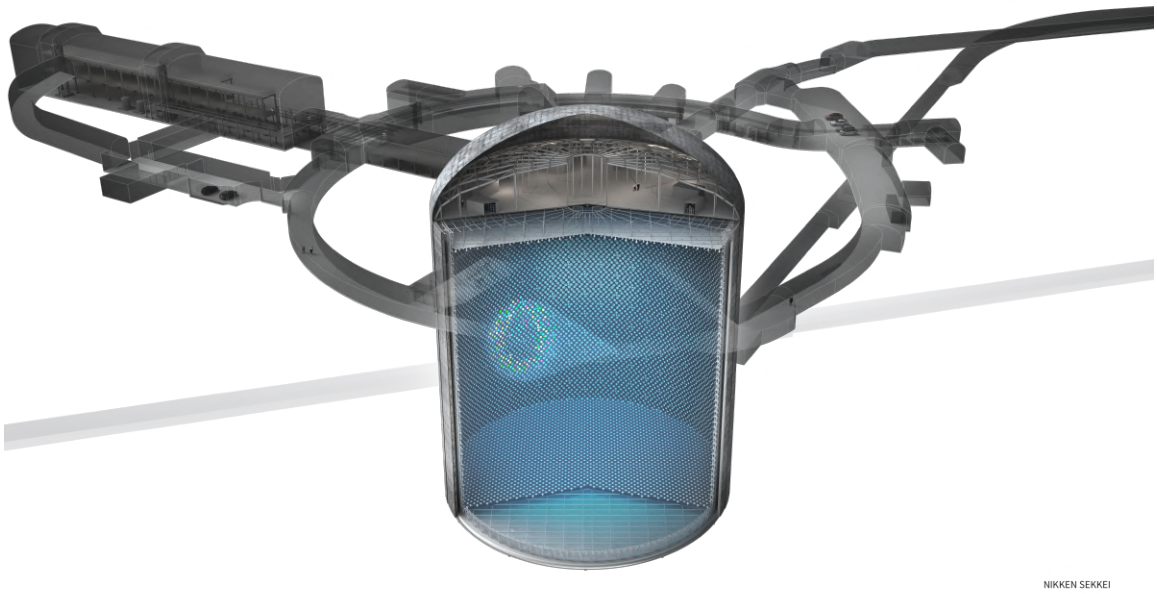


Figure 1.7. Illustration of the HK Fd. From Gallery | HK.

The Cherenkov light emitted by charged particles traversing the water volume will be detected by PMTs array. The detector is divided into two optically separated regions: an Inner detector (Id) and an Outer detector (Od).

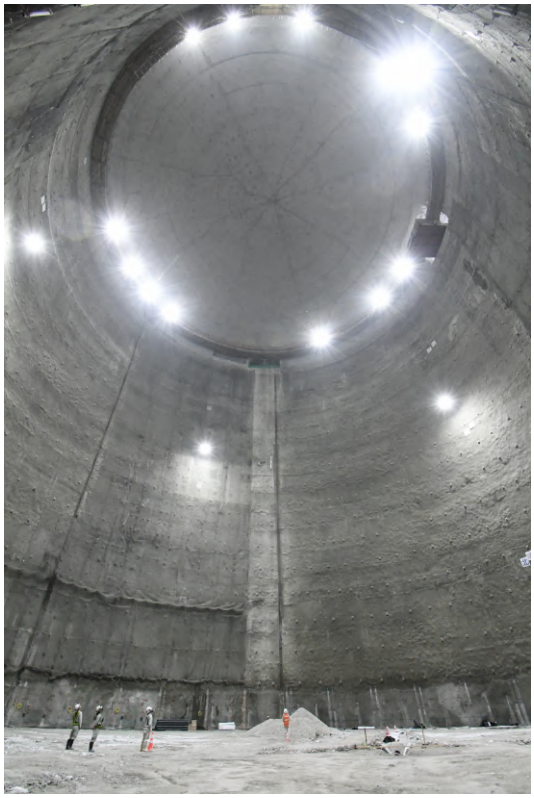
The Id serves as the primary target for neutrino interactions in both the GeV and MeV energy ranges, and it also provides the sensitive volume for the observation of proton decay products. Its photodetection system is designed to surpass the performance achieved in SK. It consists of ≈ 20 000 20-inch diameter PMTs, the same dimensions as those used in SK, together with 800 mPMTs modules pointing inward to cover 20% of the Id surface. The new 20-inch PMTs have twice the photodetection efficiency (40%) and time resolution (1.5 ns) while the dark count rate (4 kHz) is similar [24].

These improvements are expected to enhance both the reconstruction of particle momenta and the spatial resolution of the neutrino interaction vertex. To reduce the rare but not negligible risk of chain implosion over the detector operational lifetime of more than 20 years underwater, the 20-inch PMTs are put inside protective covers.

The Id system also includes 800 mPMTs, composite photosensors made of 19 3-inch PMTs that also embedded the front-end electronics. Because of their finer angular granularity, the mPMTs modules provide improved reconstruction of particle directions. They therefore complement the 20-inch PMTs, and the two photodetection systems can be cross-calibrated against one another.

The Od consists of roughly 3 600 outward-facing 3-inch diameter PMTs, each coupled to a wavelength-shifting plate to increase the overall light collection. The light yield is further enhanced by covering the cavern walls and the outer surface of the tank with highly reflective

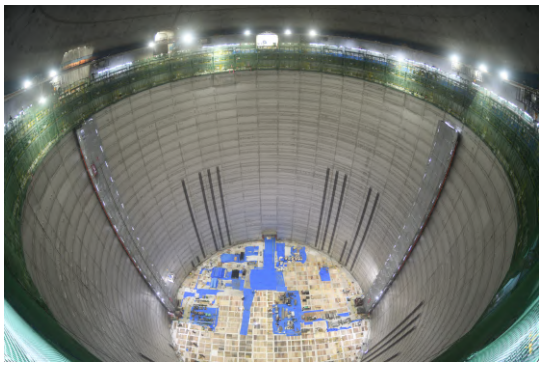
Tyvek. The Od acts as a veto from cosmic rays and generally external background and helps to identify neutrino interactions partially contained in the Id.



(a) [July 31, 2025] The main HK Fd cavern after excavation completed.



(b) [October 03, 2023] Completion of the main cavern dome section.



(c) [July 24, 2026] HK Fd tank lining completed.

Figure 1.8. Pictures from Gallery | HK.

1.3.1 Detection principle and event topologies

Physically, particle detection in water Cherenkov detectors, as suggested by the name itself, relies on the *Cherenkov effect*. The Cherenkov effect is the phenomenon that particles traversing matter with relativistic velocities emit electromagnetic radiation if their velocity exceeds the velocity of light in that medium.

- *Cherenkov effect* $\stackrel{\text{def}}{\equiv}$ "when a charged particle travels in a dielectric medium with refractive index $n > 1$, it radiates Cherenkov radiation if its velocity is larger than the phase velocity of light in the medium".

This radiation is called *Cherenkov radiation* after Pavel A. Cherenkov who received the 1958 Nobel Prize in Physics jointly with Ilya M. Frank and Igor Y. Tamm for "the discovery and the interpretation of the Cherenkov effect" [25, 26, 27].

The local disturbances to the dielectric caused by the moving charge radiate electromagnetic waves which form an electromagnetic shock wave as schematically shown in figure 1.9.

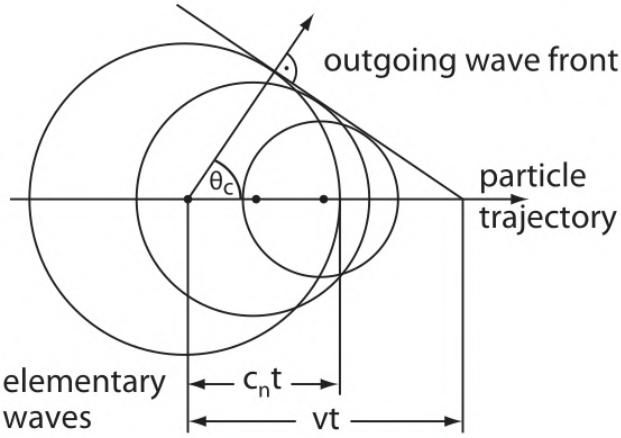


Figure 1.9. Schematic drawing of Cherenkov radiation. From [28].

This is the Cherenkov light, and it is emitted as a cone (figure 1.10) with angle θ_C relative to the particle's direction as

$$\cos(\theta_C) = \frac{1}{n\beta} \tag{1.1}$$

In water, $n \simeq 1.333$ at the typical visible light wavelength the detector is sensitive to, for a particle traveling at $\beta = 1$, Cherenkov light is therefore emitted at the angle $\theta_C \simeq 42°$. The energy threshold $E_{th} = m\,\gamma_{th}$ for the Cherenkov light emission depends on the mass of the charged particle and on the refractive index

$$\frac{E_{th}}{m} = \gamma_{th} = \frac{1}{\sqrt{1-\beta_{th}^2}} = \frac{n}{\sqrt{n^2-1}} \simeq 1.512 \tag{1.2}$$

In water the energy thresholds are $\simeq 0.7726$, $\simeq 159.8$ and $\simeq 211.0$ MeV for electrons, muons and pions, respectively.

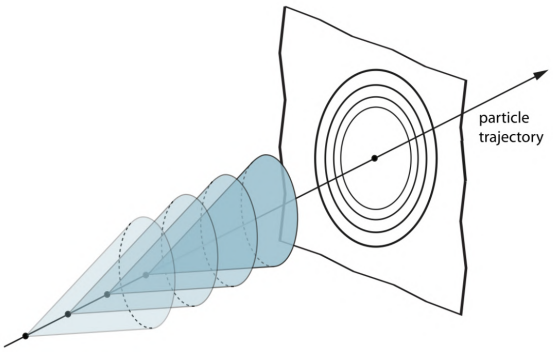


Figure 1.10. The space points of Cherenkov photons on a plane perpendicular to the particle trajectory are circles (Cherenkov rings) with different radii for different emission points. For continuous emission in a thick radiator and without focusing elements the resulting image is a broad circular structure. From [28].

When a neutrino interacts in the water volume of the HK Fd, it produces charged secondary particles, mostly electrons, muons and pions, through the processes summarized in table 1.1.

Charged-Current (CC)		
Energy	**Interaction**	**Primary process**
$\lessapprox$ 1 GeV	CC-QE	$\nu_l + n \rightarrow l^- + p$
[1, 5] GeV	CC-RES	$\nu_l + N \rightarrow l^- + N^*$
$\gtrapprox$ 5 GeV	CC-DIS	$\nu_l + q \rightarrow l^- + q'$
Neutral-Current (NC)		
Treated as *background*: no charged leptons produced in the final-state		

Table 1.1. Primary neutrino interaction channels (energy sorted).

The produced charged particles, when traveling faster than the speed of light in water, emit Cherenkov radiation as a cone of light at a characteristic angle of ($\approx 42°$) relative to their direction.

The Cherenkov photons propagate through the water and are detected by the array of PMTs on the Fd walls. Because the photons from a single particle arrive on the PMT surface in a ring-shaped pattern (figures 1.7 and 1.10), each charged particle in the final-state produces therefore a distinct Cherenkov ring. The sharpness of these rings depends on the particle type: electrons develop electromagnetic showers that produce *fuzzy* rings, while muons and pions, which do not shower, produce *sharp* rings. The number of rings observed in the detector therefore corresponds to the number of charged particles produced at the neutrino interaction vertex.

1.4 Physics programme

The primary goal of HK is the discovery of CPV in the leptonic sector and precision measurements of the neutrino oscillation parameters in the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix.

While water Cherenkov detectors cannot distinguish positron from electron rings through electromagnetic shower topologies alone, the main source of uncertainty in $\Delta(\sigma_{\nu_e}/\sigma_{\bar{\nu}_e})$ is related to the asymmetric contribution of inelastic scattering channels. At the energy scales of HK, neutrino interactions frequently occur via CC-RES and CC-DIS which produce multi-ring topologies. The corresponding final-states of these interaction processes, which determine the expected ring multiplicity and the associated PID challenges, are listed in table 1.2. The cross sections of these inelastic interactions are different between ν_e and $\bar{\nu}_e$.

The modeling of the electron neutrino to antineutrino cross section ratio relies on high purity samples, in particular isolating single-ring CC-QE events from CC-RES and CC-DIS backgrounds. Therefore, multi-ring reconstruction is crucial because a misclassification would affect the neutrino and antineutrino samples asymmetrically, due to their different cross sections, resulting in a background contamination that enters as a systematic error.

In light of this, the optimization and enhancement of multi-ring reconstruction performance in ring counting would result in an improvement of the sample purity for the measurement of

electron neutrino to antineutrino cross section ratio.

Process	final-state
CC-RES / CC-DIS	a charged lepton + one or more charged pions
NC	one or more charged pions (no leptons)
CC with π^0	a charged lepton + $\pi^0 \rightarrow$ lepton + two rings e-like
NC with π^0	$\pi^0 \rightarrow 2\gamma \rightarrow$ two rings e-like

Table 1.2. Common final-state topologies that give rise to multi-ring signatures in water Cherenkov detectors.

1.4.1 CPV and mixing parameters

Neutrino oscillations

Historically, experimental constraints indicate that neutrino mass is very small. Direct kinematic measurements of weak decay processes [29] and the observed left-handed helicity of neutrinos [30] are consistent with zero mass. Nevertheless, the discovery of neutrino oscillations revealed the Standard Model to be incomplete: neutrino oscillations require neutrinos to have non-zero masses.

Therefore, assume that neutrinos have non-zero masses and consider free neutrinos propagating in vacuum. If massive, the lepton flavor is not conserved during the neutrino propagation, as first pointed out by Bruno Pontecorvo [31, 32] by analogy with the neutral kaon system. In the following treatment the *minimal hypothesis* for the number of massive neutrinos is assumed: three as the neutrino flavors. The number of so-called *sterile neutrinos*, n_s, is hence assumed to be zero. Although more mass eigenstates $|\nu_i\rangle$ with $i \in \{1, \ldots, 3 + n_s\}$ could in principle exist, the assumption is justified by the fact that sterile neutrinos cannot interact via weak interactions.

Follows that neutrino states can be described in two distinct orthonormal bases.

- $|\nu_\alpha\rangle$ with $\alpha \in \{e, \mu, \tau\}$ which are the *flavor eigenstates* of the weak interaction.
- $|\nu_i\rangle$ with $i \in \{1, 2, 3\}$ which are the *mass eigenstates* of the free particle Hamiltonian.

The neutrino state evolves in time on the mass basis, but the weak interactions occur in the flavor basis, therefore the phase differences induced by neutrino propagation are responsible for the neutrino oscillation in the flavor space. Since each set forms a complete orthonormal basis of the same Hilbert space, $\{|\nu_\alpha\rangle\}_\alpha$ and $\{|\nu_i\rangle\}_i$, the two bases are connected by a unitary mixing matrix U:

$$|\nu_\alpha\rangle = \sum_i U^*_{\alpha i} |\nu_i\rangle \tag{1.3}$$

$$|\nu_i\rangle = \sum_\alpha U_{\alpha i} |\nu_\alpha\rangle \tag{1.4}$$

The $U_{\alpha i}$ matrix is analogous to the Cabibbo-Kobayashi-Maskawa (CKM) matrix [33], which describes the quark mixing in weak interactions, and it is known as the PMNS matrix [34, 35]. A unitary matrix $N \times N$ has N^2 real free parameters, $N(N-1)/2$ of which can be considered

as rotation angles, while the remaining $N(N+1)/2$ real parameters can be read as complex phases. Thus, a 3×3 unitary matrix U is defined by 3 mixing angles and 6 complex phases. Assuming neutrinos as Dirac fermions [36], for the lepton mixing matrix we have

$$N = 3 \implies \begin{cases} \frac{N(N-1)}{2} = 3 & \text{mixing angles,} \\ \frac{(N-1)(N-2)}{2} = 1 & \text{physical phase.} \end{cases} \tag{1.5}$$

Therefore, the standard parametrization of the PMNS matrix is

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_{23} & \sin\theta_{23} \\ 0 & -\sin\theta_{23} & \cos\theta_{23} \end{pmatrix} \begin{pmatrix} \cos\theta_{13} & 0 & \sin\theta_{13} e^{-i\delta_{\mathrm{CP}}} \\ 0 & 1 & 0 \\ -\sin\theta_{13} e^{i\delta_{\mathrm{CP}}} & 0 & \cos\theta_{13} \end{pmatrix} \begin{pmatrix} \cos\theta_{12} & \sin\theta_{12} & 0 \\ -\sin\theta_{12} & \cos\theta_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.6}$$

where θ_{12}, θ_{13} and θ_{23} are the mixing angles and δ_{CP} is the complex physical phase. If neutrinos are *Majorana fermions* $\psi = \psi^{\mathrm{C}} = C\overline{\psi}^{\mathsf{T}}$, i.e. if a neutrino is its own antiparticle, the mixing matrix is extended by two additional complex phases α_1 and α_2, known as Majorana phases. Under this assumption the mixing matrix is modified according to $U \longrightarrow U \times \mathrm{diag}(1, e^{i\alpha_1}, e^{i\alpha_2})$ as opposed to Dirac fermions. In such case, the extra Majorana phases are $N - 1$.

In appendix A, we discuss the two-flavor approximation for neutrino oscillations in vacuum (appendix A.1) and the neutrino MO (appendix A.2).

CPV in neutrino oscillations

For antineutrinos mixing is described through the complex conjugate of the mixing matrix, as shown in equation (1.4). In particular, if δ_{CP} is neither 0 nor π, it follows that the mixing matrix is different between neutrinos and antineutrinos. For this reason, δ_{CP} is named as *CPV phase*.

In this case, the sign of the third (interference) term in equation (A.7) is reversed for antineutrinos, indicating that the oscillation probabilities for neutrinos and antineutrinos may be different. From appendix A, this difference is quantified by the CP asymmetry, defined as

$$A^{\mathrm{CP}}_{\alpha\beta} = P(\nu_\alpha \longrightarrow \nu_\beta) - P(\overline{\nu_\alpha} \longrightarrow \overline{\nu_\beta}) = 4 \sum_{i>j} \mathrm{Im}\left[U^*_{\alpha i}\, U_{\beta i}\, U_{\alpha j}\, U^*_{\beta j}\right] \sin\left(\frac{\Delta m^2_{ij}\, L}{2E}\right) \tag{1.7}$$

Thus, CPV in the lepton sector can be probed through neutrino oscillation measurements.

The oscillation analyses planned for HK relies on the framework developed by T2K, with adaptations that account for the new features of the HK design, in particular the upgraded T2K Nd and the addition of the IWCD. Having a larger fiducial mass and the higher beam power, HK will be able to collect thousands of selected events in both the ν_e ans $\bar{\nu}_e$ appearance channels.

A distinctive feature of the HK physics programme is that it will have access, within a single experimental setup, to both accelerator neutrino data and high-statistics atmospheric neutrino samples. As demonstrated by the recent joint analysis performed by the T2K and SK collaborations [37], this combination of independent datasets will allow HK to break the degeneracies between δ_{CP} and MO.

At the moment, SK provides the most sensitive measurement of MO [38]. HK is projected to determine MO with a significance of 3.8σ (6.2σ) C.L. for $\sin\theta_{23} = 0.4$ (0.6) after 10 years of

data taking. Even in the scenario where the MO remains undetermined, the sensitivity to CPV would not be significantly degraded, as shown in figures 1.11 and 1.12.

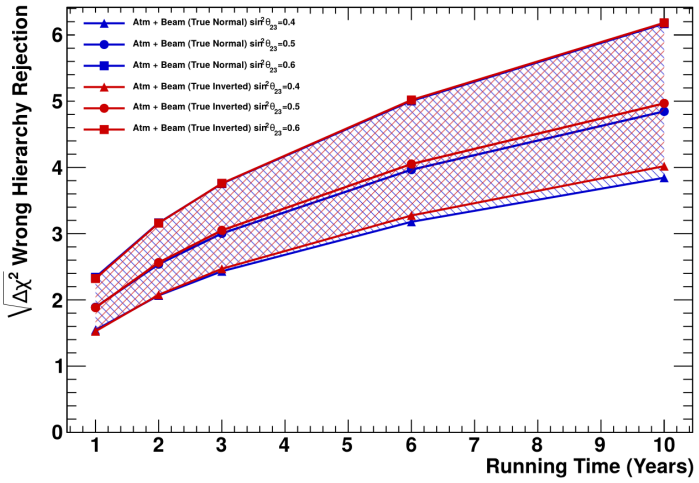


Figure 1.11. Expected sensitivity to the mass hierarchy as a function of time assuming $\sin^2(\theta_{23}) = 0.4$ (triangle), 0.5 (circle), and 0.6 (square) from a combined analysis of atmospheric and accelerator neutrinos data at HK. Blue (red) colors denote the normal (inverted) hierarchy. From [5].

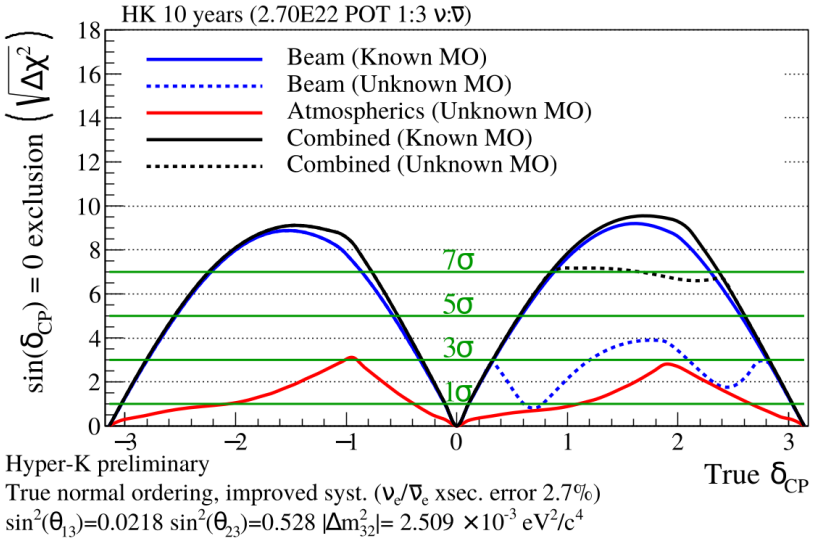
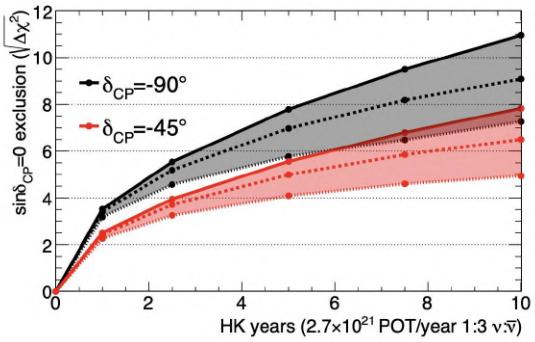


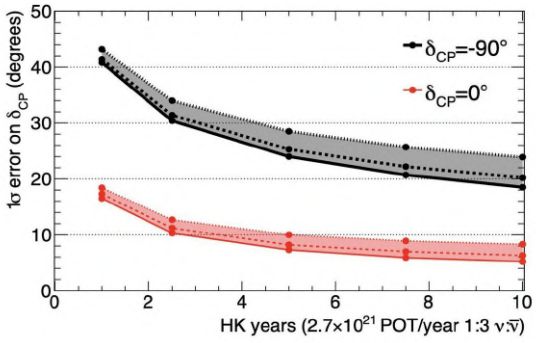
Figure 1.12. Sensitivity to exclude $\sin(\delta_{CP}) = 0$, as a function of true δ_{CP} value, for 10 HK years and true normal MO. The large sample of atmospheric neutrinos improves the sensitivity, particularly in the case of unknown MO. The sensitivity for the opposite MO is analogous after swapping curves between positive and negative values of true δ_{CP}. From [10].

To isolate the contribution of the accelerator neutrino beam, a dedicated sensitivity model was constructed by modifying the uncertainties assigned to the individual systematic parameters of the T2K framework, while preserving the correlation structure between them. Particular attention was given to the improvement in the uncertainty in $\sigma(\nu_e)/\sigma(\bar{\nu}_e)$ which is a major systematic for HK. For comparison in the following plots, sensitivities obtained using the

original T2K uncertainties and those assuming purely statistical errors are also reported. These results highlight the impact that the systematic error model has on the projected physics reach of HK. In all projections, a neutrino-to-antineutrino beam running ratio of 1:3 is assumed.



(a) Sensitivity to exclude $\sin(\delta_{CP}) = 0$, as a function of HK years, for true $\delta_{CP} = -\pi/2$ and $\delta_{CP} = -\pi/4$. The shaded areas show the span of possible values, for varying systematic error models.



(b) 1σ resolution of δ_{CP} as a function HK years for $\delta_{CP} = -\pi/2$ or 0.

Figure 1.13. In both plots, the three lines denote different systematic error scenarios: the *solid* line indicates statistical errors only; while the *dashed* line indicates improved systematic uncertainties, assuming a 2.7% $\nu_e/\overline{\nu}_e$ cross-section error; finally the *dotted* line indicates T2K systematic uncertainties, assuming a 4.9% $\nu_e/\overline{\nu}_e$ cross-section error. From [10].

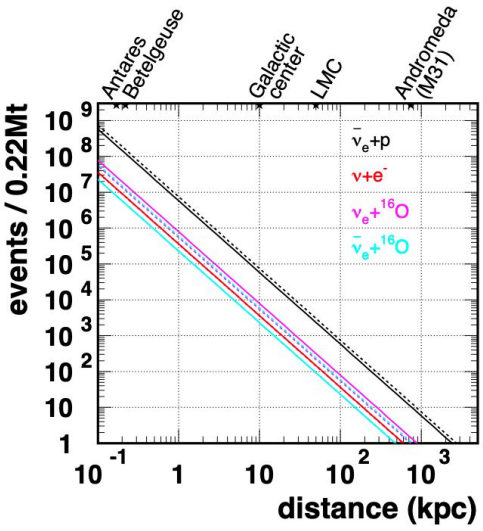
In the scenario of maximal CPV in the leptonic sector, as shown in figure 1.13a, HK is expected to establish CPV at a confidence level exceeding 5σ after 3 or 4 years of data taking, for $\delta_{CP} = -\pi/2$ and normal ordering, thanks to its large statistics and reduced systematic errors. After 10 years of data taking, CPV would be observed at more than $> 5\sigma$ C.L. for 60% of the possible values of δ_{CP}. The precision in measuring $\delta_{CP} = -\pi/2$ will range from 20°, in case of maximal CPV, to 6°, for the CP conserving values, as shown in figure 1.13b.

1.4.2 Astrophysical observatory

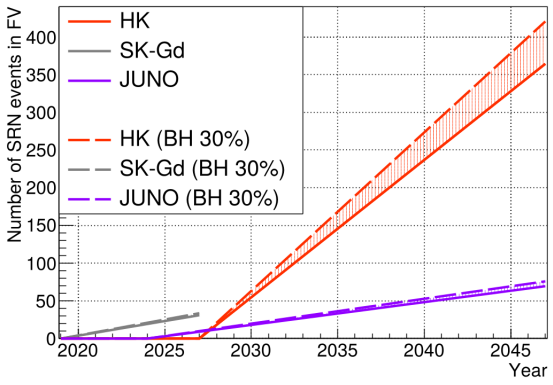
Neutrinos are among the most abundant particles in the Universe, produced in a wide range of astrophysical phenomena including stellar interiors, supernova explosions, and the accretion processes of compact binary systems. In particular, Core-Collapse SuperNovae (CCSN) represent one of the most compelling targets for neutrino astronomy. These events mark the terminal stage in the evolution of massive stars, occurring when the iron core can no longer sustain further fusion and undergoes gravitational collapse. The energy released in such an event is estimated to be $\approx 3 \times 10^{53}$ erg of which nearly 99% is carried away by neutrinos. The detection of these neutrinos therefore provides direct information about the energy flow during the explosion, information that is inaccessible through electromagnetic observations alone.

HK will be able to observe the neutrino flux from the next galactic CCSN in unprecedented detail. Once a galactic supernova happens, this ability will be a powerful tool for guiding simulations towards a precise reproduction of the explosion mechanism observed in nature. The detector low-energy threshold of 3 MeV and its event-by-event directional sensitivity will enable precise reconstruction of the supernova direction, with an accuracy of about $1^\circ \sim 1.3^\circ$.

HK will observe the antineutrino emitted in the explosion of supernovae through the inverse β-decay, which constitutes $\approx 90\%$ of the expected signal, $\bar{\nu}_e + p \rightarrow e^+ + n$, and the neutrinos through their scattering with electrons, $\nu + e^- \rightarrow \nu + e^-$, which provides strong directional information, since electrons are mostly scattered in the forward direction. [39]. The number of expected events as a function of the distance of the supernova explosions are shown in figure 1.14a.



(a) Expected number of supernova burst events as a function of the distance in the HK tank.



(b) Expected number of DSNB interactions in the HK fiducial volume as a function of the years compared with SK and JUNO.

Figure 1.14. From [10].

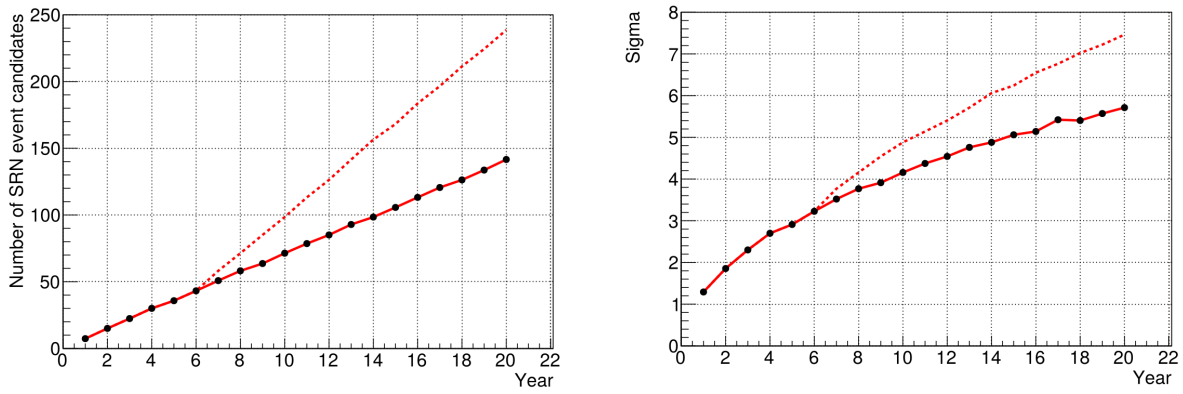
HK will have the unprecedented ability to detect neutrinos from supernovae beyond the Milky Way: for a supernova in the Large Magellanic Cloud, at 50 kpc distance, HK would detect about 3000 events, while for a supernova in the Andromeda galaxy (M31), at 780 kpc, $O(10)$ events are expected.

In addition, HK will be sensitive to the Diffuse Supernova Neutrino Background (DSNB), also known as Supernova Relic Neutrinos (SRN). This background consists of the integrated flux of neutrinos emitted by all supernovae throughout the history of the Universe. Its detection would provide constraints on the history of stellar collapse, the formation of heavy elements, and the fraction of optically failed supernovae that collapse directly to black holes. The energy spectrum of the DSNB is expected to peak in the range of a few MeV, making it a challenging signal to extract from the backgrounds of reactor neutrinos, atmospheric neutrinos, and spallation products. The expected number of events as a function of the years of data taking is shown in figure 1.14b. HK aims to discover and characterize the diffuse supernova flux owing to a high statistics sample of antineutrino inverse β-decay interactions. With a 10 years exposure, about 70 DSNB events are expected in the energy range from 16 to 30 MeV, figure 1.15a, corresponding to a discovery significance of 4.2σ, figure 1.15b.

Solar neutrino measurements will also achieve high precision. The detector will be sensitive to the day-night asymmetry of solar neutrinos flux caused by terrestrial matter effects (as

indicated by SK [40]), and will contribute to the resolution of the tension between the value of Δm^2_{21} extracted from solar neutrino data and that measured by KamLAND.

Through the observation of neutrinos with energies $\sim$ 10 MeV, with time, energy, and directional information, HK will serve as multi-messenger observatory as well. It has the potential to detect thermal neutrinos from nearby, $<$ 10 Mpc, neutron star mergers events in coincidence with gravitational waves. Finally, additional astrophysical targets include solar flares, magnetars, pulsar wind nebulae, active galactic nuclei, and gamma-ray bursts.



(a) Number of observed SRN events as a function of observation period.

(b) SRN discovery sensitivity as a function of observation period.

Figure 1.15. The solid red curve assumes continuous operation of a single detector tank, while the dashed red curve corresponds to a staged scenario in which a second tank is added after the initial running phase. From [5].

1.4.3 Proton decays

The search for nucleon decays is one of the main scientific goals of the physics programme. Therefore, HK has been designed accordingly: the huge fiducial mass of the detector, which corresponds to an unprecedented number of target protons and neutrons, is intended to push HK sensitivity beyond the current limits established by SK, as shown in figure 1.16.

While the decay mode $p \rightarrow e^- \pi^0$ is predicted to be dominant in many Grand Unified Theory (GUT) models, a variety of other decay modes are possible with different mesons and leptons final-state, e.g. $p \rightarrow \nu K^+$, each with a significant branching fraction [5].

Since the predicted branching fractions differs strongly between models, a discovery is unlikely to come from a single mode. Instead, it will be important to search across many possible decay signatures in parallel. The relative strengths of any observed signals would then help discriminate between competing theoretical frameworks, potentially revealing the underlying symmetry structure of the unification group.

The $p \rightarrow e^- \pi^0$ channel has a clean experimental signature: since all final-state particles are visible, the invariant mass of the decay products can be fully reconstructed and should peak at the proton mass for signal events. If the proton lifetime is near the current experimental bound, HK could observe a signal at the 3σ level within a few years of operation. In the absence of

a signal, the limit will be extended to beyond 10^{34} years, placing stringent constraints on the simplest GUT models.

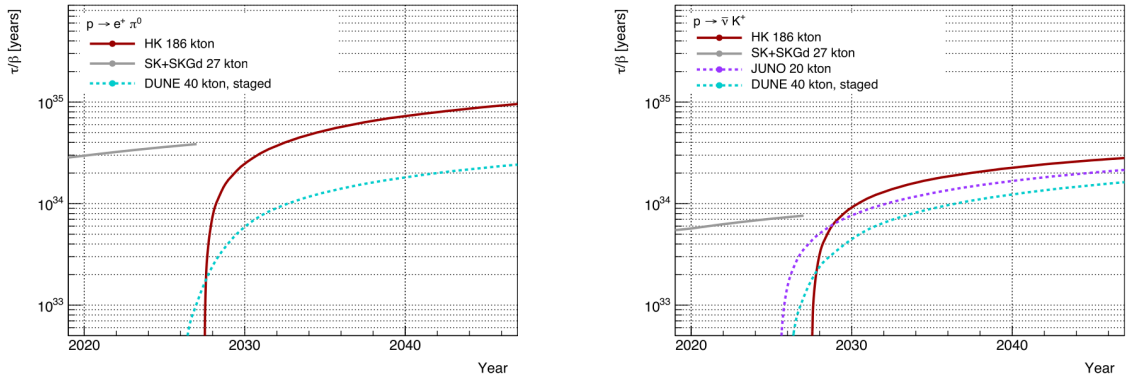


Figure 1.16. HK predicted sensitivity to the $p \rightarrow e^- \pi^0$ (left) and to the $p \rightarrow \nu K^+$ (right) channels for proton decay compared with SK, DUNE and JUNO experiments. From [10].

The $p \rightarrow \nu K^+$ channel presents a different set of experimental challenges. The kaon is produced with a momentum of only 340 MeV/c, below the Cherenkov threshold in water, and therefore cannot be observed directly. Instead, it must be identified through its decay products $K^+ \rightarrow \mu^+ \nu_\mu$ (branching fraction $\approx$ 64%) and $K^+ \rightarrow \pi^+ \pi^0$ (branching fraction $\approx$ 21%). Both are two-body decays, producing mono-energetic final-state particles. Furthermore, the 12 ns lifetime of the kaon makes it possible to observe prompt gamma-ray emission produced when the proton hole leftover in the oxygen nucleus from a bound proton decay is filled by the de-excitation of another proton. For ^{16}O, the probability of producing a 6 MeV gamma-ray from this process is $\approx$ 40%, making it a powerful tool for identifying the kaon decay products and suppressing atmospheric neutrino backgrounds.

Chapter 2

HK reconstruction software stack

The four main design rules driving the development of the HK software are: *adaptability*, *reliability*, *understandability* and *low overhead* [5].

The HK experiment is expected to run for at least 20 years from the start of operations. This period of time is typically longer than one generation of software and infrastructure; therefore the code development should be flexible enough to accommodate changes and updates in the software tools. *Adaptability*.

Then, each component of the software should exhibit a well-known and expected behavior on the control samples. *Reliability*.

To make it easier for newcomers to start working with the HK software, it is crucial to write code that is as clear as possible, with enough comments and precise documentation. The documentation should explain the main purpose of the programme, what each of its components does, point out every single dependency and describe all the inputs and outputs, which are essential for using it correctly. *Understandability*.

Finally, the main focus of the experiment should not be lost: extracting extremely high-quality physics measurements. Therefore, the collaborators should spend as little time as possible on the management and maintenance of the code, thus requiring the effort to automate these two activities. *Low overhead*.

The code language for the HK software is `C++`, while the output files are written in `ROOT` data format [41]. The software is built on a collection of loosely-coupled packages. Each package is hosted on a third-party *software forge* platform, GitLab [42].

Some packages are open source and commonly used in other high-energy physics experiments, whilst others have been developed specifically for HK. An important aspect of the code management and development is the "distributed" nature and loose-coupling between packages. Each package should be as independent as possible from the others, so that researchers can develop their own versions of the software separately. This modularity and scope separation between packages finally allows the easy replacement of "end-of-life" packages with better alternatives, minimizing the impact on the rest of the system.

Usually, the flow of a simulation proceeds as follows.

First, the neutrino interaction event topologies are generated by an event generation package (e.g. GENIE [43] and NEUT [44]); second, the detector response to these topologies is modeled via a Monte Carlo (MC) simulation package: the "Water Cherenkov Simulation" `WCSim` [45].

Finally, the reconstruction of event information is performed depending on their energy: for low-energy events the package used is `BONSAI`, while for high-energy it is `fiTQun`. This is the general picture of the event simulation and reconstruction workflow; in the next sections a deeper and more detailed description of the software involved in this thesis work will be provided. In particular, the focus will be oriented to the `WCSim` simulation package and the `fiTQun` reconstruction one, emphasizing the core topic of this work: the topology of *multi-ring* fit reconstruction.

2.1 WCSim

The Water Cherenkov Simulation package WCSim is a flexible, Geant4-based, open source software designed to simulate the geometry and physics response of user-defined water Cherenkov detector configurations. The source code of WCSim is free and open access at WCSim | GitHub.

The user-defined detector configuration should provide the following WCSim inputs: the type and number of PMTs, the detector geometry, radius and height and whether the water is doped or not with gadolinium. Then, WCSim relies on these inputs to simulate the detector response, needed to determine the physics potential. Indeed, the final performance of the HK detector depends on the detector geometry, the type of PMTs, and the photocoverage.

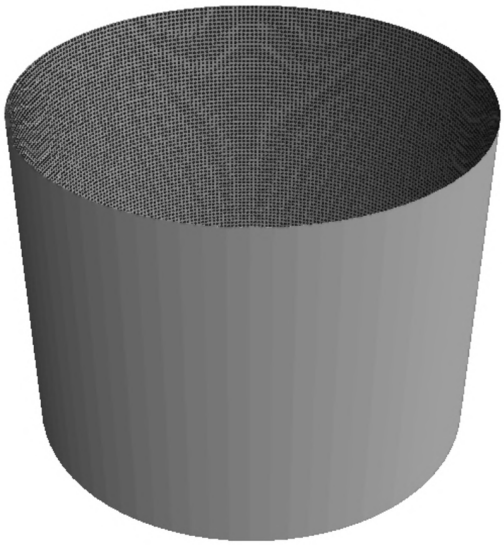
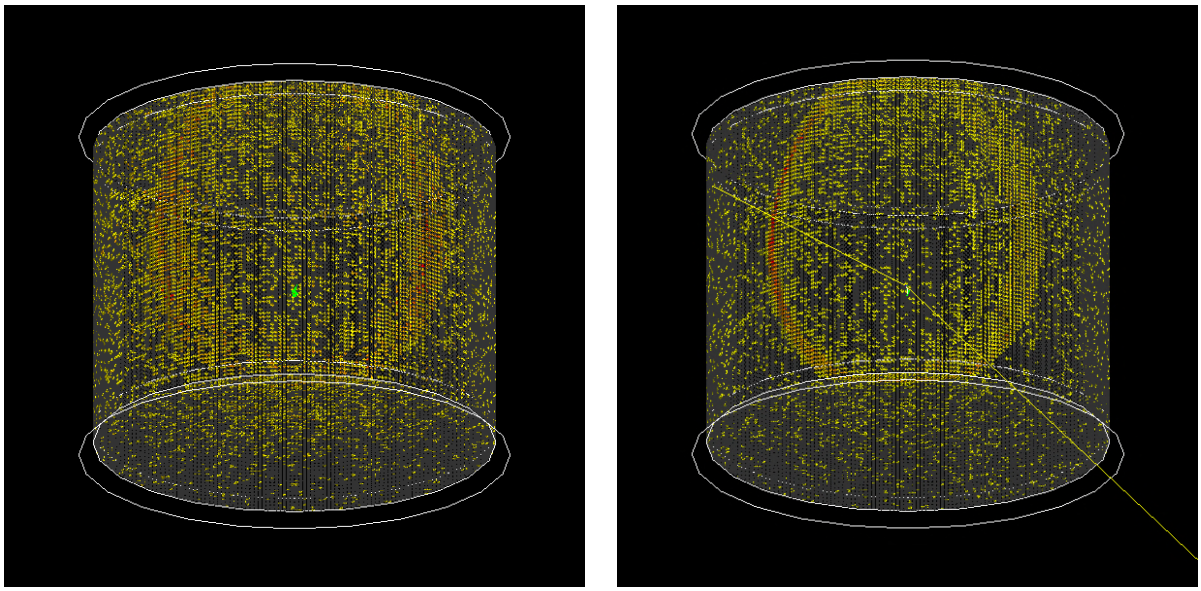


Figure 2.1. Geant4 visualization of the HyperK detector configuration; the top cap has been removed for visualization purposes. PMTs are shown in black, while the barrel side walls in grey.

For the simulations used in this thesis, the Id of HK is modelled with the 50 cm Hamamatsu R12860 PMTs (labeled as HKPMT, R12860), that are the upgraded version of the SK PMTs (SKPMT, R3600 [46]) developed specifically for HK. The simulated HK Id geometry is shown in figure 2.1. The simulation of PMTs parameters includes the time resolution, dark noise rate and the overall efficiency for a photon to register a charge (quantum efficiency, collection efficiency and hit efficiency). The parameters are taken from SK simulation code SKDETSIM. The main dependency of WCSim is Geant4 software: Geant4 is used to simulate the kinematics of the event, track the particles crossing the detector and compute the final deposited energy. When Cherenkov photons reach the PMTs, a certain fraction of them is registered as raw hits, taking into account the quantum efficiency and collection efficiency of the PMTs. The (raw) hits are then digitized based on the old electronics scheme used in SK until 2001, SKI [47]. The main disadvantage of this scheme compared to the new HK one is its inability to tag neutrons. The consequence is that the background rate obtained with the older SKI electronics is approximately twice as high as that achieved with QBEE, the upgraded SK front-end electronics system, which provides neutron tagging capability through the detection of delayed neutron-capture signals [48].

Both the raw and the digitized hits are stored in the WCSim output file. An example event display of simulated hits in HK is shown in figure 2.2. The raw hits data include the information of which tubes were hit and how many times; whilst the digitized hits count the number of hits in a certain trigger time window, as well as the charge and time of the hit tubes. As mentioned before and explained in the next section, for the sake of this thesis work the WCSim output has been given as input to fiTQun, to carry out the event reconstruction performance study.



(a) 1 GeV electron ring. (b) 1 GeV muon ring.

Figure 2.2. Event display in the HK detector.

2.2 fiTQun

When an event is recorded in HK, the main goal of the reconstruction is to identify the particle type and the kinematics, so the vertex of the interaction and the particle momentum. This section describes the event reconstruction algorithm `fiTQun`.

The `fiTQun` algorithm represents the "state of the art" for water Cherenkov reconstruction. It performs a multi-ring maximum likelihood fit, based on the time and charge information recorded by the PMTs. The algorithm was initially developed for SK and T2K experiments, and it was based on the method used in the MiniBooNE experiment [49]. The previous reconstruction tool was `APfit` [50], whilst the new SK version of `fiTQun` implements additional features such as the advanced multi-ring fit.

The `fiTQun` algorithm is based on the maximization of a likelihood function. Operatively, the reconstruction procedure consists of the following steps:

1. section 2.2.2: vertex pre-fit;
2. section 2.2.3: single-ring fit;
3. section 2.2.4: multi-ring fit.

This section provides only a brief overview of the likelihood function and the successive reconstruction steps. A comprehensive technical description of the `fiTQun` algorithm can be found in [51, 52, 53].

Assuming that the PMTs behave independently, the likelihood function to be maximized for an observed set of PMT measurements given track parameters $\boldsymbol{x}$ can be expressed as

$$L(\boldsymbol{x}) = \prod_{j}^{\text{unhit}} P_j(\text{unhit}|\boldsymbol{x}) \cdot \prod_{i}^{\text{hit}} \{1 - P_i(\text{unhit}|\boldsymbol{x})\} \cdot f_q(q_i|\boldsymbol{x}) \cdot f_t(t_i|\boldsymbol{x}) \tag{2.1}$$

where $\boldsymbol{x}$ is a vector of seven parameters characterizing an event hypothesis:

- the starting point: x_0, y_0, z_0
- the starting time: t_0
- the direction: θ_0, ϕ_0
- the kinetic energy: E_0

The index j runs over all unhit PMTs, while the index i runs over all hit PMTs. For each of the hit ones, the information about the produced charge and the time of arrival of the first incident photon is recorded in the event data structures.

- $P_j(\text{unhit}|\boldsymbol{x})$ is the probability that the jth PMT does not register a hit for the given hypothesis $\boldsymbol{x}$;
- q_i, t_i are the measured charge and time on the ith hit PMT;
- $f_q(q_i|\boldsymbol{x})$ is the Probability Distribution Function (PDF) for the measured charge on the ith PMT, given $\boldsymbol{x}$, evaluated at q_i;
- $f_t(t_i|\boldsymbol{x})$ is the PDF for the measured time given $\boldsymbol{x}$, evaluated at t_i.

fiTQun looks for the hypothesis $\boldsymbol{x}$ which maximizes $L(\boldsymbol{x})$ and uses it as the best estimate for the particle configuration in the event.

2.2.1 Predicted charge

In order to be able to compute the likelihood, the particle processes and the optical photon propagation are decoupled from the PMTs response and the electronics by the introduction of the *predicted charge* μ_i.

> *Predicted charge* $\overset{\text{def}}{\equiv}$ "the mean number of photoelectrons that are produced at the ith PMT for a given hypothesis".

For a single particle hypothesis with parameters vector $\boldsymbol{x}$, the predicted charge contribution from direct μ_i^{dir}, indirect light due to scattering with water μ_i^{sct}, and charge due to dark count μ^{drk} are taken into account separately.

For a multi-particle hypothesis, the predicted charges for each particle are calculated separately and then summed. If n is the number of particles, the total predicted charge is calculated as

$$\mu_i = \sum_n (\mu_{i,n}^{\text{dir}} + \mu_{i,n}^{\text{sct}}) + \mu^{\text{drk}} \tag{2.2}$$

For a single particle hypothesis, μ^{dir} and μ^{sct} are given as integrals along the particle track length s, defined as "the distance the particle has traveled from the point where it began to emit Cherenkov light"

$$\mu^{\text{dir}} = \Phi(p) \int g(p, s, \cos\theta)\ \Omega(R)\ T(R)\ \epsilon(\eta)\ \mathrm{d}s \tag{2.3}$$

$$\mu^{\text{sct}} = \Phi(p) \int \frac{1}{4\pi}\ \rho(p, s)\ \Omega(R)\ T(R)\ \epsilon(\eta)\ A(s)\ \mathrm{d}s \tag{2.4}$$

where p is the momentum of the particle at the position where it started to emit Cherenkov light; R, θ and η are variables to indicate the relative position between the PMT and the particle as shown in figure 2.3. Φ is the photon yield, a normalization factor proportional to the average total number of photons for a particle with initial momentum p. The Cherenkov emission profile is g, while Ω is the PMT solid angle factor, T is the light transmission factor, ϵ is the PMT angular acceptance, ρ is the photoemission fraction[1] and A is the ratio of charge generated by direct light to that generate by indirect light.

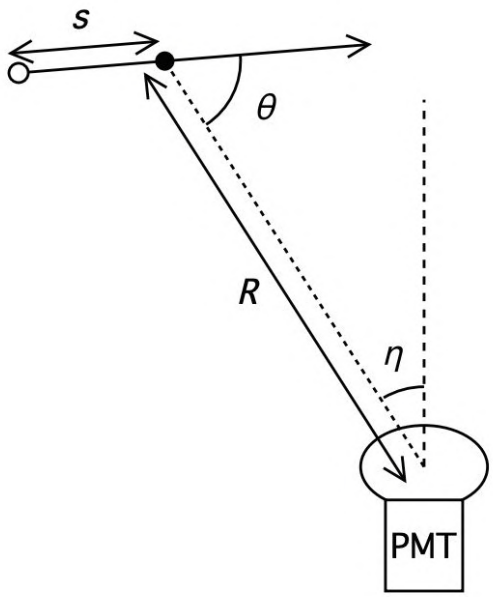


Figure 2.3. Schematic diagram illustrating the relationship between the variables for the predicted charge calculations in equation (2.3) and equation (2.4). The vertex position of the particle is shown as the white dot. From [52].

2.2.2 Vertex pre-fit

As mentioned before, `fiTQun` searches for a particle event hypothesis $\boldsymbol{x}$ which maximizes the likelihood function defined in equation (2.1). In order to ensure that the minimization of the *negative logarithmic likelihood* `nll` $(-\ln(L(\boldsymbol{x})))$ with respect to $\boldsymbol{x}$ converges, proper initial *seed* parameters are needed. The vertex pre-fitter is a fast algorithm which uses only the hit time information from the PMTs in a time window of $(-100\,\text{ns},\ 400\,\text{ns})$ around the event trigger to roughly estimate the vertex position $\bar{x} = (x_0, y_0, z_0)$ and time t which can then be used as the parameter seed for the `nll` minimization. It is run at the beginning of processing an event before the minimization is run. The estimation is done by maximizing the *goodness* function G

[1] The photoemission fraction ρ is defined as "the fraction of photons emitted per unit track length, at position s along the particle trajectory"

$$\rho(p, s) \overset{\text{def}}{\equiv} \int g(p, s, \cos\theta)\ \mathrm{d}\Omega \tag{2.5}$$

while the Cherenkov emission profile $g(p, s, \cos\theta)$ describes "the number of photons emitted per unit track length per unit solid angle at angle θ with respect to the particle direction, at the instance when the particle with initial momentum p has traveled a distance s along the particle track".

defined as

$$G(\bar{x}, t) = \sum_{i}^{\text{hit}} \exp\left[-\frac{1}{2}\left(\frac{T^i_{\text{res}}}{\sigma}\right)^2\right] \tag{2.6}$$

where the residual hit time T^i_{res} is computed as

$$T^i_{\text{res}} = t_i - t - \frac{|\boldsymbol{R}^i_{\text{PMT}} - \bar{x}|}{c/n} \tag{2.7}$$

where t_i and $\boldsymbol{R}^i_{\text{PMT}}$ are the hit time and position of the ith PMT. When $\bar{x}$ and t get close to the true time and position of the event, T^i_{res} gets close to 0 and $G(\bar{x}, t)$ becomes larger.

To locate the global maximum of this goodness function, the vertex pre-fitter begins by maximizing the goodness through an iterative grid search over the event time, progressively reducing both the grid spacing and the width parameter σ. Once the grid search is completed, the negative goodness function $-G(\bar{x}, t)$ is minimized using a fixed resolution of $\sigma = 4\,\text{ns}$. The resulting parameters are passed as the minimization seeds for $-\ln(L(\boldsymbol{x}))$. Often, a single event trigger captures multiple distinct clusters of PMT hits caused by delayed secondary interactions. If there are multiple hit clusters by the secondary interactions such as Michel electrons from muon decay near one event trigger, they are counted as subevents through the following procedure. In order to identify and isolate these subevents, the goodness function is scanned over t from $-200\,\text{ns}$ to $15000\,\text{ns}$ around the event trigger, while the vertex position $\bar{x}$ is held fixed at the value obtained from the pre-fitter. Figure 2.4 shows an example of the distribution of goodness as a function of the time t. In order to select subevent peaks, a *goodness threshold curve* $F(t)$, defined below, is used. This temporal profile reveals two discrete peaks corresponding to the distinct subevents.

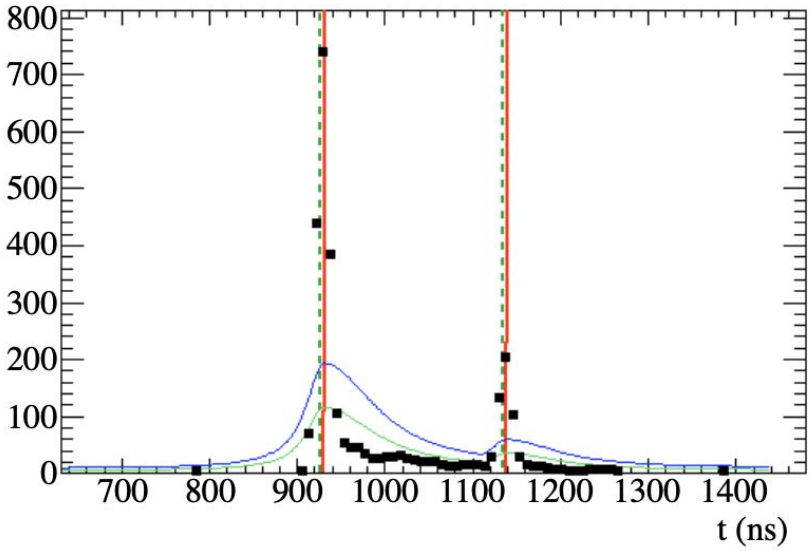


Figure 2.4. Distribution of the scanned goodness G as a function of time t, for an example event with a muon primary and a decay electron. The vertical dashed lines indicate true particle time, the black dots are the goodness scan points, and the blue and green curves represent the threshold curves which are used for subevent peak selection. The red vertical lines indicate the time of the subevent peaks which is found by the algorithm. From [52].

The *goodness threshold curve* is defined as

$$F(t) \overset{\text{def}}{\equiv} 0.25 \max_{i \in M} \{G(\boldsymbol{x}, t_i)\, f(t - t_i)\} + \eta \tag{2.8}$$

where M represents all local maxima of goodness scan points, which has a goodness value larger than the constant threshold $\eta = 9$, and

$$f(\tau) \overset{\text{def}}{\equiv} \frac{1}{1 + (t/\gamma)^2}, \quad \gamma = \begin{cases} 25 \text{ ns} & (\tau < 0) \\ 70 \text{ ns} & (\tau > 0) \end{cases} \tag{2.9}$$

Once the peaks have been identified, a time window of $-180\,\text{ns} < T < 800\,\text{ns}$ is set around each peak, where T represents the peak position. If the time windows for different peaks are overlapped, they are merged together into a single one subevent. Within each resulting time window, the residual time T^i_{res} is re-evaluated.

The vertex pre-fit and the peak search are then repeated using the updated time windows. Finally, the final time windows are fixed for each remaining peak, and the maximum likelihood reconstruction is done in each of them.

2.2.3 Single-ring fit

Once the vertex pre-fitter has provided approximate vertex positions and times for each subevent, the single-ring fit is performed to determine the direction, momentum, vertex position and time of the particle for each subevent with single-ring hypothesis. The single-ring fit works under the assumption that the observed light pattern is dominated by one ring, and it considers three possible particle hypotheses: e^-, μ^-, and π^+. For each subevent, this fit determines the direction, momentum, vertex position, and time of the particle.

Operationally, the evaluation begins with the electron hypothesis, minimizing the `nll`. The parameters obtained from the electron fit are then used as the seed for the subsequent muon fit, and those obtained from the muon fit are used as the seed for the charged pion. Differently, the charged pion hypothesis requires a distinct treatment due to the hadronic interaction with water nucleus and the production of two rings. The track segment prior to the hadronic scatter generates an *upstream ring* and the one after a *downstream ring*. Therefore, single-ring charge pion fitter is performed considering both the upstream and downstream ring. The predicted charge of the upstream track is obtained by subtracting the predicted charge of the downstream track from the predicted charge of an imaginary full track without any hadron interaction happening. Finally, the charged pion hypothesis contains the same parameters as the electron and muon ones, plus an additional fit parameter describing the *energy lost in the upstream track segment*, defined as

$$E_{\text{loss}} \overset{\text{def}}{\equiv} \text{“the kinetic energy deposit in the upstream track”}.$$

How the single-ring fit result is used depends on the subevent's position in the event sequence.

- Primary subevent: for the first subevent, the result of the single-ring fit above is used as the seed of the multi-ring fitter, and the particle type is identified by the multi-ring fitter.

- Delayed subevents: for subsequent subevents, PID is based only on the single-ring fit, since the multi-ring fitter is applied exclusively to the first subevent. However, the PID parameters for these later subevents have not yet been tuned for HK [51].

The work developed in this thesis is based only on the analysis of the first subevent, focusing on the multi-ring fitter optimization for the first subevent (primary).

2.2.4 Multi-ring fit

This section is devoted to the detailed explanation of the multi-ring fit in fiTQun. As mentioned before, fiTQun is a maximum likelihood event reconstruction which searches for the particle event hypothesis $\boldsymbol{x}$ that maximizes the likelihood function defined in equation (2.1). The core idea driving this search is the minimization of the nll with respect to $\boldsymbol{x}$. This operation is computationally performed by the MINUIT [54] minimization package.

In a multi-ring fit, the best fit hypothesis is generally *the one with the lowest* nll *value*. To be fully clear, this definition formally holds only within hypotheses with the same number of rings and different PIDs. Indeed, if the algorithm is comparing the results from a fit performed testing a (n)-rings hypothesis against a $(n+1)$-rings, it is crucial to pay attention to the fact that the likelihood function score of the $(n+1)$-ring hypothesis would be lower (so "better") than the one of the (n)-rings hypothesis:

$$L_n > L_{n+1} \tag{2.10}$$

This follows from the fact that the $(n+1)$-rings fit relies on more parameters than the (n)-rings one, and so it would always have a lower ("better") likelihood score.

It is important to clarify this concept before delving deeply into the fiTQun logic explanation. As we have just pointed out, it is not enough to just compare the (n) and $(n+1)$-rings hypotheses to decide which one is the "best", and therefore the most probable one. From the knowledge of the behavior of the likelihood function score, when the number of fitting parameters is increased, we should fix a certain *threshold* value (or function) to discriminate between valid and fake multi-ring hypotheses. If the improvement in the $(n+1)$-rings likelihood score compared to the (n)-rings is above that certain threshold, it means the $(n+1)$-rings is valid; otherwise not, and the hypothesis has to be rejected, falling back to the (n)-rings.

This is the core concept this thesis work was focused on: the improvement of fiTQun performance through the improvement in the *threshold* selection criteria.

In atmospheric neutrino analyses, a large fraction of events in the GeV energy range exhibits a multi-particle final-state, i.e., multi-ring topologies. Particularly for some oscillation analyses, the key to reducing systematic errors is the distinction between single and multi-ring events, making it therefore essential to correctly reconstruct and classify multi-ring events.

fiTQun had been designed to identify and reconstruct up to six-ring hypotheses in an event. It is applied only to the first subevent (the set of particles directly produced by the neutrino interaction), while the subsequent subevents are not considered as multi-ring events, because they usually do not contain multi-particle states. The reconstruction process of a multi-ring event is a step-by-step fit procedure that starts from the result of the single-ring fitter and tests an increasing number of fit hypotheses. The basic flow of steps is: *8-digit fit* and *9-digit fit*. The first one is the core procedure to identify the best multi-ring hypothesis, whilst the second consists of a series of re-fits to refine the final result and the PID identification.

The primary objective of this work was to improve the `fiTQun` multi-ring reconstruction performance during the 8-digit fitting stage. This focus is motivated by the sequential nature of the software fit procedure. To effectively optimize the algorithm, modifications must follow the natural execution flow-specifically, addressing the 8-digit step before the subsequent 9-digit step. Reversing this order would be counterproductive, as any improvements made at the 9-digit stage could be easily invalidated by subsequent alterations to the preceding 8-digit algorithm.

Multi-ring: 8-digit fit

Every multi-ring fit starts from the single-ring hypothesis. The single-ring hypothesis is tested with two PIDs: e^- and π^+. This results in two single-ring hypotheses ("top branches"), where the electron branch is intended to fit the electron and gamma-ray rings, while the upstream-track pion rings are expected to fit muon or pion rings. An example of the electron hypothesis branch can be visualized in the tree diagram in figure 2.5.

For each branch the multi-ring fit performs an iterative search for additional rings, increasing one-by-one the number of rings. For the two-ring hypotheses, we have:

1st ring: e^-	1st ring: π^+
• ee	• πe
• $e\pi$	• $\pi\pi$

The same logic is iterated sequentially for all $(n+1)$-ring hypotheses, always adding one e^- and π^+ to the previous (n)-ring hypothesis, as long as there are *valid* hypotheses.

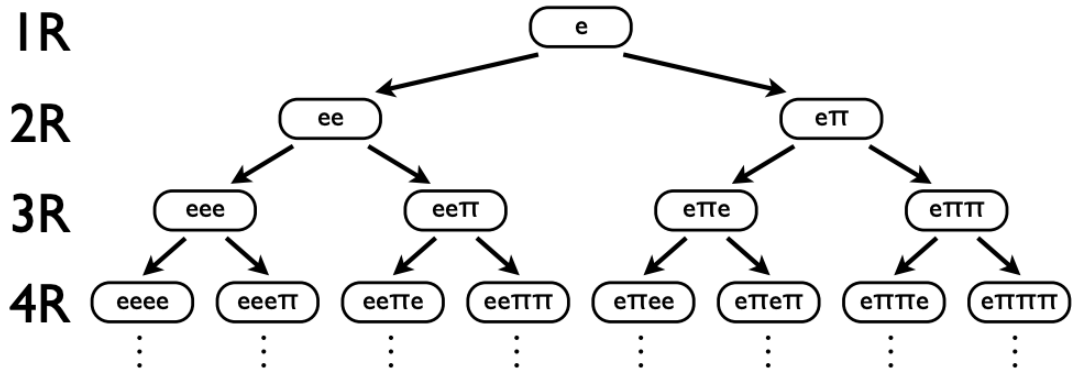


Figure 2.5. Tree diagram of multi-ring fit logic for the electron branch. Equivalently, we have the same tree structure for the pion branch.

Starting from the single-ring fit hypotheses, `fiTQun` moves to the four two-ring hypotheses adding the 2nd ring (e^- or π^+) at the same vertex position as the 1st ring. Then the likelihood functions L_2 of the four hypotheses are evaluated at different directions for the 2nd ring at 400 points, uniformly distributed on the unit sphere surface. The momentum of the 2nd ring is arbitrarily assumed to be 50 MeV/c. Then, for each hypothesis, the scan point with the minimum `nll` is chosen. Finally, the momentum of the 2nd ring is fitted,

while keeping all the other parameters fixed. This gives a rough estimation of the 2nd ring momentum.

Then, using that rough momentum estimation, the fit is refined:

1. The momentum and direction of the 2nd ring are simultaneously fit, keeping the common vertex and the direction and momentum of the 1st ring fixed.
2. Next, with the vertex and ring directions held fixed, the momenta of the two rings are fitted simultaneously. This ensures the *visible energy* E_{vis} is accurately distributed between them, correcting the single-ring algorithm's tendency to overestimate the 1st ring's momentum by absorbing the charge from the 2nd.
3. The final step consists of the simultaneous fitting of all directions, momenta and common vertex of the two rings to get the final best fit for the two-ring hypotheses.

Visible energy $\overset{\text{def}}{\equiv}$ "the energy of an electron which would produce the same number of Cherenkov photons" [55].

This fit procedure is repeated for both branches, e^- and π^+, separately.

For the hypotheses where the 2nd ring being added is a π^+, so $e\pi$ and $\pi\pi$, the parameter E_{loss}, defined in section 2.2.3, is added to the simultaneous fit.

Once the fits on the $(n+1)$-ring hypothesis have been completed, the algorithm has to decide whether to accept or reject the new ring. In the current official version of fiTQun the criterion used is based on the comparison of the likelihood values L_n and L_{n+1}, for the hypotheses of (n) and $(n+1)$-ring respectively.

The currently implemented selection criteria are:

$$\begin{cases} \ln\left(\dfrac{L_2}{L_1}\right) > a_0^e + a_1 E_{\text{vis}} & \text{if the 1st reconstructed ring PID is } e^- \quad (2.11) \\ \ln\left(\dfrac{L_2}{L_1}\right) > a_0^\pi + a_1 E_{\text{vis}} & \text{if the 1st reconstructed ring PID is } \pi^+ \quad (2.12) \end{cases}$$

$$\ln\left(\frac{L_{n+1}}{L_n}\right) > a_0 + a_1 E_{\text{vis}} \quad \text{if } n > 1 \qquad (2.13)$$

where equations (2.11) and (2.12) are used as criteria for the transition from the single-ring to the two-ring hypotheses, according to the reconstructed PID of the 1st ring. It is therefore important to observe that to discriminate between single-ring and multi-ring (i.e. two-ring) hypotheses, fiTQun relies on more refined and sophisticated criteria, dependent on the PID.

Then, if the multi-ring topology turned out to be the best one (two-ring hypothesis over the single-ring), it still has to count how many multi-rings there are in the event. In this case the criterion is unique and not dependent on the PID, equation (2.13). This criterion is used in both branches of the tree diagram to test the hypothesis validity for the cases of $n > 1$.

In figure 2.5 the criterion defined by equation (2.11) is used to test the two two-ring hypotheses ee and πe against the single-ring with e^- PID; similarly equation (2.12) is used

to test the two two-ring hypotheses $e\pi$ and $\pi\pi$ against the single-ring with π^+ PID. Then, from the second row down to the sixth, the used criterion is equation (2.13), the same for both branches.

Also, for the 5th and 6th rings, only the new ring electron PID hypothesis is considered in order to save computation time.

The value of the parameters in the selection criteria are defined in the following `fiTQun v6r0` source code file:

Short commit SHA: `d92e855b`
`ParameterOverrideFiles/HK.parameters.dat`

$$\begin{aligned}
\texttt{fiTQun.RCCut1Ea0WCSim} &\equiv a_0^e = 301;\\
\texttt{fiTQun.RCCut1Mua0WCSim} &\equiv a_0^\pi = 483;\\
\texttt{fiTQun.RCCuta0WCSim} &\equiv a_0 = 241;\\
\texttt{fiTQun.RCCuta1WCSim} &\equiv a_1 = 0.
\end{aligned}$$

It should be noted that because $\texttt{fiTQun.RCCuta1WCSim} \equiv a_1 = 0$, the selection criteria at the software level are completely independent of E_{vis}. Consequently, the three conditions defined in equations (2.11) to (2.13) simplify to the following general form

$$\ln\left(\frac{L_{n+1}}{L_n}\right) > \text{constant} \tag{2.14}$$

As previously pointed out, the core of this work was the effort to improve the performance at the 8-digit fitting step level; therefore the following example will be focused more on that part.

To better explain and understand the multi-ring reconstruction logic at the 8-digit fitting step, let us look at the listing 2.1. The variable `fqmrifit` is the "`fiTQun` multi-ring fit ID", and it can have 8 or 9 digits, depending on the fitting step.

For the case of an 8-digit `fqmrifit`, the variable has the structure `N0...ZYX`, where the most significant digit `N` is the number of rings (1-6), and `X`, `Y` and `Z` are the particle type (as in single-ring fit, 1 for e^- electron-like ring, 3 for π^+ pion-like ring) of the 1st, 2nd, and 3rd ring respectively. A negative fit indicates that the ring which is newly added in the fit is considered as a *fake* ring by the fake ring reduction algorithm.

Therefore, the two starting single-ring fit hypotheses are the ones on the rows with `Instance` values 11 and 20, 10000001 and 10000003. The first one is the single ring electron-like hypothesis, while the second is the single-ring pion-like hypothesis. For each fit hypothesis (`fqmrifit`), a likelihood score is associated with it: `fqmrnll`.

```
1  ***********************************************
2  * Instance *   fqmrnll *   fqmrifit * true_pid *
3  ***********************************************
4  *        0 * 31979.312 *  320000011 *       11 *
5  *        1 * 31979.437 *  220000011 *       11 *
6  *        2 * 16999.117 *  801000000 *          *
7  *        3 * 18061.255 *  803000000 *          *
8  *        4 * 17808.109 *  802000000 *          *
9  *        5 * 6299.5478 *  811000000 *          *
10 *        6 * 6697.4389 *  813000000 *          *
11 *        7 * 6682.7563 *  812000000 *          *
12 *        8 * 31997.523 *  120000011 *          *
13 *        9 * 32205.660 *  130000131 *          *
14 *       10 * 37481.175 *  140001113 *          *
15 *       11 * 36058.863 *   10000001 *          *
16 *       12 * 31997.527 *   20000011 *          *
17 *       13 * 32930.890 *   20000031 *          *
18 *       14 * 31920.066 *  -30000111 *          *
19 *       15 * 31959.878 *  -30000311 *          *
20 *       16 * 32087.660 *   30000131 *          *
21 *       17 * 32806.644 *  -30000331 *          *
22 *       18 * 32012.015 *  -40001131 *          *
23 *       19 * 32044.142 *  -40003131 *          *
24 *       20 * 38177.843 *   10000003 *          *
25 *       21 * 37630.871 *   20000013 *          *
26 *       22 * 37917.824 *  -20000033 *          *
27 *       23 * 37339.679 *   30000113 *          *
28 *       24 * 37446.089 *  -30000313 *          *
29 *       25 * 37064.726 *   40001113 *          *
30 *       26 * 37268.128 *  -40003113 *          *
31 *       27 * 36975.101 *  -50011113 *          *
32 ***********************************************
```

Listing 2.1. `ROOT Scan(...)` output of an example from the dataset used in this thesis work. The variable `fqmrnll` corresponds to `nll`, while `fqmrifit` is the fit ID. The fit IDs are blue and green colored to easily distinguish between 8-digit fit e^- and π^+, respectively, whilst the 9-digit fit steps are orange colored. In order to better distinguish the two sections, the same colors have been used for the vertical lines on the left delimiting the corresponding sections: Multi-ring: 8-digit fit and Multi-ring: 9-digit fit respectively. (The 8-digit vertical line left delimiter is blue for both e^- and π^+). The variable `true_pid` stores the `WCSim` simulation output: in this case the truth is two-ring, both with PID e^- (11).

A more comprehensive example is provided in listing B.1.

The variable `fqmrnll` is the `fiTQun` multi-ring `nll`, corresponding to the scores of `36058.863` and `38177.843` for the single-ring hypotheses e^- and π^+, respectively. Then the four two-ring hypotheses fits are performed and the likelihood score of each one is stored in the corresponding row. These are the `Instance` 12, 13 (ee, $e\pi$) and 21, 22 (πe, $\pi\pi$); two for each branch. At this point the algorithm has to decide if the two-ring hypotheses are valid (*accepted*) or not (*rejected*): following the selection criteria of the likelihood ratios

equations (2.11) and (2.12),

$$\begin{aligned} \ln\left(\frac{L_{n+1}}{L_n}\right) &= \ln(L_{n+1}) - \ln(L_n) \\ &= -\ln(L_n) - (-\ln(L_{n+1})) \\ &= \texttt{fqmrnll}_n - \texttt{fqmrnll}_{n+1} \end{aligned} \tag{2.15}$$

can be computed as follows

- $ee \longrightarrow 36058.863 - 31997.527 = 4061.335 \quad (> 301)$, accepted $\longrightarrow$ 20000011;
- $e\pi \longrightarrow 36058.863 - 32930.890 = 3127.973 \quad (> 301)$, accepted $\longrightarrow$ 20000031;
- $\pi e \longrightarrow 38177.843 - 37630.871 = 546.972 \quad (> 483)$, accepted $\longrightarrow$ 20000013;
- $\pi\pi \longrightarrow 38177.843 - 37917.824 = 260.019 \quad (< 483)$, rejected $\longrightarrow$ -20000033.

Once the `fqmrifit` fit IDs for the two-ring hypotheses have been assigned, the algorithm moves to the next step. For each *accepted* hypothesis it adds an e^- and a π^+ ring, and repeats the same procedure as before, but using the equation (2.13) criterion for both branches. This sequential fit scheme is repeated until there are no more *accepted* hypotheses or the multi-ring hypotheses have reached the six-ring hypothesis level.

In the specific case of the example listing 2.1, the electron-like branch stops at two-ring level with 20000011 and at three-ring level with 30000131 as the accepted hypotheses, while the pion-like branch reaches the four-ring level with the only accepted hypothesis being 40001113.

In the following figures 2.6 and 2.7 is shown the tree structure representation of the 8-digit fit step from the above listing 2.1.

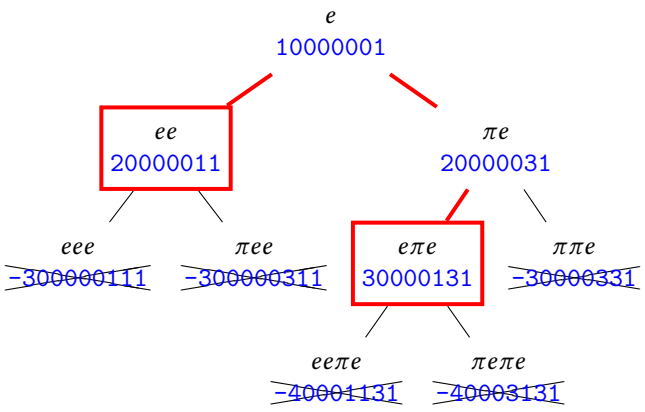


Figure 2.6. Tree structure corresponding to the electron branch of the 8-digit fit is shown for the example presented in listing 2.1. Fit IDs marked with a strikethrough indicate hypotheses that were rejected, while the thick red lines and boxes highlight the surviving branches and the final accepted hypotheses, for the next "9-digit 1st step" fit.

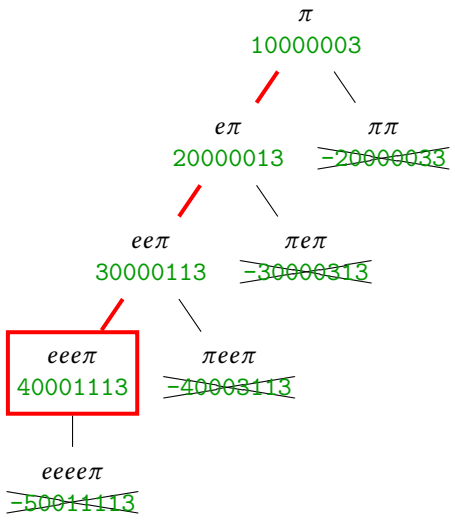


Figure 2.7. Tree structure corresponding to the pion branch of the 8-digit fit is shown for the example presented in listing 2.1. Fit IDs marked with a strikethrough indicate hypotheses that were rejected, while the thick red lines and boxes highlight the surviving branches and the final accepted hypotheses, for the next "9-digit 1st step" fit. As previously pointed out, for the 5th and 6th rings, only the electron hypothesis is considered. Hence, this is the reason why no hypothesis like $\pi eeee\pi$ | `-50031113` has been tested.

From the above example, the chosen multi-ring hypotheses for the next fit step are `20000011`, `30000131` and `40001113`. The accepted hypotheses will therefore be used as the *seeding multi-ring hypotheses* for the next stage of the multi-ring fit: "9-digit 1st step".

Multi-ring: 9-digit fit

The *raw* accepted multi-ring hypotheses from the 8-digit (`NO...ZYX`) provide an initial set of candidates. Due to the limitations of the seeding procedure, it can still contain spurious rings and misidentified particle types. In order to improve the fit result, the 9-digit fit level is therefore introduced as successive three refinement steps: each ring is refitted sequentially, and its particle type is re-evaluated following the procedures below.

The 1st refinement stage of the 9-digit fit level applies a *fake ring reduction* to the raw 8-digit fit output and the accepted results are labeled by the (`1NO...ZYX`) fit ID, "9-digit 1st step". The accepted ring hypotheses are then rearranged in descending order of their visible energy E_{vis}. Then, for each candidate ring, the following steps are performed. The ring momentum is temporarily set to zero, and the `nll` is recomputed without that ring. The quantity

$$\Delta \ln(L) = \ln(L_{\text{without}}) - \ln(L_{\text{full}}) \tag{2.16}$$

measures how much the likelihood changes when the ring is removed. A large positive value means the ring is relevant for the fit and is therefore accepted. If the difference is close to zero or negative, the ring is considered fake and thus rejected. Because the ring PIDs can shift during later stages, the particle type at this step and beyond is no longer

read from the fit ID, but from the `fqmrpid` variable. As in the previous step, a negative fit indicates that the newly added ring is considered a *fake* ring by the fake ring reduction algorithm.

In the 2nd refinement, the (`1N0...ZYX`) hypothesis is processed by *fake ring merging* and *sequential ring re-fitting*, resulting in the (`2N0...ZYX`) fit IDs. Because the initial seeding procedure can artificially split single physical energy depositions into multiple overlapping fits, a *collinearity* check is performed. Starting with the most energetic ring, the opening angle $\Delta\theta$ between the directions of this ring and each lower energy ring is calculated.

> *Opening angle* $\Delta\theta \overset{\text{def}}{\equiv}$ "the angle between the reconstructed directions of two particles that produce Cherenkov radiation, where the direction of each particle coincides with the axis of its corresponding Cherenkov cone".

If the opening angle for a given lower energy ring is equal or smaller than 20 degrees, $\Delta\theta \leq 20^\circ$, that ring is merged into the most energetic one by adding its visible energy, (*merging procedure*). Successively, the highest energy ring is re-fitted assuming three particles type: e^-, μ^- and π^+, while the other non-merged rings are held fixed in their original configurations. This step is essentially a single-ring fit in the presence of other fixed rings. All parameters of the ring being re-fitted, including its vertex, are varied simultaneously, so the resulting multi-ring hypothesis does not require all rings to originate from a single common vertex. The type of the most energetic ring, whether it is a showering, such as e^- and γ, or a non-showering, such as μ^- and π^+ ring, is then re-evaluated by comparing the best-fit likelihoods between the hypotheses assuming the most energetic ring as e^- or π^+. The ring is classified as showering if

$$\ln\left(\frac{L_{e^-}}{L_{\pi^+}}\right) > -10 \tag{2.17}$$

The pion ring hypothesis is used as the non-showering reference.

The ring type selected by this cut is accepted as the final type of the ring (showering or non-showering), and the ring in the original multi-ring hypothesis is replaced by the selected re-fitted hypothesis. If the most energetic ring is classified as non-showering, *pion-like*, it is assigned the muon hypothesis; this reflects the expectation that a non-showering leading ring is predominantly a true muon from ν_μ CC interactions, whereas a true charged pion as the leading ring is rare in NC events. For the other lower-energy rings, a non-showering classification is instead always assigned to the charged-pion hypothesis. After the most energetic ring has been re-fitted, the above procedure is repeated with e^- and π^+ hypotheses for the subsequent lower energy rings which remained after ring merging, in the descending order of the ring energy until all the remaining rings are re-fitted. At this stage, the above e^-/π^+ likelihood cut is the only PID criterion applied in the sequential fit procedure; no separate μ^-/π^+ identification is performed. The particle type assigned after the sequential re-fit can therefore differ from the type before the re-fit.

In the 3rd refinement, the (`2N0...ZYX`) hypothesis is processed by a *simultaneous fit* of the longitudinal vertex position and the visible energy of all rings. The resulting IDs are (`3N0...ZYX`). According to the official documentation this stage is still experimental [56]. Indeed, a known limitation of the sequential re-fitting strategy is that it optimizes each ring

independently, which does not guarantee a common interaction origin and could result in multi-ring events with disjoint vertex positions. Performing a *simultaneous fit* at the final 9-digit 3rd step is intended to resolve these spatial discrepancies and restore vertex coherence across the event.

For hypotheses containing more than one ring, the 9-digit IDs with 1st digit 8, (8NX000000), provide the nll values for ring N under PID hypothesis X. For example, (813000000) and (811000000) can be compared for the pion and electron hypotheses of the 2nd ring.

The fit hypothesis obtained after the entire fit procedure is the final fiTQun reconstruction result.

Chapter 3

WCSim events data and fiTQun optimized reconstruction

This chapter details the contributions of this study to the development of the fiTQun software. The main goal and effort were to enhance the performance of the multi-ring counting algorithm, specifically targeting single- and two-ring topologies at the 8-digit fit level. This scope was constrained by the six-month timeframe of a Master's thesis; consequently, the optimization strategy and results presented here represent only the tip of the iceberg of a broader project. Indeed, future work will extend this research to more complex topologies, involving more than two rings at the 8-digit level, before eventually advancing to the 9-digit fit level.

The subsequent sections first introduce the dataset utilized in this study, followed by a presentation and analysis of the fiTQun reconstruction results. Building upon these findings, the core idea underlying the optimization strategy is discussed. Finally, the specific software modifications and implementation details are thoroughly documented from a technical perspective.

3.1 WCSim dataset simulation

The dataset produced and utilized in this study comprises ~50 000 events (precisely, 49 980), distributed across 1 000 independent ROOT output files. Hereafter, we refer to this dataset as "dataset50k". The underlying physics processes and detector response were simulated utilizing the WCSim framework (version: 1.12.20). To evaluate the multi-ring counting algorithm performance, the simulation pipeline was configured to generate complex multi-particle topologies within a precise model of the HK detector environment. The configuration details and kinematic parameters governing the dataset generation defined in the simulation files wcsim_execute_4rings.sh and tuning_parameters.mac are documented below.

The flow of the simulation script is:

- **Initialization:** the script first establishes execution fail-safes. It then initializes environment variables to dynamically assign job identification tags, ensuring seamless integration and tracking when deployed across High-Performance Computing (HPC) batch scheduling systems (e.g. "LSF" or "Slurm").
- **Variables declaration:** global variables are then explicitly defined for different scopes. CFG_DIR and OUT_DIR are the variable paths pointing to the configuration and output directories. LOG, MAC, ROOTFILE are the input/output files, while N_EVENTS is used to define the number of events to be simulated. Finally, SEED=$((300000 + BATCH_ID)), is the Geant4 seed simulation variable which guarantees that the events are all different from one another, and avoids generating the same event more than once.
- **Detector geometry:** the simulated environment is based on the realistic HK geometry model with the Od structure.
  In the simulation script: `/WCSim/WCgeom HK_HybridmPMT_WithOD_Realistic' the radius and height of the cylinder are defined, together with an additional margin to avoid

particles being generated on the detector walls. Cylinder radius $r_c = 32\,423$ mm; cylinder height $h_c = 67\,020$ mm; cylinder margin $m_c = 50$ cm.

- **PMT configuration:** the PMT percentage coverage of the inner detector is set to ~20.215% while for the mPMT of the inner detector it is set to zero. The Quantum Efficiency (QE), probability that a photon produces an electrical signal, is applied only when the photon hits the sensitive part of the PMT; whilst the Collection Efficiency (CE) is turned on, to simulate whether the internal electronics of the PMT successfully amplifies or not the electron cascade.
- **Data acquisition (DAQ):** the simulation uses the SKI digitizer that is the old electronics scheme used in SK until 2001 [47]. The trigger module is set to a specific Number of Digitized hits (NDigits) within a time window: if the number of fired PMTs crosses the defined threshold within that time window, the detector triggers and records the event data. The "multi digits per trigger" option is turned on: if a single PMT fires multiple times within the readout window (which can happen due to delayed light scattering, reflections, or PMT after pulsing), all of those distinct hits will be saved. If disabled, only the first hit on each PMT would be recorded for the event.
- **Dark noise:** dark mode is enabled (1), an average dark rate of 4.2 kHz is applied to the tank detector elements, utilizing a conversion factor of 1.126, and an integration window of 2000 ns.
- **Simulation setup and generator:** the primary generator is set to gps (Geant4 General Particle Source), and saving to RooTracker is disabled (0). Visual tracking for optical photons is turned off by setting the draw fraction to 0.0.
- **Vertex generation:** a loop iterates over the total number of events and for each event a Python script calculates a random interaction vertex V_x, V_y, V_z uniformly distributed within a cylindrical volume. This volume is defined by the detector's radius and height, reduced by the margin.
- **Multi-ring particles production:** for each interaction vertex, a random number of particles between 1 and 4 is generated, all above their Cherenkov energy threshold. A secondary Python script computes isotropic direction cosines p_x, p_y, p_z utilizing a spherical distribution. The particle type is then randomly assigned as either a μ^- ("pion-like") or an e^- ("electron-like"), with equal probability.
- **Multi-ring particles kinematics:** the computed kinematics are written to the macro (`.mac`) file for each particle in the event. The source is configured as a point source emitting linearly distributed energies within the range from 100 MeV to 1.5 GeV. Each event includes between 1 and 4 primary particles (above the Cherenkov threshold), with equal probability for each multiplicity; the primary particles may be electrons or muons with equal probability. The particle directions are generated isotropically, and the vertices are uniformly distributed within the Id volume.
- **Environment setup and execution:** after the macro file has been written, the `ROOT` environment variables for dynamic libraries are exported. The `WCSim` setup script is sourced, and the simulation executable is launched using the generated macro and the tuning parameters, appending all output to a log file.

To accurately model photon propagation and scattering within the water volume, the following tuning parameters, defined in the file `tuning_parameters_iwcd.mac` (GitHub link), are used.

- Absorption length scale factor (`abwff`): `1.30`.

The absorption length (λ_{abs}) is the characteristic distance a photon travels before being absorbed (in this case, by a water molecule). A scale factor of 1.3 means the baseline absorption length is increased by 30%. This makes the simulated water more transparent than the default model, allowing photons to travel farther before disappearing, which ultimately increases the total photon yield reaching the PMTs.

- Rayleigh scattering coefficients (`rgcff` and `rayff`): `0.32` and `0.75`, respectively.
  Rayleigh scattering occurs when light interacts with particles much smaller than the photon's wavelength such as water molecules. It has a strong dependency on the wavelength ($\sim\lambda^{-4}$), resulting in a stronger scattering for shorter (blue) than longer (red) wavelengths. These parameters downscale the default Rayleigh scattering probability. This means photons undergo fewer large-angle deflections, preserving their original directional information longer and reducing the *fuzziness* of electron-like Cherenkov rings.
- Black sheet reflection factor (`bsrff`): `2.50`.
  It is used to adjust the reflectivity of the black sheets covering the detector walls, and used to isolate the detector volumes. This parameter increases the probability of a photon being reflected by 2.5 times.
- Mie scattering factor (`mieff`): disabled `0.0`.
  Mie scattering occurs when photons interact with particles whose size is comparable to their wavelength (e.g. microscopic impurities, dust particles or organic matter inside the detector water). This is different from Rayleigh scattering where the particle size is much smaller than the wavelength of the radiation.
- Top veto (`topveto`): disabled `0`, with a nominal PMT spatial configuration of 100 cm.
  A "veto" is a shielding region used to flag and reject background events, such as cosmic ray muons entering from the Od. In particular, the top veto uses an array of PMTs to detect these incoming particles before they enter the Id.

This tuning is derived from SK: the parameters were specifically calculated by comparing `WCSim` simulation outputs against the optical calibration data from SK. Currently, these are the default optical parameters, and are used as a starting point for HK simulations as well.

3.1.1 True data selection

In water Cherenkov detectors, a primary electron or muon produced at the neutrino interaction vertex generates Cherenkov radiation along its trajectory, if its energy is above Cherenkov threshold. Both particle types ionize the medium, liberating secondary electrons (delta-rays).

Electrons, in addition, can lose energy via bremsstrahlung, producing energetic photons. Those photons with energy above the pair-production threshold can convert into e^+e^- pairs in the electromagnetic field of a nucleus; the positron may subsequently annihilate with an atomic electron, yielding two back-to-back photons. Photons whose energies fall below the pair-production threshold interact predominantly through Compton scattering, ejecting further electrons. This chain of processes constitutes an *electromagnetic cascade*.

Although the total Cherenkov light emitted within the characteristic cone remains roughly proportional to the energy of the primary particle, the spatial extent of the shower degrades the definition of the Cherenkov ring. Consequently, an electron-like ring appears significantly more diffuse ("fuzzy") than the ring produced by a muon (pion-like ring), which traverses the

detector without developing an electromagnetic shower. PID in fiTQun exploits this difference in ring sharpness to identify the rings.

The ring-counting algorithm aims to identify the primary charged particles that originate at the neutrino vertex. The observed ring multiplicity must reflect only these primary particles, not the secondary shower electrons that contribute to the detector response.

Therefore, to briefly resume the main WCSim simulation properties:

- ~50 000 events split across 1000 ROOT files simulated with WCSim 1.12.20;
- each event includes between 1 and 4 primary particles (above the Cherenkov threshold), with equal probability for each multiplicity;
- the primary particles may be electrons or muons with equal probability;
- kinetic energies are uniformly distributed from 100 MeV to 1.5 GeV;
- the particle directions are generated isotropically;
- generation vertices are uniformly distributed within the Inner Cylinder (IC) volume.

The simulation output files, e.g. wcsim_output.root, store the full particle track hierarchy, from the primary particles through the complete secondary development. In this study, we are concerned with the primary charged leptons (e^- or μ^-) that are directly generated at the neutrino-nucleus interaction vertex (these correspond to tracks 1-4 in the simulation truth record).

Some of the most relevant variables in the WCSim output are: the Track ID that labels the tracks in hierarchical order, assigning the lowest indices to the primary particles and reserving higher indices for the secondary particles produced along the simulated chain. This variable differs from the Track number, which simply reports an internal index used by the simulation and does not reflect the particle hierarchy. The PDG ID identifies the particle type according to the standard PDG encoding, 11 for electrons and 13 for muons. The PDG code parent stores the particle type of the parent track; for primary particles this is set to zero, corresponding to the target nucleus. Finally, the Time variable record the creation time respect to the primary vertex, so the primaries are assigned a zero value.

In table 3.1 is shown an example of an event with four primaries, together with the corresponding variables.

Track ID	Track number	Particle	PDG ID	PDG code parent	Track ID parent	Time [ns]
0	1	target nucleus, massless, no kinematics	0	0	0	0.00
1	not fixed	e^-	11	0	0	0.00
2	not fixed	μ^-	13	0	0	0.00
3	not fixed	μ^-	13	0	0	0.00
4	not fixed	e^-	11	0	0	0.00

Table 3.1. Record example of simulated primary particles at the neutrino interaction vertex.

It follows, for example, that a track whose PDG code parent or Time differs from zero corresponds to a secondary particle and is thus not relevant for the final purpose of this study.

To compare the truth from the WCSim simulation with the reconstructed values from fiTQun, it was necessary to scrape the WCSim output files and select only the primary particle data. This selection was performed using the script "ringred.C".

To select the primary data out of the dataset50k, the script loops over all the particle tracks and select only those satisfying the following conditions:

- the Track ID is positive;
- both the PDG code parent and Track ID parent are zero;
- for each spatial coordinate, the difference between the event vertex and the start position of the track is smaller than 10^{-4};
- the Time is zero;
- the Mass different from zero;
- the particle energy is greater above the Cherenkov threshold increased by 30 MeV extra as an additional safety check, energy > E_th+30.

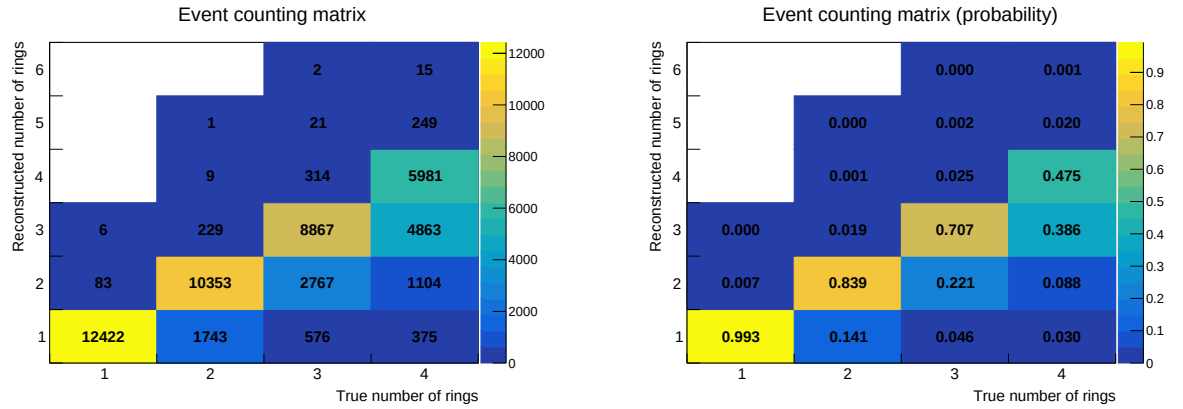
A more comprehensive example is provided in listing B.2, where the full data stored by WCSim for an event with three primaries is shown.

3.2 fiTQun events reconstruction

The dataset50k has been used as the benchmark sample to evaluate the multi-ring performance of fiTQun. The fiTQun framework was deployed on the INFN Pisa HPC cluster. Subsequent analysis of the reconstruction output was carried out, and it provided the starting point to refine the multi-ring counting algorithm.

3.2.1 Multi-ring performance

In order to compute the efficiency and purity of the reconstruction results, it is necessary to count how many rings were correctly reconstructed and how many not. Therefore, a ROOT script was written with the scope of taking as input the fiTQun output file fitqunOutput.root and returning as output the *confusion matrices* shown in figure 3.1.



(a) *Confusion matrix.* (b) Column-normalized *confusion matrix.*

Figure 3.1. *Confusion matrices* counting the number of true rings (x-axis) against the number of fiTQun reconstructed rings (y-axis) for each event from the dataset50k.

Here, the number of reconstructed rings is the one returned by fiTQun at the end of the multi-ring fit procedure, so after both 8 and 9-digit steps. It is worth noting that the ring counting performance decreases as the number of true rings increases. Indeed, looking at the correctly reconstructed rings, that is, the values on the matrix antidiagonal, one would observe that the percentage of one true ring correctly reconstructed as one ring is greater than the percentage for the correctly classified two rings, and so on (0.993, 0.839, 0.707, 0.475).

For brevity, hereafter the following notation will be used: "tnrm", where "t" refers to the true number of rings and "r" to the reconstructed number of rings, while $n \in \{1, 2, 3, 4\}$ and $m \in \{1, 2, 3, 4, 5, 6\}$. Hence, for example, t1r2 is the topology of events where the WCSim simulation produced one ring, but the fiTQun multi-ring algorithm returned two rings as the best fit hypothesis. Then, with the notation tn and rm we will refer to:

$$\mathrm{t}n = \sum_{m=1}^{6} \mathrm{t}_n \mathrm{r}_m \tag{3.1}$$

$$\mathrm{r}m = \sum_{n=1}^{4} \mathrm{t}_n \mathrm{r}_m \tag{3.2}$$

where tn denotes the sum over matrix columns and rm the sum over rows.

The variable tn was used to validate the simulated data in dataset50k:

$$\begin{aligned}
\mathrm{t1} = 12511 \sim 12495 &\implies \Delta_{\text{sim-pred}} = +16 \implies +0.17\sigma \\
\mathrm{t2} = 12335 \sim 12495 &\implies \Delta_{\text{sim-pred}} = -160 \implies -1.66\sigma \\
\mathrm{t3} = 12547 \sim 12495 &\implies \Delta_{\text{sim-pred}} = +52 \implies +0.54\sigma \\
\mathrm{t4} = 12587 \sim 12495 &\implies \Delta_{\text{sim-pred}} = +92 \implies +0.96\sigma
\end{aligned}$$

12495 is the *predicted* number of events in each true topology: since the probability of each ring multiplicity has been set to be equal, it then follows that $\mathrm{t}n_{\text{pred}} = 49980/4 = 12495$ events. The standard deviation was calculated across the population of the four distinct true-ring categories to evaluate the uniformity of the generated event totals.

$$\sigma \overset{\text{def}}{\equiv} \sqrt{\frac{\sum_i (x_i - \mu)^2}{N}} = \sqrt{\frac{(16)^2 + (-160)^2 + (52)^2 + (92)^2}{4}} \simeq 96.2081 \tag{3.3}$$

Finally, the sigma deviations were computed for each category from $Z_i = (x_i - \mu)/\sigma$, resulting in the values listed above.

Another important detail to pay attention to in figure 3.1 is the tendency of the algorithm to reconstruct fewer rings rather than more. This can be seen looking at the super-antidiagonal[1] and sub-antidiagonal[2] elements: for each of the four categories the sub elements have greater values than the super ones, proving therefore that most of the wrong reconstructions actually have fewer rings than the true simulated ones.

Subsequently, we define the *efficiency* and *purity* as

$$\epsilon_i \overset{\text{def}}{\equiv} \frac{\mathrm{t}_i \mathrm{r}_i}{\sum_m \mathrm{t}_i \mathrm{r}_m} = \frac{\mathrm{t}_i \mathrm{r}_i}{\mathrm{t}_i} \tag{3.4}$$

[1] "The super-antidiagonal of a square matrix is the set of elements directly above the elements comprising the antidiagonal".

[2] "The sub-antidiagonal of a square matrix is the set of elements directly under the elements comprising the antidiagonal".

$$p_i \overset{\text{def}}{\equiv} \frac{t_i r_i}{\sum_n t_n r_i} = \frac{t_i r_i}{r_i} \tag{3.5}$$

These definitions are fundamental to measuring the performance of the fiTQun ring counting algorithm, in particular to compute the selection *efficiency* and *purity*. The efficiencies and purities obtained with the standard selection criteria at the end of the full reconstruction procedure are summarized in table 3.2.

Efficiency	Purity
$\epsilon_1 = 0.993 \pm 0.001$	$p_1 = 0.822 \pm 0.003$
$\epsilon_2 = 0.839 \pm 0.003$	$p_2 = 0.724 \pm 0.004$
$\epsilon_3 = 0.707 \pm 0.004$	$p_3 = 0.635 \pm 0.004$
$\epsilon_4 = 0.475 \pm 0.004$	$p_4 = 0.949 \pm 0.003$

Table 3.2. Efficiencies and purities computed at the end of the full reconstruction process (8 and 9-digit), from the values shown in figure 3.1a.

These values provide the starting point for the optimization procedure developed in this work, whose final goal is to enhance the efficiency and purity achieved at the intermediate transition between 8 and 9-digit fit steps.

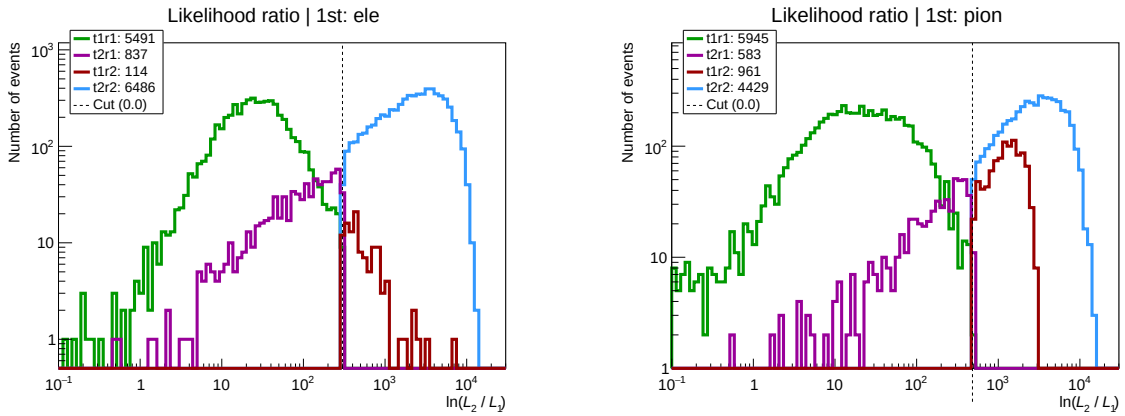
The standard deviation σ_x has been computed treating the efficiency and purity as binomial distributions. The variance of a binomial distribution is defined mathematically as $x(1-x)$ and therefore the standard deviation is

$$\sigma_x = \sqrt{\frac{x(1-x)}{N}} \tag{3.6}$$

where x, either efficiency or purity, is the probability of positive outcome, $(1-x)$ the probability of the negative outcome and N the total number of events in that category.

Then, the analysis focused specifically on the four topologies t1r1, t2r1, t1r2 and t2r2 to study more deeply and understand how to improve the selection criteria for the transition from single to multi-ring. This was done, first by studying the distributions of events from these four topologies. As explained before, in section 2.2.4, the new ring hypotheses being added are divided according to the 1st ring PID branch and therefore also all the plots produced were separated according to this classification.

In figure 3.2 the distributions of events for the two true categories t1 and t2 as a function of the nll ratio are shown, following the classification into the four topology categories: t1r1, t2r1, t1r2, t2r2. The vertical flat dashed black lines correspond to the two parameters $a_0^e = 301$ and $a_0^\pi = 483$ from equations (2.11) and (2.12), respectively. Looking closely at the two distributions plots, one might wonder why in the "pion" plot (1st reconstructed ring: π^+) the red t1r2 does not decrease as in the "electron" plot, but instead it peaks. This is due to the shapes of electron-like and pion-like rings. Indeed, the former, due to the electromagnetic shower processes, are fuzzy, while the latter are sharp and more spatially defined. Therefore, it can happen that the t1 ring is electron-like, but fiTQun gets a better likelihood score for the two rings hypothesis (one pion and one electron-like) than the single one, because of the higher number of parameters for the multi-ring hypothesis.



(a) 1st reconstructed PID: e^-. (b) 1st reconstructed PID: π^+.

Figure 3.2. Distributions of events from the t1 and t2 categories among the four classification topologies as a function of `nll` ratio.

3.2.2 Performance enhancement strategy

The main goal of this study is to enhance the efficiencies and purities $\epsilon_{1,2}$ and $p_{1,2}$ of the t1 and t2 topologies. Ideally, this involves improving the selection criteria to distinguish and select single and multi-ring hypotheses to reach the perfect classification of the events in the "correct" categories t1r1 and t2r2 only, leaving empty the "wrong" t2r1 and t1r2 categories.

The core idea driving this study and the strategy implemented to look for better selection cuts was to use a bi-dimensional cutting function. The original selection criteria were dependent on one variable only: `nll` ratio. The core contribution of this study consisted of extending the selection criteria to a higher dimension, 1D⟶2D. Defining a generic cutting function as

$$f_{\text{cut}}^{\text{old}}\left(\ln\left(\frac{L_2}{L_1}\right)\right) \longrightarrow f_{\text{cut}}^{\text{new}}\left(\ln\left(\frac{L_2}{L_1}\right), z\right) \tag{3.7}$$

where z is a new generic cutting variable added to the `nll` ratio. It was necessary to determine which variable should be used in place of z. The approach was therefore to study the two-dimensional distributions of the combinations of `nll` ratio and the following list of variables of interest: (ΔP), $(\min P)$, (P_1), (P_2), $(P_1 + P_2)$, $(\Delta\theta)$.

$(P_{1,2})$ are the momenta of the 1st and 2nd reconstructed rings, respectively; (ΔP) is computed as the momentum of the 1st ring minus the momentum of the 2nd reconstructed ring:

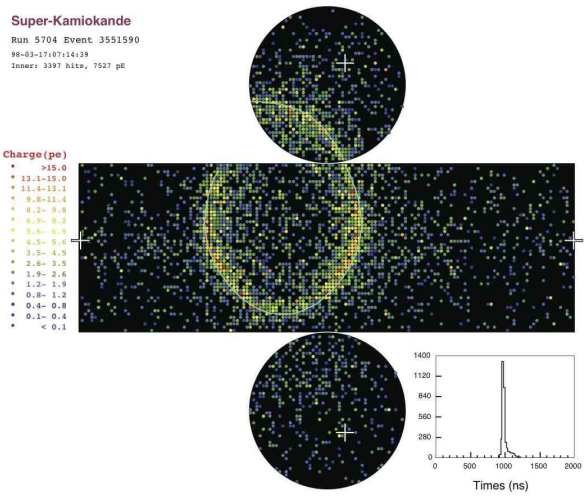
$$\Delta P \overset{\text{def}}{\equiv} P_1 - P_2 \tag{3.8}$$

Similarly, $(\Delta\theta)$ is the *opening angle* computed between the directions of the reconstructed rings, previously defined in section 2.2.4.

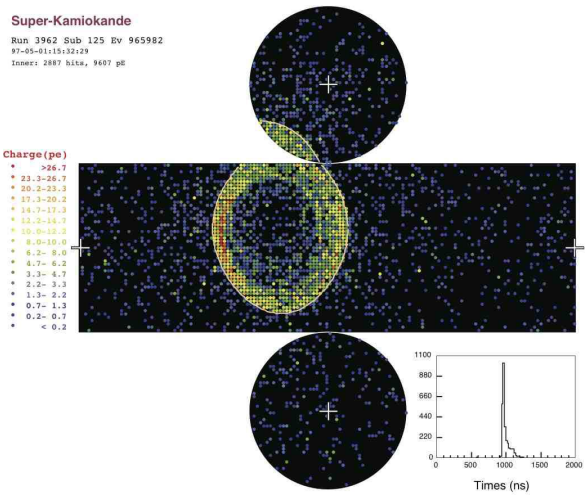
Opening angle $\Delta\theta$

A few observations can be made regarding the *opening angle* variable.

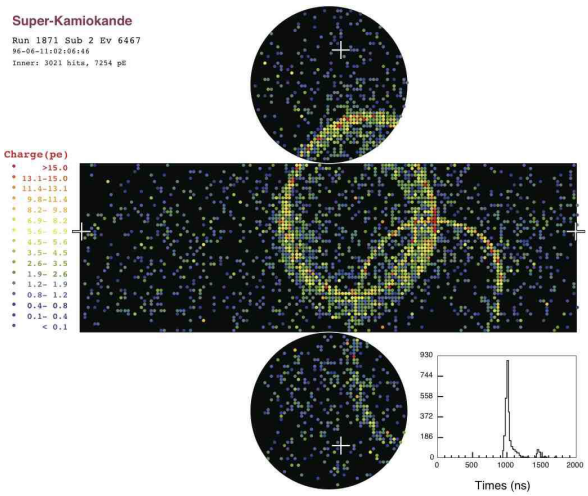
Firstly, it should be noted that the opening angle is always defined between the reconstructed directions, not the true ones. The reason behind this choice is due to the fact that for the events in the categories t1r1 and t1r2 it is actually impossible to define a true opening angle, because there is only one ring in the simulated data. It should therefore be observed that any fit at the 8-digit level will always evaluate both the two single-ring hypotheses (electron and pion-like) and then the four two-ring hypotheses. This is the bare minimum. After this, if there is at least one accepted two-ring hypothesis (positive `fqmrifit`), the fit would proceed with three or more-ring hypotheses. Since the multi-ring fit for two-ring hypotheses is always attempted, for any event the directions of the two rings in each two-ring hypothesis are computed, even if the hypothesis is ultimately rejected.



(a) Single electron event



(b) Single muon event



(c) Two-ring event display

Figure 3.3. Event displays from SK's phase-I period. Top: single electron (left) and muon (right) events. Electrons produce fuzzy rings due to electromagnetic shower, while muons, which do not shower, produce clear rings with sharp edges. Bottom: unrolled view of a two-ring event. Colored points indicate the number of detected photoelectrons in each PMT. From [57].

The idea is therefore to look for differences in the distributions of the reconstructed opening angle ($\Delta\theta$) values across the four different topologies, and use this knowledge to define a more precise selection cut.

For example, one of the reasons that could possibly explain the misclassification into category t1r2 is the *collinearity* of reconstructed rings. Two rings are said to be *collinear* when the opening angle between the directions of the two related Cherenkov cones is less than or equal to twenty degrees, $\Delta\theta \leq 20°$. This results in an overlap of rings, that causes the failure of the fiTQun ring counting algorithm. Specifically, the most common failure case leading to classification as a t1r2 event is electron ring fuzziness, a representative case of which is displayed in figure 3.3a. Indeed, it can happen that the likelihood score improvement between the single-ring hypotheses and the two-ring ones exceeds the threshold for accepting the multi-ring hypothesis. Despite being a true electron single-ring, because of the diffuse shape of the ring, fiTQun could actually give a better fit result using two almost collinear reconstructed rings rather than just one, resulting in a t1r2 classification.

The collinearity of the two reconstructed rings can be verified by looking at figure 3.4. In this plot we can observe that the blue distribution t1r2 peaks at small angle values ($\Delta\theta \leq 20°$) proving that a large fraction of t1r2 events are collinear, and that the majority of collinear events belong to the t1r2 topology.

In order to fix and recover miscounted events after the 8-digit fit step, the fiTQun algorithm performs a *merging* procedure.

Merging $\overset{\text{def}}{\equiv}$ "for any pair of rings whose relative opening angle is less than or equal to twenty degrees ($\Delta\theta \leq 20°$), fiTQun tags the two rings as fake-rings, merges them into only one, and fits the event again, performing a sequential re-fit on the results from the merging".

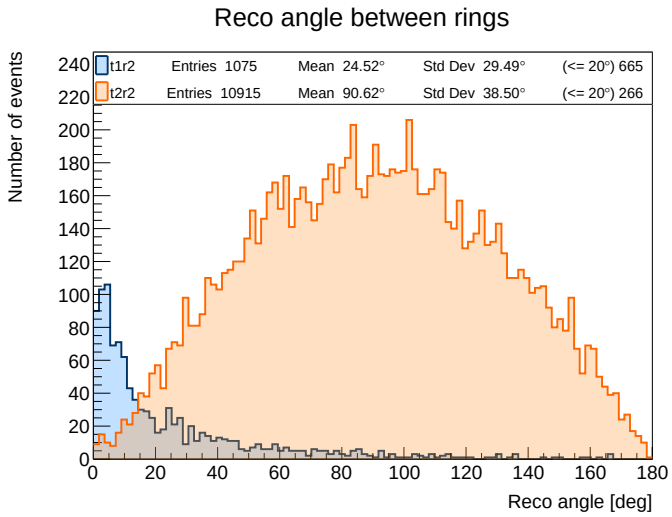


Figure 3.4. Distributions of the number of events as a function of the reconstructed opening angle ($\Delta\theta$) for the topologies t1r2 and t2r2.

This procedure is applied to all multi-ring reconstruction hypotheses r*m*, not only to r2. Therefore, it can happen that in an event reconstructed as r6 there is, for example, more than one pair with $\Delta\theta \leq 20°$ and thus the final reconstructed number of rings is reduced to r5, r4, r3, etc. Focusing only on two-ring topologies, figure 3.4, the merging procedure would be applied

to events reconstructed as r2 and with $\Delta\theta \leq 20°$. Namely,

$$\Delta\theta \leq 20°, \text{t1r2} \;\longrightarrow\; \text{t1r1}.$$
$$\Delta\theta \leq 20°, \text{t2r2} \;\longrightarrow\; \text{t2r1}.$$

Hence, it is crucial to observe that, on one hand, the merging corrects the classification of the events from t1r2 (first bullet); but, on the other hand, it introduces an error by moving events from the correct category t2r2 into t2r1 (second bullet). It follows that, in order to achieve a global improvement, it is mandatory that the number of events reclassified from t1r2 must be greater than the number moved from t2r2. That is: the merging procedure should result in an overall increase of the number of correctly reconstructed events

$$\#_{\text{evt}}(\text{t1r2},\ \Delta\theta \leq 20°) > \#_{\text{evt}}(\text{t2r2},\ \Delta\theta \leq 20°) \tag{3.9}$$

For example, in figure 3.4 this condition is satisfied (#t1r2 > #t2r2), and so the merging improves the final efficiency and purity of the reconstruction. It is precisely this observation that motivates the inclusion of the merging step in the fiTQun reconstruction chain: the procedure is justified because the gain from correcting misclassified single-ring events outweighs the loss of two-ring events.

As a final observation on the opening angle, before moving on to discuss the other variables of interest, it is worth briefly discussing the combinations of opening angles for the topologies involving more than two rings. Let N be the number of types of true primary particles ($N = 2$, e^- and μ^-) and n as the number of primaries ($n \in \{1, 2, 3, 4\}$).

Thus, if we consider an ordered set with repetitions allowed ($e\mu \neq \mu e$), the number of combinations of rings is

$$_N U_n = N^n \tag{3.10}$$

For $N = 2$

$$_2 U_n = 2^n \tag{3.11}$$

n	1	2	3	4
2^n	2	4	8	16

Conversely, if we consider an unordered set with repetitions allowed ($e\mu = \mu e$), the number of combinations of rings is

$$_N U_n^* = \binom{n+N-1}{N-1} = \binom{n+N-1}{n} = \frac{(n+N-1)!}{n!(N-1)!} \tag{3.12}$$

For $N = 2$

$$_2 U_n^* = \frac{(n+1)!}{n!} = n+1 \tag{3.13}$$

n	1	2	3	4
$n+1$	2	3	4	5

Finally, the combinations of the opening angles between two directions are given by

$$U_n = \binom{n}{2} = \frac{n(n-1)}{2} \tag{3.14}$$

n	1	2	3	4
$n(n-1)/2$	0	1	3	6

Scatter plots

Scatter plots of the following variables of interest: (ΔP), $(\min P)$, (P_1), (P_2), $(P_1 + P_2)$, $(\Delta\theta)$ have been produced and analyzed. In the following only the most relevant ones are shown; the full list is in appendix C. All the plots show the scatter distributions of the four reconstruction categories, measuring the correlation between the `nll` ratio variable on the y-axis and the variable of interest on the x-axis. In the left column are the plots "1st: ele", where the first reconstructed ring PID is e^-, while in the right column are the plots "1st: pion", where the first reconstructed ring PID is π^+ (at the 8-digit fit level). Finally, all the variables tested in this section are the reconstructed ones from `fiTQun`, not the true values from the `WCSim` simulation to avoid introducing any bias from prior external knowledge of the dataset.

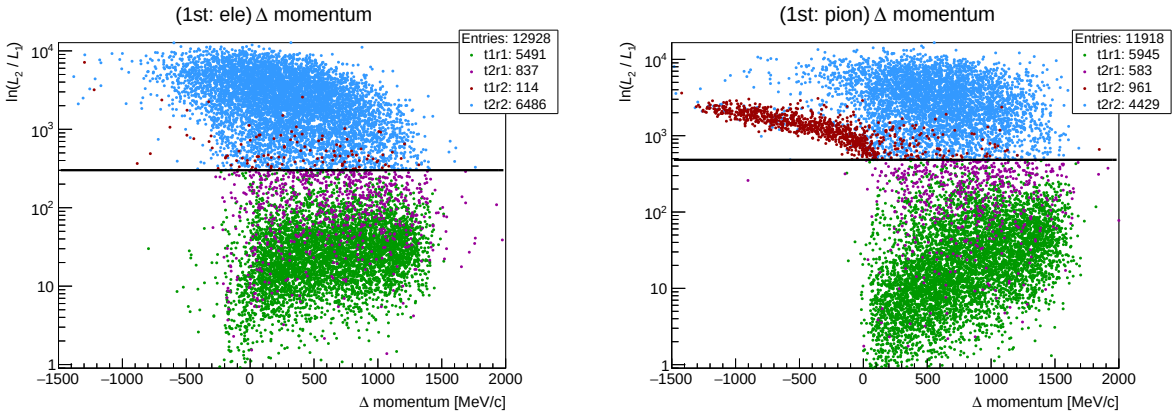


Figure 3.5. Scatter plots of the `nll` ratio versus ΔP variables.

In figure 3.5 the scatter plots of `nll` ratio versus ΔP are shown. In the left "1st: ele" plot there is no distinct clustering between the correctly and wrongly reconstructed categories, so the information provided by this scatter plot in particular was not considered sufficient to improve the reconstruction. In contrast, the right "1st: pion" plot presents a prominent "cloud" of red points t1r2, well separated from the blue t2r2. In particular, most points in the t1r2 cloud have $\Delta P < 0$. This means that the `fiTQun` algorithm has reconstructed and sorted the

rings such that the first one is less energetic than the second one. This specific clustering and shape of the two-dimensional scatter distribution has been successfully used to define a refined selection criterion, explained in the following sections.

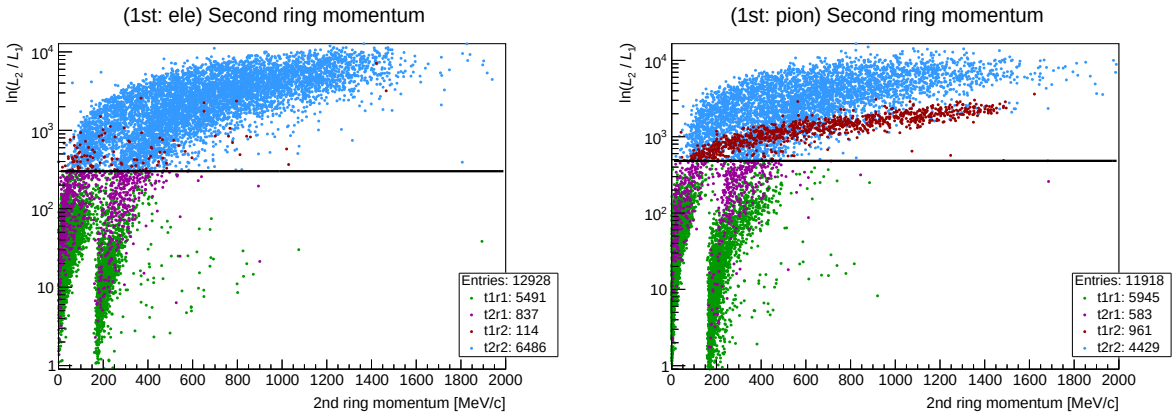


Figure 3.6. Scatter plots of the nll ratio versus P_2 variables.

The scatter plots for the variable P_2, figure 3.6, exhibit a peculiar shape. Indeed, two distinct stripes are clearly visible, with the separation being larger in the r1 region of the plots. The same "two stripes" structure, although less marked, persists in the r2 region as well.

From a physical perspective, it is worth observing that the clear separation into two stripes is a direct consequence of the hard cutoff imposed by the Cherenkov momentum threshold

$$p_{\text{th}} = \beta_{\text{th}}\,\gamma_{\text{th}}\,m = \frac{m}{\sqrt{n^2-1}} \tag{3.15}$$

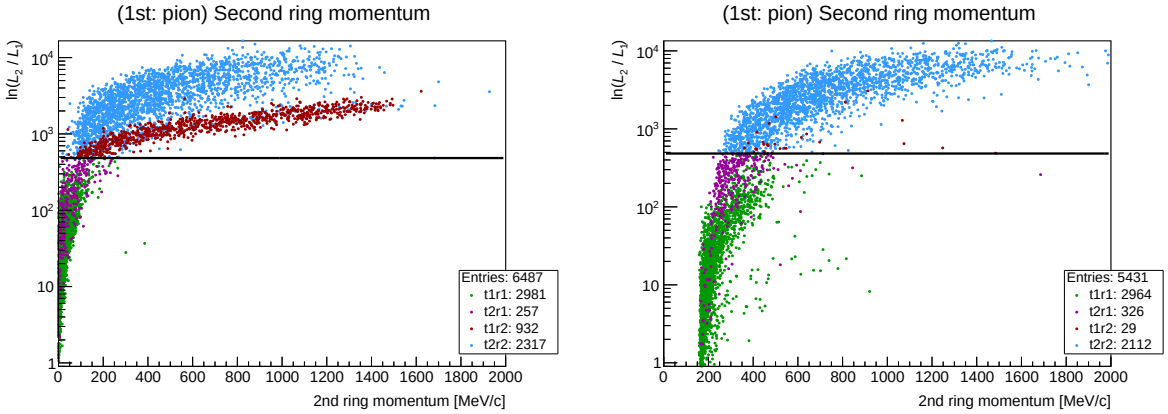
The electron mass is $m_e \simeq 0.511$ MeV/c^2 and the pion mass is $m_\pi \simeq 139.570$ MeV/c^2, hence the Cherenkov momentum thresholds correspond to $p^e_{\text{th}} \simeq 0.580$ MeV/c and $p^\pi_{\text{th}} \simeq 158.348$ MeV/c respectively. If a pion has a momentum less than p^π_{th}, it is completely invisible to the detector because it does not produce a Cherenkov ring. Because fiTQun is fitting physical Cherenkov light profiles, it fundamentally cannot reconstruct a 2nd pion ring below this momentum threshold, which explains why the scatter plot in figure 3.7b is empty below this value.

Therefore, this separation into two distinct stripes is a direct consequence of the two PID hypotheses. The left stripe arises from 2nd rings that are reconstructed as e^-, while the right stripe from 2nd rings reconstructed as π^+.

Despite the presence of these two stripes, the distributions do not provide a clear separation between the four event categories (no color clustering). Thus, unlike ΔP, this variable was not used to improve the selection efficiency and purity of the cut. Nevertheless, it is important to note that the fact that this variable, as well as others, was not used in this study, does not imply that it could not be used at a later stage for a more advanced and sophisticated study.

It is also important to note that the t1r1 and t2r1 topologies are those in which fiTQun reconstructed only one ring; the plotted P_2 momentum is therefore that of a fake ring. This ring is truly fake for the t1r1 topology, and wrongly reconstructed as fake for the t2r1 topology. However, it follows that in both cases the momentum of the (fake) 2nd ring is not "large" enough to produce a sufficient amount of Cherenkov light and accept the two-ring hypothesis.

The first category t1r1 consists of true single-ring events. However, the fiTQun algorithm evaluates a two-ring hypothesis for all events to determine if an additional ring yields a better fit. In the absence of a physical 2nd ring, the minimizer attempts to reconstruct a secondary ring from background sources such as scattered photons, PMT dark noise, or optical reflections. Consequently, this fake ring is associated with a minimal amount of detected light. Because the Cherenkov light yield is proportional to a particle's momentum, this low photon count translates to a very "small" reconstructed momentum for the 2nd ring. The second category t2r1 represents events that possess two true primary particles, but fiTQun rejects the secondary ring and classifies the event under a single-ring hypothesis. This misclassification typically occurs when the 2nd particle is kinematically weak, often possessing a momentum just above the Cherenkov threshold p_{th}. Because this low-energy particle produces a limited amount of light, the two-ring fitter assigns it a correspondingly low reconstructed momentum. Finally, the presence of this faint 2nd ring does not provide a sufficient improvement to the overall fit likelihood to convince the algorithm that the signal is a distinct physical particle rather than background noise.



(a) $1^{\text{st}} : \pi^+ \mid 2^{\text{nd}} : e^-$. (b) $1^{\text{st}} : \pi^+ \mid 2^{\text{nd}} : \pi^+$.

Figure 3.7. Scatter plots of the nll ratio versus P_2 variables, with 1st reconstructed ring PID as a π^+ and 2nd as an e^- (left) and as a π^+ (right).

The vast majority of the t1r2 events have $\Delta P < 0$ (figure 3.5) and reconstructed PIDs (figures 3.6 and 3.7) as

$$1^{\text{st}}\text{: } \pi^+ \mid 2^{\text{nd}}\text{: } e^-$$

Indeed, out of a total of 961 t1r2 events,

- 96.98% are 1st: π^+ | 2nd: e^-;
- 3.02% are 1st: π^+ | 2nd: π^+.

Finally, from the analysis of the opening angle scatter plots, figure 3.8, a clear and distinct red cloud of t1r2 points emerges for the 1st: π^+ ring case. As previously discussed, all the events reconstructed as two-ring topologies with a $\Delta\theta \leq 20°$ will be merged in a later fitting step.

However, despite solving the t1r2 misclassification, the merging procedure would also introduce a new error by reclassifying t2r2 events as t2r1 events. Therefore, it is crucial to look for a finer selection cut at the 8-digit fit step to avoid this overly broad merging procedure.

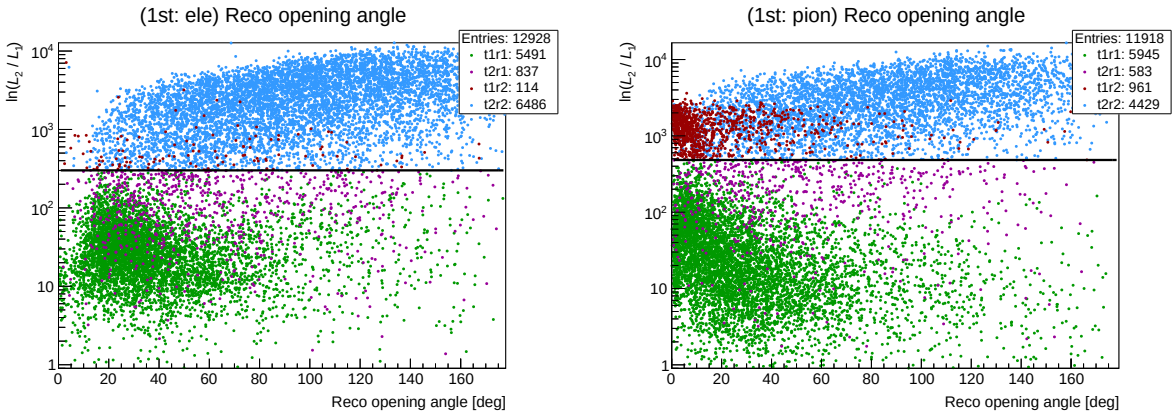


Figure 3.8. Scatter plots of the `nll` ratio versus $\Delta\theta$ variables.

3.3 New selection criteria

Following the strategy introduced by equation (3.7), we worked on the implementation of a new bi-dimensional family of cuts dependent on `nll` ratio and $\Delta\theta$

$$\ln\left(\frac{L_{n+1}}{L_n}\right) > \text{constant} \longrightarrow \ln\left(\frac{L_{n+1}}{L_n}\right) > A\ \exp(B\,\Delta\theta) \tag{3.16}$$

In particular, according to the log-scale of the y-axis, it follows that

$$\ln(y) = Bx + \ln(A) \tag{3.17}$$

where the straight line parameters are the *slope* B and the *y-intercept* $\ln(A)$. A is dimensionless, while B has dimension $\left[\text{deg}^{-1}\right]$.

From the shape of the scatter plot distributions in figure 3.8, we decided to implement "double line" selection cuts, distinguishing between 1st reconstructed PID. This choice was driven by the observation that a steep cut is necessary at "small" opening angles ($\Delta\theta \leq 20°$), while a shallower slope is more appropriate at "big" opening angles ($\Delta\theta > 20°$).

The "double line" selection was formally defined as

$$\begin{cases} \text{If } \Delta\theta \leq 25°, & \ln\left(\dfrac{L_{n+1}}{L_n}\right) > A_1^e\ \exp\left(B_1^e\,\Delta\theta\right) \\ \text{If } \Delta\theta > 25°, & \ln\left(\dfrac{L_{n+1}}{L_n}\right) > A_2^e\ \exp\left(B_2^e\,\Delta\theta\right) \end{cases} \tag{3.18}$$

$$\begin{cases} \text{If } \Delta\theta \leq 22°, & \ln\left(\dfrac{L_{n+1}}{L_n}\right) > A_1^\pi\ \exp\left(B_1^\pi\,\Delta\theta\right) \\ \text{If } \Delta\theta > 22°, & \ln\left(\dfrac{L_{n+1}}{L_n}\right) > A_2^\pi\ \exp\left(B_2^\pi\,\Delta\theta\right) \end{cases} \tag{3.19}$$

Hence, we decided to keep the distinction based on the 1st ring reconstructed PID, so to use equation (3.18) for 1st: e^- and equation (3.19) for 1st: π^+. Each criterion depends on two exponential cutting functions, with the transition between them occurring at the corresponding "splitting angle". The numerical values of 25° and 22° have been computed and fine-tuned as explained in the following. Therefore, for each PID there is a set of four parameters (A_1, B_1, A_2, B_2): three of those are independent and need to be optimized, while the fourth is given by the *continuity condition* between the two cutting lines as

$$B_2 = B_1 + \frac{\ln(A_1/A_2)}{\Delta\theta_{\text{split}}} \tag{3.20}$$

To measure and compare the performance of different selection cut criteria, it was decided to use the following metric

$$J = \epsilon_1 \, \epsilon_2 \, p_1 \, p_2 \tag{3.21}$$

This metric computes the product of the efficiencies, equation (3.4), and the purities, equation (3.5), of the categories t1r1 and t2r2. Henceforth, the new optimal cut parameters are defined as those that maximize the J metric

$$(A_1^*, \, B_1^*, \, A_2^*, \, B_2^*) \mid J = \max\{\boldsymbol{J}\} \tag{3.22}$$

Finally, the score obtained by the standard selection criteria defined in equations (2.11) and (2.12) under the J metric was computed and presented in table 3.3.

Std cut	e^-	π^+
ϵ_1 (%)	97.97	86.08
p_1 (%)	86.77	91.07
ϵ_2 (%)	88.57	88.37
p_2 (%)	98.27	82.17
J index	0.740	0.569

Table 3.3. Electron and pion efficiencies, purities, and resulting J values computed at the end of the 8-digit fit level, using the default standard selection cuts.

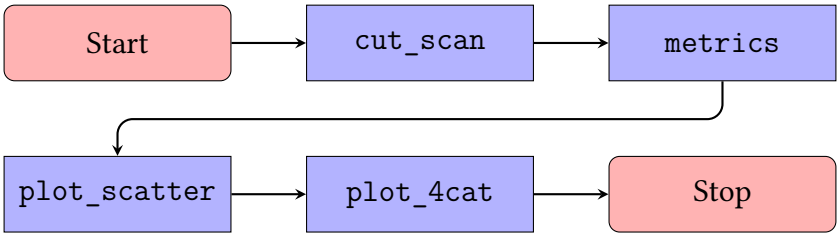
3.3.1 Strategy for optimal selection cuts

Before looking for the strategy to find the new optimal cuts, it is worth to spend some time to point out the logic path followed and the main caveats to keep in mind. Indeed, the main concept that should be fully understood is that, in order to perform the classification with the new cuts, we need to know the value of $\Delta\theta$ *a priori*, meaning before deciding whether the best 8-digit fit hypothesis is one or two-ring. The reason behind this is that the new cuts depend on both `nll` and $\Delta\theta$ values.

In order to better illustrate the difference between this new approach and the standard one, consider the `nll` ratio-$\Delta\theta$ scatter plots in figure 3.8. These plots were produced using the standard selection criteria, dependent solely on `nll`, and not on $\Delta\theta$. It is therefore clear that the

standard cuts need "to know" only the nll values in order to classify the multi-ring hypotheses. Hence, the $\Delta\theta$ values showed in the plots were not used to perform the classification, but are just the opening angle values of the "best" hypotheses selected by the nll-only classification.

That being said, let's outline the general strategy, referring to the structure implemented in the C++ ROOT script twocut.C



A parameter scan is performed by the function cut_scan to determine the optimal cut thresholds for the two PID hypotheses. The full parameter space can be described as a single eight-dimensional vector $\boldsymbol{v}$

$$\boldsymbol{v} = (A_1^e,\ B_1^e,\ A_2^e,\ B_2^e,\ A_1^\pi,\ B_1^\pi,\ A_2^\pi,\ B_2^\pi) \tag{3.23}$$

Alternatively, depending on the chosen optimization strategy, it can be decoupled into two four-dimensional vectors $\boldsymbol{v}^e$ and $\boldsymbol{v}^\pi$

$$\boldsymbol{v}^e = (A_1^e,\ B_1^e,\ A_2^e,\ B_2^e) \quad \boldsymbol{v}^\pi = (A_1^\pi,\ B_1^\pi,\ A_2^\pi,\ B_2^\pi) \tag{3.24}$$

The metric used is the J index (*figure of merit*) defined in equation (3.21). For each PID case, the scan produces a plot of the J index distribution as a function of the cut parameters, and updates the variables out_bestA1_e, out_bestB1_e, out_bestA2_e, out_bestB2_e, and out_bestA1_pi, out_bestB1_pi, out_bestA2_pi, out_bestB2_pi with the best values found for the electron and pion branches, respectively. The results are written to a .txt file and exported for integration into the reconstruction pipeline.

At this stage, where we only deal with the MC truth, the algorithm is not provided with any classification criteria, thus no hypothesis can be identified as the "best". It follows therefore that, when looking for new cuts, the distinction between the two 1st ring PIDs, previously used to draw the scatter plot distributions, cannot be used anymore.

It is therefore crucial to decide: *how to perform the scan in a way that is both logically sound and computationally efficient*, and *which subsample of the data should be selected for the scan*?

Before any cut is available, and therefore without a classification rule, no hypothesis can be labeled as the "best", by definition. It follows that there cannot be any "best" hypothesis among the two *sons* of the same *father* with 1st ring PID e^- or π^+, unless we use the "best" hypothesis previously classified by the standard cuts. For example, when dealing with the true number of ring t1 and t2, there is no way to decide which is the "best" reconstructed hypothesis among the two, r1 and r2. Actually, there should not be, since it is what we are looking for. Selecting one hypothesis a priori might compromise the purpose of the search itself. Therefore, during the cut scan, or we rely on the "best" from the previous standard cuts fiTQun reconstruction, or we use both 2nd ring hypotheses that belong to the same parent PID.

Henceforth, for a given trial cut the cut_scan function should

- classify the event as r1, if both hypotheses lie below the cut;
- classify the event as r2, if at least one hypothesis lies above the cut.

In principle, the most general and independent approach to the problem would be to compute the four two-ring hypotheses opening angles ($\Delta\theta_{11}$, $\Delta\theta_{31}$, $\Delta\theta_{13}$, $\Delta\theta_{33}$) and produce a single scatter plot with all of them. This would result in a scatter plot containing both e^- and π^+ 1st-ring distributions, and four points for each event, corresponding to the four two-ring hypotheses. Starting from this scatter distribution the strategy would be to use the knowledge of the true number of rings and tag each hypothesis (represented by a point in the plot) accordingly to his true number of rings, one or two. Finally, test many different possible selections cuts configurations, varying the $\boldsymbol{v}$ parameters to find the optimal one, maximizing J.

However, this way is too generic and not precise enough; in particular it does not take advantage of the splitting depending on the 1st ring reconstructed PID. Moreover, one should also pay attention to the fact that the likelihood ratios are invariant with respect to the selection threshold. The underlying likelihood values for the two single-ring and four two-ring hypotheses are fixed *a priori* by `fiTQun` calculation and remain independent of the chosen selection criterion. Consequently, the resulting likelihood ratios are constant; adjusting the selection cut-off alters only the classification threshold, determining whether an event is reconstructed under a single or multi-ring topology, without modifying the foundation (`nll` ratios) upon which this decision is based.

3.3.2 Cut scan

Two strategies have been considered for scanning the parameters of the new selection cuts. Before introducing the two possible scan approaches, it is helpful to explain the general structure of the scan procedure. The scan is performed over a two-dimensional region, chosen as a rectangular box, defined by x-range = $[\theta_{\text{min}}, \theta_{\text{max}}]$ and y-range = $[\Delta\texttt{nll}_{\text{min}}, \Delta\texttt{nll}_{\text{max}}]$. Then, a fixed number of *iterations*, N_{it}, and *steps*, N_{st}, is set. For every iteration, all N_{st} steps are evaluated. At the end of an iteration, the boundaries of the search range are updated and reduced around the region of interest. This procedure continues until all iterations and steps have been completed.

At each iteration, the range is discretized into a "grid" pattern, whose *density*[3] of points is determined by the number of steps along the x and y axes. Naturally, a larger number of steps produces a finer grid and therefore a more precise scan.

The scan can be configured to either explore the full eight-dimensional parameter space, described by equation (3.23), or to carry out two independent four-dimensional scans for each hypothesis, described by equation (3.24).

When evaluating these two approaches, it should be observed that even when the electron and pion branches are examined separately, the final step always consists in selecting the best hypothesis among all candidates. It follows that and independent scan of the two branches is not logically consistent. Indeed, while a given electron cut is being tested, the pion branch

[3] The definition of *density* is

$$\textit{Density} \overset{\text{def}}{\equiv} \frac{\text{Total number of points in the grid}}{\text{Range area}} \tag{3.25}$$

must also be taken into account, because the optimal electron cut depends on how the pion hypotheses behave under the corresponding pion cut.

For example, consider a true two-ring event t2, where both the reconstructed electron hypotheses fall below the electron cut being tested; they would be assigned to the single-ring category r1. In an isolated electron scan, the algorithm would discard these cut parameters, in favour of a cut that accepts at least one electron two-ring hypothesis r2. However, this reasoning ignored the pion branch. If at least one of the two pion hypotheses lies above the pion cut, the event is already correctly reconstructed as r2, through the pion branch. The event would therefore be correctly reconstructed as a two-ring even if the electron branch rejects both electron hypotheses. In such a case, there is no advantage in forcing the electron cut to recover the r2 classification. This shows why the electron and pion cut parameters should not be optimized independently. Finally, it is crucial to observe that, at this stage, our main goal is "multi-ring", hence to correctly *count* the number of rings; while improving PID efficiency is left for later optimization.

The total number of steps $\mathrm{Tot}_{\mathrm{st}}$ required by this strategy can be computed as

$$\begin{aligned}\mathrm{Tot}_{\mathrm{st}} = {} & [N_{\mathrm{st}}(A_1)\, N_{\mathrm{st}}(B_1)\, N_{\mathrm{st}}(A_2)\, N_{\mathrm{st}}(B_2)]_e \\ & \times [N_{\mathrm{st}}(A_1)\, N_{\mathrm{st}}(B_1)\, N_{\mathrm{st}}(A_2)\, N_{\mathrm{st}}(B_2)]_\pi\end{aligned} \tag{3.26}$$

where N_{st} is the number of steps for a single parameter. From equation (3.26) emerges the limitation of this approach. Indeed, although logically consistent, the computational cost of scanning the full 8D parameter space is extremely high. Even with just $N_{\mathrm{st}} = 10$ for each of the eight parameters, the total number of possible configurations would require $\mathrm{Tot}_{\mathrm{st}} = 10^8$ steps.

It follows that the second strategy therefore represents a compromise between logical correctness and computational feasibility. Despite the previously established need to evaluate both PID branches simultaneously, this decoupled approach performs the parameter scan twice: once for the electron hypothesis and once for the pion hypothesis. While not fully rigorous, it reduces the computational time and so the number of required steps to

$$\begin{aligned}\mathrm{Tot}_{\mathrm{st}} = {} & [N_{\mathrm{st}}(A_1)\, N_{\mathrm{st}}(B_1)\, N_{\mathrm{st}}(A_2)\, N_{\mathrm{st}}(B_2)]_e \\ & + [N_{\mathrm{st}}(A_1)\, N_{\mathrm{st}}(B_1)\, N_{\mathrm{st}}(A_2)\, N_{\mathrm{st}}(B_2)]_\pi\end{aligned} \tag{3.27}$$

Therefore, with $N_{\mathrm{st}} = 10$, $\mathrm{Tot}_{\mathrm{st}} = 2 \cdot 10^4$ steps. This is the strategy implemented in the analysis. The electron and pion branches are scanned independently: operatively, for each point in the four-dimensional parameter space vector $(A_1,\ B_1,\ A_2,\ B_2)$, the algorithm builds the corresponding "double-line" selection cut, and applies it to the input data (t1 and t2 event samples). The reconstruction is then evaluated in terms of efficiency ϵ and purity p of reconstruction, assigning a J index score to each parameter space configuration. Once the scan has finished testing all the possible configurations, so all the points in the space grid, the configuration yielding the maximum J index value is chosen as the new best cut $(A_1^*,\ B_1^*,\ A_2^*,\ B_2^*)$.

Because the optimization relies on this decoupled scan search procedure, it follows that the choice of the input sample is crucial. The final improvement on the selection cuts efficiency and purity directly depends on the input data. In particular, this brings us to the other question from before: *which subsample of the data should be selected for the scan*?

Below we list the main criteria used to select the input sample for the scan.

- "True PID": an event is included in the cut search only if it matches the 1st ring true PID (t1 or t2). For example if t1 PID is e^-, the hypothesis reconstructed as $e^- \pi^+$ would not be used for the new electron cut search.
- "Best `nll`": for each PID branch, only the "best" two-ring hypothesis is plotted and used. (When this option is selected, the following "both hypotheses" option is not used, and vice versa).
- "Both hypotheses": for each branch, both hypotheses are used, thereby doubling the number of entries with respect to the previous criterion.
- "Visible momentum P_{vis} sorting": for the case of true multi-ring, i.e. t2, the `WCSim` output is not sorted. So, in order to define a 1st true ring, two momentum-based sorting schemes were therefore tested, based on P or on P_{vis}.
- "ΔP cut on pion events": a dedicated cut in the `nll` ratio-ΔP plane is applied to refine and increase the pion branch reconstruction.

The `WCSim` simulation software does not sort the rings. A sorting procedure is thus required, in particular because the true PID selection criterion depends on defining which ring is the 1st true simulated ring. The ring sorting was done according to decreasing momentum P. In parallel, the visible momentum P_{vis} was computed as well

$$P_{\text{vis}} = \sqrt{E_{\text{vis}}^2 - m^2} \tag{3.28}$$

where E_{vis} is the visible energy previously defined in section 2.2.4.

These criteria were combined in 16 different versions of the `twocut.C` script.

`twocut`	No true PID selection, "best" `nll` hypothesis.
`v1CASE1`	True PID selection (P sorted), "best" `nll` hypothesis.
`v1CASE2`	True PID selection (P sorted), 2nd ring matching hypothesis[4].
`v2NOTRUE`	No true PID selection, both hypotheses.
`v2TRUE`	True PID selection (P sorted), both hypotheses.
`v3`	No true PID selection, "best" `nll` hypothesis and ΔP pion cut applied after scan.
`v4CASE1`	Same as `v1CASE1` with P_{vis} sorting.
`v4CASE2`	Same as `v1CASE2` with P_{vis} sorting.
`v5NOTRUE`	Same as `v2NOTRUE` sorting.
`v5TRUE`	Same as `v2TRUE` with P_{vis} sorting.
`v6NOTRUE`	Same as `v2NOTRUE` with ΔP pion cut.
`v6TRUE`	Same as `v2TRUE` with ΔP pion cut.
`v7CASE1`	Same as `v1CASE1` with ΔP pion cut.
`v7CASE2`	Same as `v2CASE1` with ΔP pion cut.
`v7CASE3`	Same as `v3` with ΔP pion cut applied before the scan.
`v8`	Final optimal cuts combining `v7CASE1` for electron and `v7CASE3` for pion branch.

All the 16 configuration were executed with all the same initial cutting parameters set to zero, $\boldsymbol{v} = \mathbf{0}$, and with initial parameter values, ranges and steps defined in table 3.4.

[4] If t1 → "best" `nll` hypothesis; if t2 → r2 matching both the `WCSim` true PIDs.

Parameters	**Range**		N_{st} **of 1st iteration**	N_{st} **of 2nd iteration**
θ range	$[0°,\ 180°]$	**A1**	30	30
Δnll range	$[1,\ 10000]$	**B1**	30	30
Split angle ($e^- \mid \pi^+$)	$10° \mid 20°$	**A2**	30	30

Table 3.4

For each version the efficiency ϵ, purity p and J index were measured and compared in the following tables; with the version that maximizes the J index highlighted in bold.

e^-	twocut	v1 CASE1	v1 CASE2	v2 NOTRUE	v2 TRUE	v3	v4 CASE1	v4 CASE2
ϵ_1 (%)	96.96	94.95	94.91	96.16	95.15	97.05	95.12	94.99
ϵ_2 (%)	90.63	93.04	92.94	91.43	92.95	90.23	92.99	92.87
p_1 (%)	89.46	91.17	91.09	90.20	91.14	89.54	91.13	91.07
p_2 (%)	97.32	96.05	96.00	96.67	96.17	97.26	96.18	96.03
J index	0.765	0.774	0.771	0.767	0.775	0.763	0.775	0.771

e^-	v5 NOTRUE	v5 TRUE	v6 NOTRUE	v6 TRUE	v7 CASE1	v7 CASE2	v7 CASE3	v8
ϵ_1 (%)	96.16	94.58	96.35	95.16	95.65	95.51	97.11	**95.61**
ϵ_2 (%)	91.43	93.20	91.15	92.68	92.40	92.30	90.23	**92.16**
p_1 (%)	90.20	91.38	90.40	91.76	91.49	91.41	89.57	**91.35**
p_2 (%)	96.67	95.75	96.65	95.72	96.14	95.99	97.31	**96.03**
J index	0.767	0.771	0.767	0.775	**0.777**	0.774	0.764	**0.773**

π^+	twocut	v1 CASE1	v1 CASE2	v2 NOTRUE	v2 TRUE	v3	v4 CASE1	v4 CASE2
ϵ_1 (%)	89.17	82.53	83.77	88.38	83.07	94.41	83.28	84.39
ϵ_2 (%)	95.24	95.73	94.82	94.61	95.57	95.16	95.23	94.94
p_1 (%)	96.05	96.40	95.73	95.52	96.28	96.04	96.03	95.82
p_2 (%)	87.13	79.82	80.85	86.23	80.37	93.20	80.42	81.54
J index	0.711	0.608	0.615	0.689	0.614	0.804	0.613	0.626

π^+	v5 NOTRUE	v5 TRUE	v6 NOTRUE	v6 TRUE	v7 CASE1	v7 CASE2	v7 CASE3	v8
ϵ_1 (%)	88.38	83.23	93.68	93.07	93.29	94.30	94.33	**94.64**
ϵ_2 (%)	94.61	95.46	94.56	95.42	95.18	94.88	95.21	**95.25**
p_1 (%)	95.52	96.20	95.52	96.20	96.03	95.82	96.05	**96.10**
p_2 (%)	86.23	80.49	92.36	91.70	91.89	93.04	93.14	**93.50**
J index	0.689	0.615	0.782	0.783	0.784	0.798	**0.804**	**0.810**

3.3.3 Classification

Once identified the best parameters for each of the 16 configurations, performance metrics (efficiency, purity and J index) are computed and printed by the function `metrics`. The results are compared to the standard selection cuts ones. Finally, a new branch `new_fitId` is added to the `TTree` to store the "best" fit result.

The function `plot_scatter` produces a scatter plot of `nll` ratio versus $\Delta\theta$ for the true single-ring t1 (green) and two-ring t2 (blue) event categories in figure 3.9, with both the standard horizontal cut and the newly optimized selection boundary overlaid. It is invoked twice, once for 1st reconstructed PID as electron and once as pion.

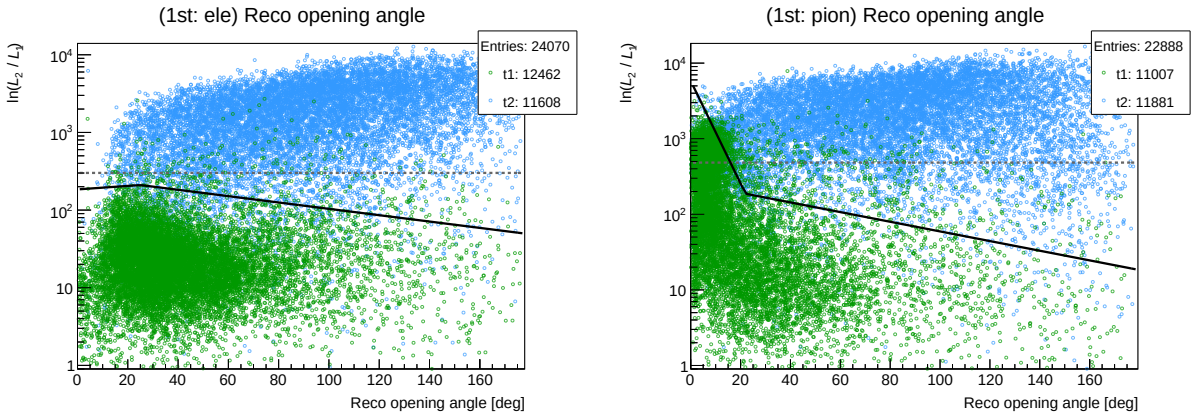


Figure 3.9. Two-color scatter plots for the new cuts search scan, obtained with the v8 configuration introduced in section 3.3.1. The t1 category represents the reconstructed hypotheses associated with one true ring, while the t2 the hypotheses associated with two true rings.

For example, the logic followed by the `twocut` configuration (no true PID selection, "best" `nll` hypothesis) would be to independently evaluate both the electron and pion hypotheses for each of the 24846 t1 and t2 events in the dataset. Consequently, each scatter plot would contain the same number of events (24846), colored by true ring multiplicity, because every event is tested under both PID hypotheses. This is an important difference to observe in the chain of functions of the script. Indeed, this step is performed before the "classification" with the new criteria and so the number of selected events for the scan varies according to the configuration being used. On the opposite, after the classification, a unique best-fit PID hypothesis is assigned to each event, so the electron and pion distributions together account for the total number of events.

In summary, the optimal configuration for the electron branch is given by the `v7CASE1`, while for the pion branch is given by the `v7CASE3`. These results also show that imposing the true PID condition does not improve the scan. Thus, we combined these two configurations into a single script, `v8twocut.C`. As we pointed out before, since the electron and pion scans are not independent, combining the two separately optimized cuts changes the final result, as can be seen in the v8 columns of the tables. In particular, the electron J index decreases, while the pion J index improves.

Successively, a nested loop over $\Delta\theta$ was used to identify the optimal splitting angle for the

two branches.

$$\Delta\theta^{e}_{\text{split}} = 25^{\circ}$$
$$\Delta\theta^{\pi}_{\text{split}} = 22^{\circ}$$

The ΔP cut applied to the pion branch was motivated by the t1r2 scatter point distribution shown in figure 3.5. In the code, this cut is implemented before the `cut_scan` function: specifically, events whose 1st ring is reconstructed as a pion are rejected if their `nll` is below 3000 and their ΔP value is negative.

Once the classification of the multi-ring hypotheses is complete, the reconstruction finally proceeds to select the "best" hypothesis. This is done by the function `plot_4cat`, which attempts to reproduce and predict as accurately as possible the `fiTQun` final results after the 9-digit steps using the new criteria. It draws a scatter plot for each 1st ring PID hypothesis, showing the distribution of events across the four reconstruction categories (t1r1, t2r1, t1r2, t2r2). It computes and adds to the legend the efficiency and purity. When the horizontal axis corresponds to the opening angle $\Delta\theta$ the new cut value is overlaid as a reference line. Note: here, differently from `plot_scatter`, the classification has already been applied, and the function input is the "best" fit hypothesis for each event. Consequently, the events are partitioned according to the assigned 1st ring PID, and the total number of events in the two plots sums to the full dataset.

For example, consider a true two-ring event where the two hypotheses with electron parent PID are both rejected, while one of the two with pion parent PID is rejected and the other (say, both rings identified as pions) is accepted. Hence, we have that the electron branch is reconstructed as r1, while the pion branch as r2. Finally, `fiTQun` compares the `nll` of the pion two-ring accepted hypothesis with the `nll` of the electron single-ring hypothesis: the smaller is selected as the "best" reconstructed hypothesis. Similarly, if both pion parent PID hypotheses are accepted, the `nll` values of the two pion parent two-rings accepted hypotheses and the single-ring electron hypothesis are compared (three values), again picking the minimum.

The selection logic for the new cut criteria is set out in detail below.

a) If 1st ring is electron:

$$1^{\text{st}}\ e^- \longrightarrow e^-\, e^-$$
$$1^{\text{st}}\ e^- \longrightarrow \pi^+\, e^-$$

1. if both $e^-\, e^-$ and $\pi^+\, e^-$ pass the electron cut $\longrightarrow$ r2, the one with the lower `nll` is chosen as the "best" two-ring hypothesis (r2 event with 1st ring e^-);
2. if only one of them passes the cut $\longrightarrow$ r2, that hypothesis is selected;
3. if neither passes the cut $\longrightarrow$ r1, the event is classified as single-ring r1, with the 1st ring PID electron. For the $\Delta\theta$ calculation, the better of the two rejected two-ring hypotheses (smallest `nll`) is used as the 2nd ring.

b) If 1st ring is pion:

$$1^{\text{st}}\ \pi^+ \longrightarrow e^-\, \pi^+$$
$$1^{\text{st}}\ \pi^+ \longrightarrow \pi^+\, \pi^+$$

1. if both $e^-\, \pi^+$ and $\pi^+\, \pi^+$ pass the pion cut $\longrightarrow$ r2, the one with the lower `nll` is chosen as the "best" two-ring hypothesis (r2 event with 1st ring π^+);

2. if only one of them passes the cut → r2, that hypothesis is selected;
3. if neither passes the cut → r1, the event is classified as single-ring r1, with the 1st ring PID pion. For the $\Delta\theta$ calculation, the better of the two rejected two-ring hypotheses (smallest `nll`) is used as the 2nd ring.

c) The final "best" hypothesis is chosen by comparing the results from steps (a) and (b):

- if both branches yielded a two-ring hypothesis r2, the one with the lowest `nll` is kept;
- if both yielded a single-ring hypothesis r1, the single ring with the lowest `nll` is chosen;
- if one branch gave r1 and the other r2, the hypothesis with the lowest `nll` is selected, (independently of the ring multiplicity).

3.3.4 Final results

Finally, the new best cut parameters are given by the `v8` configuration

- $A_1^e = 184.6$
- $B_1^e = 0.0052\ \text{deg}^{-1}$
- $A_2^e = 265.9$
- $B_2^e = -0.0094\ \text{deg}^{-1}$
- $\Delta\theta^e_{\text{split}} = 25°$
- $A_1^\pi = 5815.2$
- $B_1^\pi = -0.1562\ \text{deg}^{-1}$
- $A_2^\pi = 258.4$
- $B_2^\pi = -0.0147\ \text{deg}^{-1}$
- $\Delta\theta^\pi_{\text{split}} = 22°$

The impact of the new selection criteria on the `nll` ratio versus ΔP distribution is presented in figure 3.10. Comparing it with figure 3.5 reveals a clear improvement due to the ΔP cut applied on the pion branch. Indeed, almost all the misclassified t1r2 events are removed.

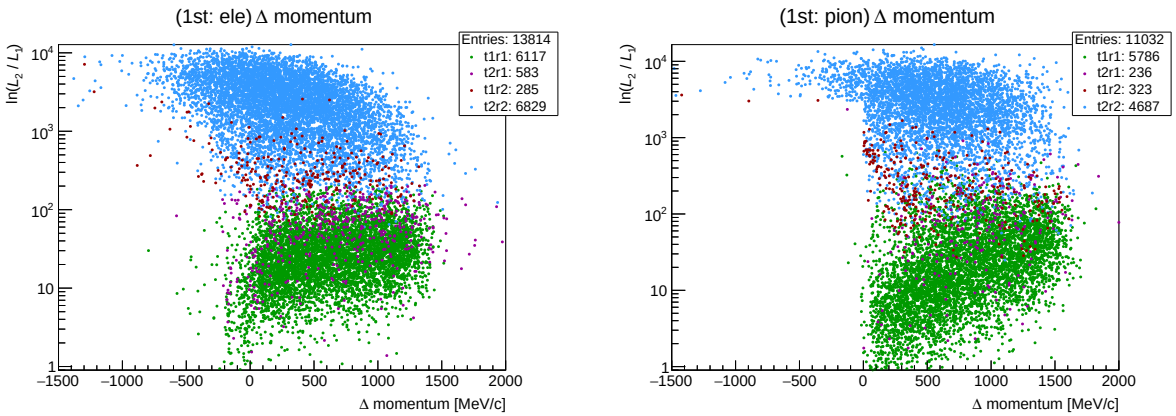


Figure 3.10. Scatter plots of the `nll` ratio versus ΔP variables, using the new cut parameters.

The comparison of figure 3.11 with figure 3.8 highlights the distinct behavior of the electron and pion branches. For the electron branch, the double line selection cuts are optimized to improve the correct classification of two-ring events, although this comes at the cost of a slight increase in contamination within the single-ring category. For the pion branch, by contrast, the steep cut below $\Delta\theta^\pi_{\text{split}} = 22°$ corrects most of the t1r2 events.

Finally, it is crucial to observe that, in particular for the pion branch, this new selection criteria solve the *collinearity* problem between rings, without requiring the rough merging procedure that was previously necessary.

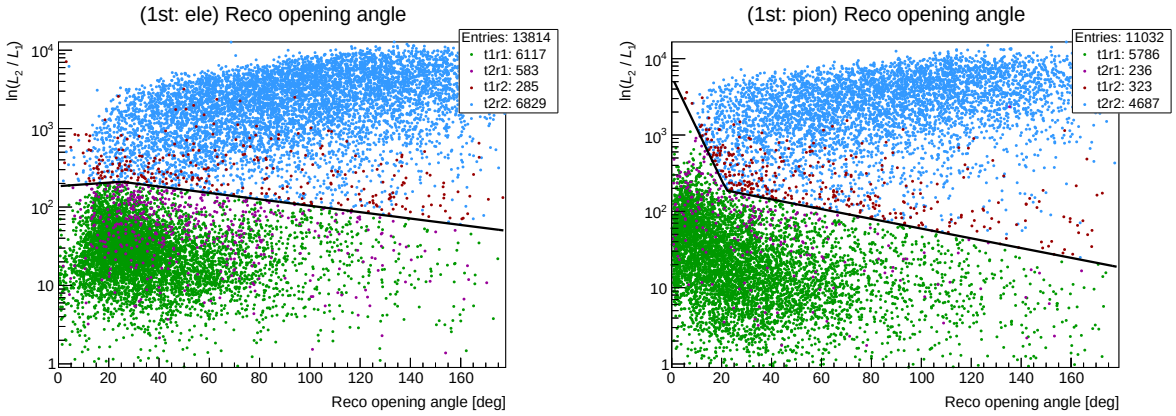


Figure 3.11. Scatter plots of the `nll` ratio versus $\Delta\theta$ variables, using the new cut parameters.

The result presented in the tables 3.5 to 3.10 reveal an asymmetry in the impact of the new selection cuts on the electron and pion branches.

e^-	**Std cut**	**New cut**
t1r1	5491	6117
t2r1	837	583
t1r2	114	285
t2r2	6486	6829

Table 3.5. Comparison of standard and new cuts.

π^+	**Std cut**	**New cut**
t1r1	5945	5786
t2r1	583	236
t1r2	961	323
t2r2	4429	4687

Table 3.8. Comparison of standard and new cuts.

e^-	**Std cut**	**New cut**
ϵ_1	0.980 ± 0.002	0.955 ± 0.003
p_1	0.868 ± 0.004	0.913 ± 0.003
ϵ_2	0.886 ± 0.004	0.921 ± 0.003
p_2	0.983 ± 0.002	0.960 ± 0.002

Table 3.6. Efficiency and purity.

π^+	**Std cut**	**New cut**
ϵ_1	0.861 ± 0.004	0.947 ± 0.003
p_1	0.911 ± 0.004	0.961 ± 0.003
ϵ_2	0.884 ± 0.005	0.952 ± 0.003
p_2	0.822 ± 0.005	0.936 ± 0.003

Table 3.9. Efficiency and purity.

e^-	**Net change**	
Net ϵ_1 loss	-0.024 ± 0.003	$(7.6\ \sigma)$
Net p_1 gain	$+0.045 \pm 0.005$	$(8.3\ \sigma)$
Net ϵ_2 gain	$+0.036 \pm 0.005$	$(7.3\ \sigma)$
Net p_2 loss	-0.023 ± 0.003	$(8.1\ \sigma)$

Table 3.7. Changes and significance.

π^+	**Net change**	
Net ϵ_1 gain	$+0.086 \pm 0.005$	$(17.1\ \sigma)$
Net p_1 gain	$+0.050 \pm 0.004$	$(11.6\ \sigma)$
Net ϵ_2 gain	$+0.068 \pm 0.005$	$(12.5\ \sigma)$
Net p_2 gain	$+0.114 \pm 0.006$	$(18.2\ \sigma)$

Table 3.10. Changes and significance.

For the electron branch, the new classification resulted in a redistribution of the misclassified events, rather than a uniform improvement across all categories. The increase in the t1r1 category (5491 → 6117) indicates that the new criteria accept a larger fraction of true single-ring events. However, this gain is followed by a rise in t1r2 events (114 → 285), suggesting that some true single-ring events are now being classified as two-ring events. On the opposite, the decrease in t2r1 events (837 → 583) shows that fewer true two-ring events t2r2 (6486 → 6829) are being mistaken for single-ring events. These opposing trends are reflected in the efficiency and purity values (table 3.6). Indeed, the efficiency ϵ_1 decreases resulting in a negative net change, while the purity p_1 increase leads to a positive net change. Instead, the efficiency ϵ_2 increases resulting in a positive net change, and the purity p_2 decrease gives a negative net change. The values are shown in table 3.7.

For the electron branch the J index improved from 0.740 to 0.773, assessing an overall improvement (with respect to the standard cuts), despite the losses in ϵ_1 and p_2. Therefore, the new cut for electrons favours the correct counting of the two-ring at the expense of some contamination in the single-ring classification.

On the other hand, for the pion branch, the new classification resulted in a uniform improvement across all categories. The t1r1 population slightly decreases (5945 → 5786), but the t2r2 population increases (4429 → 4687) and both the mistaken categories t1r2 (961 → 323) and t2r1 (583 → 236) decrease. These trends are reflected in the efficiency and purity values (table 3.9). The efficiency ϵ_1 and the purity p_1 increase resulting in a positive net change; similarly the efficiency ϵ_2 and the purity p_2 increase leading to a positive net change as well. The values are shown in table 3.10.

For the pion branch the J index improved from 0.569 (standard cuts) to 0.810, representing a significant increase in the discrimination power. Therefore, this indicates that the new cut reduces the fraction of true single-ring events misclassified as two-ring events, while simultaneously increasing the correct classification of two-ring topologies.

This general improvement of the selection criteria for both the electron and the pion branches resulted in the reduction of the misclassified events. This represents the main contribution of the present work and demonstrates that the optimization of the multi-ring reconstruction algorithm has been successfully accomplished.

Residuals distributions

Additionally, the residual distributions for the reconstructed kinematic parameters were produced and are shown in appendix C.2. The residuals distributions are computed by measuring the difference between the true simulation values and the reconstructed ones. These provide a validation of the reconstruction quality, confirming that the improvements in ring-counting performance do not come at the expense of degraded kinematic reconstruction.

The residual distributions are presented in figures C.7 to C.10 and C.15 to C.18, and show the quality of angular and direction reconstruction under both the standard and new selection criteria. In both cases, the mean residual values remain consistent with zero, indicating that neither selection introduces a directional bias. The width of the residual distributions, however, differs systematically between particle types. Muon events show narrower distributions: this difference arises from the intrinsic sharpness of muon Cherenkov rings compared with the more fuzzy ring produced by electromagnetic showers. The observed behaviour is therefore

consistent with expectations and confirms that the reconstruction correctly accounts for the distinct topological signatures of the two particle species.

The vertex residual distributions, shown in figures C.8 to C.10 and C.16 to C.18, highlight the most significant differences between the two selection approaches. For single-ring events, the new selection yields a small improvement in vertex resolution. The standard deviation of the x-vertex residual, in figure 3.13, is reduced from 18.7 cm to 18.5 cm for electrons, and from 29.3 cm to 29.0 cm for muons. For two-ring events, the vertex reconstruction performance remains essentially unchanged, with no measurable reduction introduced by the new selection criteria.

The momentum residual distributions, displayed in figures C.14 and C.22, indicate that the new selection performs at least as well as the standard selection for single-ring events, with slightly tighter distributions and a reduced number of outliers. For two-ring events, the momentum resolution is preserved, suggesting that the revised criteria maintain the core reconstruction performance while simultaneously improving the ring-counting accuracy.

A few examples of residuals distributions are presented in figures 3.12 to 3.14 on the next page.

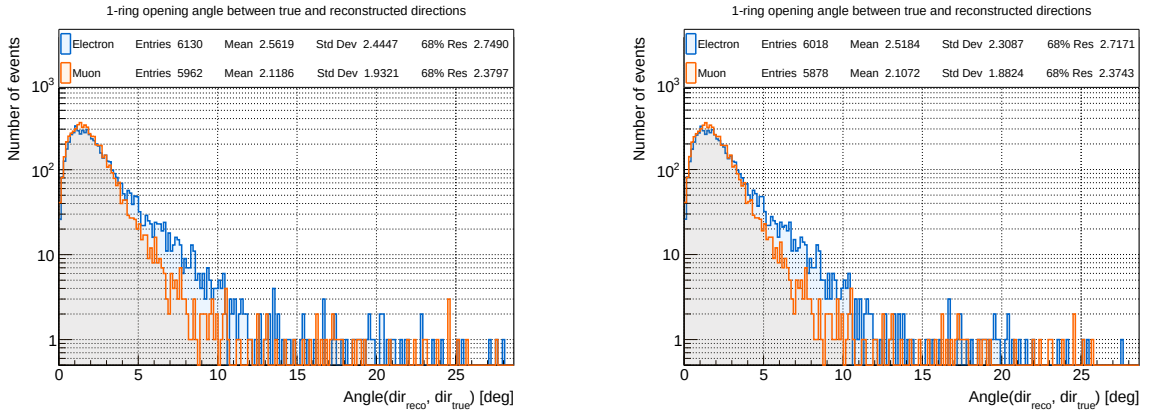


Figure 3.12. Residuals distributions of the one-ring data with correctly identified PID for $\Delta\theta$ opening angle variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

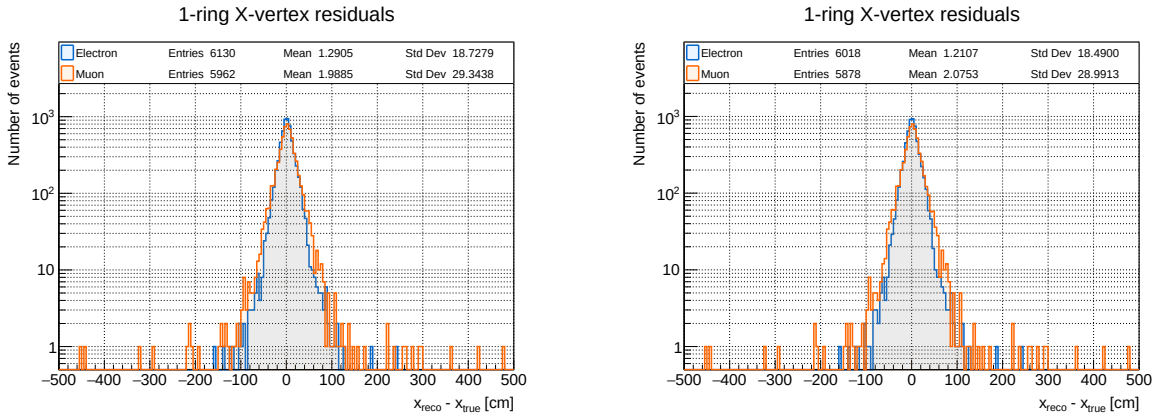


Figure 3.13. Residuals distributions of the one-ring data with correctly identified PID for x position variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

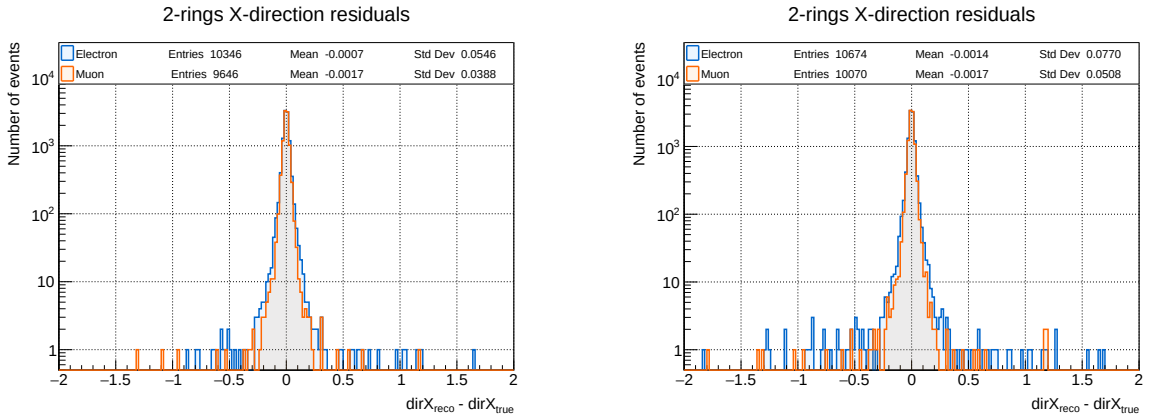


Figure 3.14. Residuals distributions of the two-ring data with correctly identified PIDs for x direction variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

Conclusion

A study of the multi-ring event reconstruction algorithm `fiTQun` for the HK experiment was presented. The study primary objective was optimizing the performance of the ring-counting algorithm at the 8-digit fit level. The work focused on the identification of the primary limitations in the currently implemented selection criteria and consequently proposed an optimized bi-dimensional approach to improve the discrimination between single-ring and two-ring topologies.

This study was based on the analysis of ~50 000 multi-particle events. The events were simulated using the `WCSim` software, with the HK detector parameters. In a first phase the `fiTQun` software was used to perform event reconstruction and analyze the results produced. The performance examination of the standard version of `fiTQun` software yielded that the ring-counting efficiency decreases significantly with increasing true ring multiplicity, with correctly reconstructed fractions of 99.3% for single-ring events, 83.9% for two-ring events, 70.7% for three-ring events, and only 47.5% for four-ring events. It was observed the tendency for the algorithm to reconstruct fewer rings than the true number. In particular, this work focused on the ring-counting at the transition between single and two-ring topologies.

The analysis of the opening angle $\Delta\theta$ distributions between the directions of the reconstructed rings shed light on a crucial insight. The distributions demonstrated that a consistent fraction of the misclassified events had *collinear* opening angle values $\Delta\theta \leq 20^\circ$, in particular those where a true single-ring t1 event was reconstructed as two-ring r2. The implemented *merging* procedure was too rough: despite correcting many of these misclassifications, it was simultaneously introducing new errors by incorrectly merging valid two-ring events with small opening angles. This limitation led to the development of more refined selection criteria. Henceforth, the main contribution of this work is the search, the implementation, and the optimization of bi-dimensional selection criteria.

Throughout the course of this project, several methodological aspects were discussed. Regarding the opening angle variable, it emerged that its computation depends on the accepted hypothesis and cannot be defined *a priori*, before applying the selection criteria. Many different cut scanning strategies were tested, looking for the most balanced one between logical consistency and computational efficiency. The strategy of decoupling the electron and pion branch optimizations has proven to be the best choice, despite introducing an interdependence that affected the final combined performance. Finally, it was also tested an additional ΔP cut on the pion branch: this targeted a specific misclassification pattern where a reconstructed pion ring with negative momentum difference $\Delta P < 0$ indicated a fake 2nd ring. Although it is a first rough estimate, this cut ΔP indicates possible developments in the selection process using several different variables, to reach a finer level of discrimination.

The new selection criteria cuts demonstrated measurable improvements in reconstruction performance. Indeed, the new criteria showed an asymmetric improvement between the two branches. For the pion branch, the classification improved uniformly across all categories, with the J index increasing from 0.569 to 0.810. For the electron branch, the J index improved from 0.740 to 0.773, reflecting a trade-off: the correct counting of two-ring events improved at the expense of some contamination in the single-ring classification.

Finally, the residual distributions confirmed that the optimized selection criteria enhance the discrimination between single and two-ring events at the 8-digit fit level without compromising the quality of the kinematic reconstruction.

Despite the progress achieved, several limitations remain. This work represents only a preliminary study, and no advanced multivariate techniques, nor deep learning methods, were employed. Only one- and two-ring topologies at the 8-digit stage were examined, due to the six-month timeframe of this Master's thesis. Moreover, the optimization was performed on a specific set of simulated data with a uniform energy distribution and isotropic particle production: validation on different physics channels (e.g. atmospheric neutrinos) would be necessary to confirm the general applicability of the results.

Future work could extend the bi-dimensional approach to higher ring multiplicities. Machine learning methods, such as Boosted Decision Tree (BDT) or Artificial Neural Network (ANN), may offer a wider framework for combining additional discriminating variables beyond `nll` and $\Delta\theta$. Finally, the effects of the modifications at the 8-digit level should be analyzed in terms of their impact on the 9-digit fit level, with the aim of achieving further optimization at this level as well.

In conclusion, this thesis demonstrates that the performance of the multi-ring reconstruction algorithm `fiTQun` can be meaningfully improved through the introduction of bi-dimensional selection criteria that rely on the correlation between the likelihood ratio and the opening angle variables. The optimized cuts enhance the discrimination between single and two-ring events at the 8-digit fit level. This work provides a first multi-ring reconstruction approach for the HK reconstruction and selection chain. The methodology developed, and the knowledge gained in this work will be useful for further develop the multi-ring reconstruction in `fiTQun`.

Appendix A

Neutrino oscillations

A.1 Neutrino oscillations in vacuum

To describe the propagation of a neutrino produced in a flavor eigenstate ν_α, consider its time evolution in vacuum ($c = \hbar = 1$ in the following). As the neutrino propagates, the transition amplitude between the flavor eigenstates ν_α and ν_β after time t is

$$\left\langle \nu_\beta \middle| e^{-i\hat{H}t} \middle| \nu_\alpha \right\rangle = \sum_i e^{-iE_i t}\, U_{\beta i}\, U^*_{\alpha i} \tag{A.1}$$

where $\hat{H}$ is the free Hamiltonian and $E_i = \sqrt{|\boldsymbol{p}|^2 + m_i^2}$ is the energy eigenvalue of the free particle with momentum $\boldsymbol{p}$. This equation shows that the transition amplitude between different eigenstates can be non-zero as the time passes. Therefore, the transition probability from the state ν_α to ν_β is

$$P(\nu_\alpha \to \nu_\beta)(t) = \left| \left\langle \nu_\beta \middle| e^{-i\hat{H}t} \middle| \nu_\alpha \right\rangle \right|^2 = \sum_{i,j} U^*_{\alpha i}\, U_{\beta i}\, U_{\alpha j}\, U^*_{\beta j}\, e^{-i(E_i - E_j)t} \tag{A.2}$$

It is important to note that the Majorana phases α_1 and α_2 mentioned before do not appear in the transition probability: these phases do not appear in the oscillation probability, since each matrix element is multiplied by its complex conjugate in the expression above, and the phases therefore cancel out. Thus, θ_{12}, θ_{13}, θ_{23} and δ_{CP} are the only parameters in the mixing matrix which affect the transition probability, regardless of whether neutrinos are Dirac or Majorana fermions. Hence, the transition probability of neutrino oscillations is insensitive to the Majorana phases, making it impossible, with an experiment based on this measurement, to resolve whether neutrinos are Dirac or Majorana fermions. In the ultra-relativistic limit of $\boldsymbol{p}^2 \gg m_i^2$,

$$E_i - E_j = \sqrt{|\boldsymbol{p}|^2 + m_i^2} - \sqrt{|\boldsymbol{p}|^2 + m_j^2} \approx \frac{\Delta m^2_{ij}}{2E} \tag{A.3}$$

where $E \approx |\boldsymbol{p}|$ is the *characteristic* energy and Δm^2_{ij} is the *mass squared splitting*

$$\Delta m^2_{ij} \overset{\text{def}}{\equiv} m_i^2 - m_j^2 \tag{A.4}$$

From equation (A.3) and $ct \approx L$, with L the distance traveled by the neutrino, the transition amplitude equation (A.2) becomes

$$P(\nu_\alpha \to \nu_\beta) = P_{\alpha\beta} = \sum_{i,j} U^*_{\alpha i}\, U_{\beta i}\, U_{\alpha j}\, U^*_{\beta j}\, \exp\left(-i \frac{\Delta m^2_{ij}\, L}{2E}\right) \tag{A.5}$$

Converting in SI unit,

$$\Delta m^2\, \frac{L}{4E} \approx 1.27 \times \left(\frac{\Delta m^2}{\mathrm{eV}^2}\right) \left(\frac{L}{\mathrm{km}}\right) \left(\frac{\mathrm{GeV}}{E}\right) \tag{A.6}$$

The unitarity of U further leads the *transition probability* equation to the following form ([58] for the full computation)

$$P(\nu_\alpha \to \nu_\beta) = P_{\alpha\beta} = \delta_{\alpha\beta} - 4 \sum_{i>j} \mathrm{Re}\left[U^*_{\alpha i}\, U_{\beta i}\, U_{\alpha j}\, U^*_{\beta j}\right] \sin^2\left(\frac{\Delta m^2_{ij}\, L}{4E}\right) + 2 \sum_{i>j} \mathrm{Im}\left[U^*_{\alpha i}\, U_{\beta i}\, U_{\alpha j}\, U^*_{\beta j}\right] \sin\left(\frac{\Delta m^2_{ij}\, L}{2E}\right) \tag{A.7}$$

The oscillation probabilities of the channels with $\alpha = \beta$ are usually called *survival probabilities*, and can be written as

$$P(\nu_\alpha \to \nu_\alpha) = P_{\alpha\alpha} = 1 - 4 \sum_{i>j} |U_{\alpha i}|^2 \left|U_{\alpha j}\right|^2 \sin^2\left(\frac{\Delta m^2_{ij}\, L}{4E}\right) \tag{A.8}$$

Henceforth, it follows that a neutrino can change its flavor as it travels in space: a neutrino produced in one flavor eigenstate may be detected in a different flavor eigenstate after propagating. Because the transition probability varies *periodically* with a phase proportional to $\propto L/E$, the phenomenon is known as *neutrino oscillation*.

Two conditions must be satisfied for oscillations to occur

- $U \neq \mathbb{I}$, the mixing matrix must differ from the identity, meaning that the flavor and mass eigenstates are not identical;
- at least one of the mass squared differences must be non-zero, indeed not all three mass eigenvalues can be equal. At least one has to be different from the others and not null $\Delta m^2_{ij} \neq 0$.

An observation of neutrino oscillations therefore provides direct evidence that neutrinos have non-zero mass.

A.2 Neutrino mass ordering

While the squared mass differences between the neutrino mass eigenstates have been measured with considerable precision, the sign of Δm^2_{32} is still to be determined. This sign ambiguity gives rise to two possible *orderings* of the neutrino mass spectrum, illustrated in figure A.1.

The case with $\Delta m^2_{32} > 0$, in which m_3 is the heaviest eigenstate, is referred to as the Normal Ordering (NO), while the case with $\Delta m^2_{32} < 0$, in which m_3 is the lightest eigenstate, is referred to as the Inverted Ordering (IO). The absolute mass scale, however, cannot be accessed through oscillation measurements alone, since these are sensitive only to mass-squared differences.

Cosmological observations provide a complementary constraint. Indeed, from cosmological measurements, the sum of the three neutrino masses is measured to be less than 0.3 eV [59], and combining the knowledge of the mass order from oscillation experiments we may be able to fully reveal the mass structure of the neutrinos in near future. Furthermore, the origin of the neutrino masses, the reason for their smallness relative to the other known fermions, and the origin of the order remain unexplained within the Standard Model.

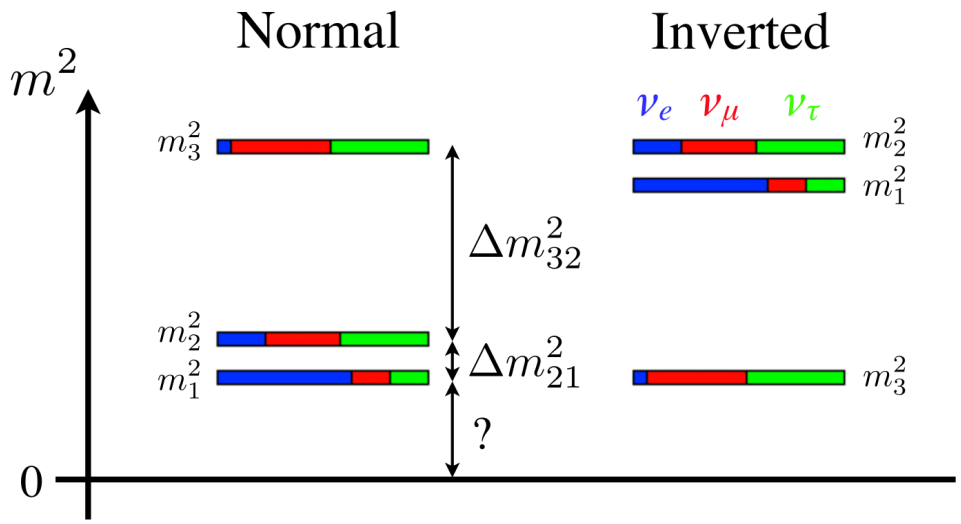


Figure A.1. Schematic representation of the two possible MO. In the NO m_3 is the heaviest mass eigenstate, while in the IO it is the lightest. The absolute mass scale remains undetermined. From [52].

Another fundamental question concerns the nature of the neutrino: whether it is a Dirac or a Majorana fermion. The next generation of neutrino-less double β-decay experiments can determine whether neutrinos are Majorana fermions in case the mass order is IO, but not necessarily if it is NO [60]. Therefore, knowledge of the MO provides valuable input in order to unveil the fundamental nature of the massive neutrinos.

The MO also plays a crucial role in oscillation phenomenology. Degeneracies between the effects of the MO and the CPV phase δ_{CP} can limit the sensitivity of experiments. The knowledge of the MO would resolve these degeneracies, improving the prospects for measuring CPV in the lepton sector.

Finally, and probably most importantly, the question of whether CP is violated for leptons is still open, i.e. if $\delta_{\text{CP}} \neq 0$. Certain theories propose that CPV in the lepton sector, possibly involving additional heavy neutrino states, could explain the observed matter-antimatter asymmetry of the Universe [61]. Although the observable CPV in the currently known leptons itself may not necessarily be directly related to such process [62, 63], the observation of CPV in neutrino oscillations would nonetheless represent a major advance in our understanding of fundamental interactions.

Appendix B

ROOT scan examples

```
1  ************************************************
2  * Instance *           fqmrnll *        fqmrifit *
3  ************************************************
4  *        0 *         43361.371 *       340003111 *
5  *        1 *         43421.535 *       240003111 *
6  *        2 *         22618.132 *       801000000 *
7  *        3 *         22951.377 *       803000000 *
8  *        4 *         22951.521 *       802000000 *
9  *        5 *         7834.4653 *       811000000 *
10 *        6 *         7817.7998 *       813000000 *
11 *        7 *         7811.0400 *       812000000 *
12 *        8 *         5550.2451 *       821000000 *
13 *        9 *         4932.2128 *       823000000 *
14 *       10 *         4959.1284 *       822000000 *
15 *       11 *         44371.699 *       140003111 *
16 *       12 *         44658.285 *       140001111 *
17 *       13 *         45505.464 *       130000331 *
18 *       14 *         46188.820 *       130000131 *
19 *       15 *         49879.820 *       110000003 *
20 *       16 *         49007.168 *        10000001 *
21 *       17 *         45364.949 *        20000011 *
22 *       18 *         46825.402 *        20000031 *
23 *       19 *         45109.550 *        30000111 *
24 *       20 *         45127.906 *       -30000311 *
25 *       21 *         44569.972 *        40001111 *
26 *       22 *         44249.882 *        40003111 *
27 *       23 *         44532.921 *       -50011111 *
28 *       24 *         44215.453 *       -50013111 *
29 *       25 *         46188.820 *        30000131 *
30 *       26 *         45505.464 *        30000331 *
31 *       27 *         46162.171 *       -40001131 *
32 *       28 *         46158.621 *       -40003131 *
33 *       29 *         45496.984 *       -40001331 *
34 *       30 *         45480.785 *       -40003331 *
35 *       31 *         49879.820 *        10000003 *
36 *       32 *         49713.347 *       -20000013 *
37 *       33 *         49544.668 *       -20000033 *
38 ************************************************
```

Listing B.1. `ROOT Scan(...)` output of an example from the dataset used in this thesis work. The variable `fqmrnll` corresponds to `nll`, while `fqmrifit` is the fit ID. The fit IDs are blue and green colored to easily distinguish between 8-digit fit e^- and π^+, respectively, whilst the 9-digit fit steps are orange colored.

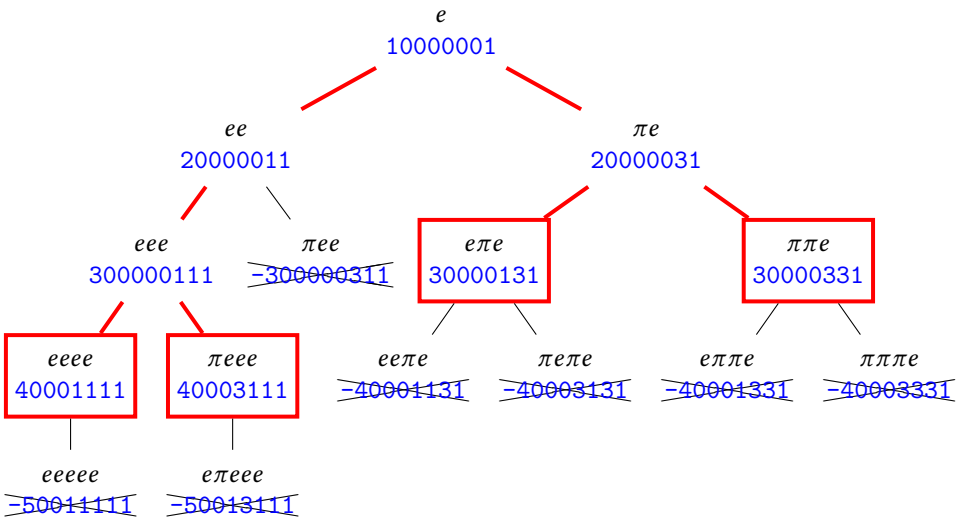


Figure B.1. Tree structure corresponding to the electron branch of the 8-digit fit is shown for the example presented in listing B.1. Fit IDs marked with a strikethrough indicate hypotheses that were rejected, while the thick red lines and boxes highlight the surviving branches and the final accepted hypotheses, for the next "9-digit 1st step" fit. For the 5th and 6th rings, only the electron hypothesis is being tested.

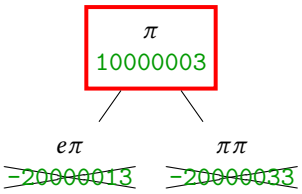


Figure B.2. Tree structure corresponding to the pion branch of the 8-digit fit is shown for the example presented in listing B.1. Fit IDs marked with a strikethrough indicate hypotheses that were rejected, while the thick red lines and boxes highlight the surviving branches and the final accepted hypotheses, for the next "9-digit 1st step" fit.

```
Start of loop event processing
*******************************************************
Trigger Number: 1
Event Number (from loop): 0
Event Number (from WCSimRootEventHeader): 0
Trigger Time [ns]: 25
Trigger Type: 0 NDigits
Interaction Nuance Code: 0
Number of Delayed Triggers (sub events): 0
Primary particle Vertex Geometry Volume Code: -1
Primary particle Vertex Location [cm]: 1597.609985 2145.260010 -
    1631.290039
Index of muon in WCSimRootTracks 0
Number of final state particles 5
SAVED TRACKS
Number of Saved WCSimRootTracks: 3493
Track: 1
  Target nucleus track
  Track ipnu (PDG code): 0
  PDG code of parent particle (0 for primary, 999 for parent PDG code not
     found in the specific logic used): 0
  Track ID of parent particle (0 for primary): 0
  Track initial pos [unit 3-vector]: (0, 0, 0)
  Track initial dir [unit 3-vector]: (0, 0, 0)
  Track initial relativistic energy [MeV]: 0.000000
  Track initial momentum magnitude [MeV/c]: 0.000000
  Track mass [MeV/c2]: 0.000000
  Track ID: 0
  Track creator process: hFritiofCaptureAtRest
  Track time [ns]: 0.000000
  Neutral/massless particle. Cherenkov threshold: N/A (-1)
  Number of ID/OD crossings: 0
Track: 1099
  final state particle track
  Track ipnu (PDG code): 11
  PDG code of parent particle (0 for primary, 999 for parent PDG code not
     found in the specific logic used): 0
  Track ID of parent particle (0 for primary): 0
  Track initial pos [unit 3-vector]: (1597.61, 2145.26, -1631.29)
  Track initial dir [unit 3-vector]: (0.0812518, 0.991039, 0.10602)
  Track initial relativistic energy [MeV]: 1335.025146
  Track initial momentum magnitude [MeV/c]: 1335.025024
  Track mass [MeV/c2]: 0.510999
  Track ID: 2
  Track creator process: hFritiofCaptureAtRest
  Track time [ns]: 0.000000
  Over Cherenkov threshold: 1335.03 > 0.772806 [MeV] (1)
  Number of ID/OD crossings: 0
Track: 3185
  final state particle track
  Track ipnu (PDG code): 11
  PDG code of parent particle (0 for primary, 999 for parent PDG code not
     found in the specific logic used): 0
  Track ID of parent particle (0 for primary): 0
```

```
51   Track initial pos [unit 3-vector]: (1597.61, 2145.26, -1631.29)
52   Track initial dir [unit 3-vector]: (-0.905766, -0.349147, -0.240177)
53   Track initial relativistic energy [MeV]: 776.324219
54   Track initial momentum magnitude [MeV/c]: 776.324036
55   Track mass [MeV/c2]: 0.510999
56   Track ID: 3
57   Track creator process: hFritiofCaptureAtRest
58   Track time [ns]: 0.000000
59   Over Cherenkov threshold: 776.324 > 0.772806 [MeV] (1)
60   Number of ID/OD crossings: 0
61 Track: 3445
62   final state particle track
63   Track ipnu (PDG code): 13
64   PDG code of parent particle (0 for primary, 999 for parent PDG code not
       found in the specific logic used): 0
65   Track ID of parent particle (0 for primary): 0
66   Track initial pos [unit 3-vector]: (1597.61, 2145.26, -1631.29)
67   Track initial dir [unit 3-vector]: (-0.61295, -0.768886, 0.181953)
68   Track initial relativistic energy [MeV]: 649.710388
69   Track initial momentum magnitude [MeV/c]: 641.061523
70   Track mass [MeV/c2]: 105.658371
71   Track ID: 1
72   Track creator process: hFritiofCaptureAtRest
73   Track time [ns]: 0.000000
74   Over Cherenkov threshold: 649.71 > 159.792 [MeV] (1)
75   Number of ID/OD crossings: 0
```

Listing B.2. Full WCSim output for an event with three primaries. (Only the primaries data are printed).

In listing B.2 the Trigger Number corresponds to the Track number. It can be observed that the target nucleus is assigned a Track number of 1, while the three primary particles are labelled with the values 1099, 3185 and 3445. All of these entries agree with the values reported in table 3.1.

Appendix C

Graphs

C.1 Scatter plots 73
C.2 Residuals 76

C.1 Scatter plots

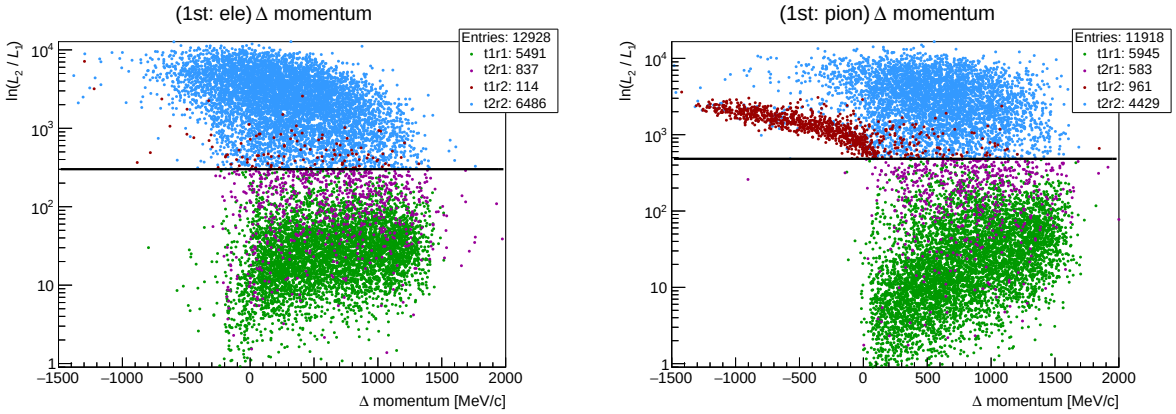


Figure C.1. Scatter plots of the `nll` ratio versus ΔP variables.

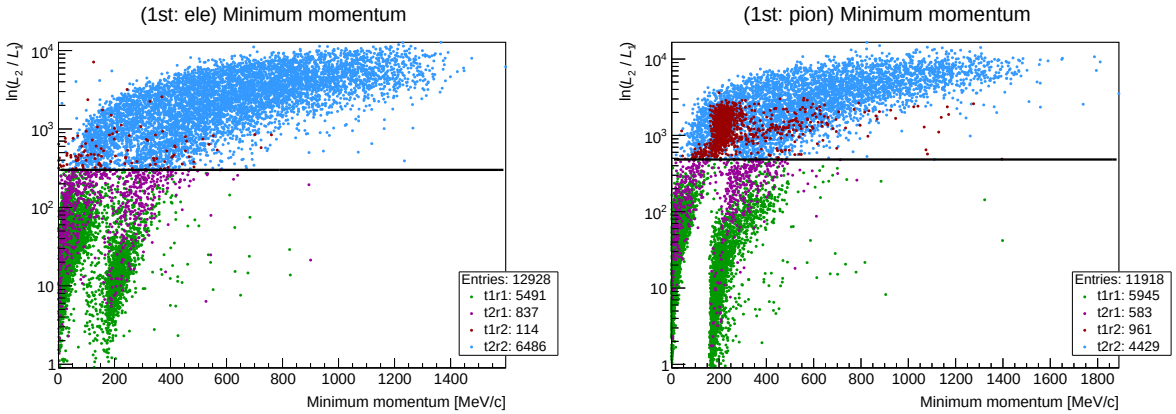


Figure C.2. Scatter plots of the `nll` ratio versus min P variables.

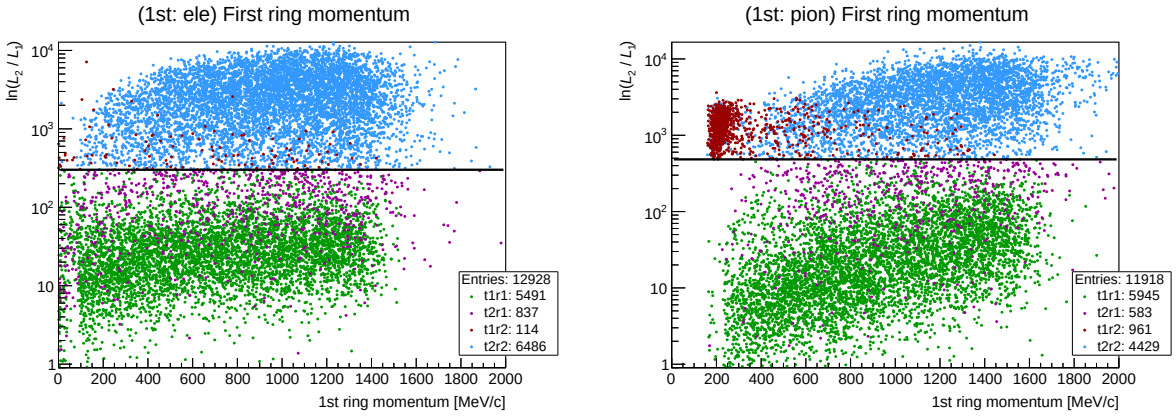


Figure C.3. Scatter plots of the `nll` ratio versus P_1 variables.

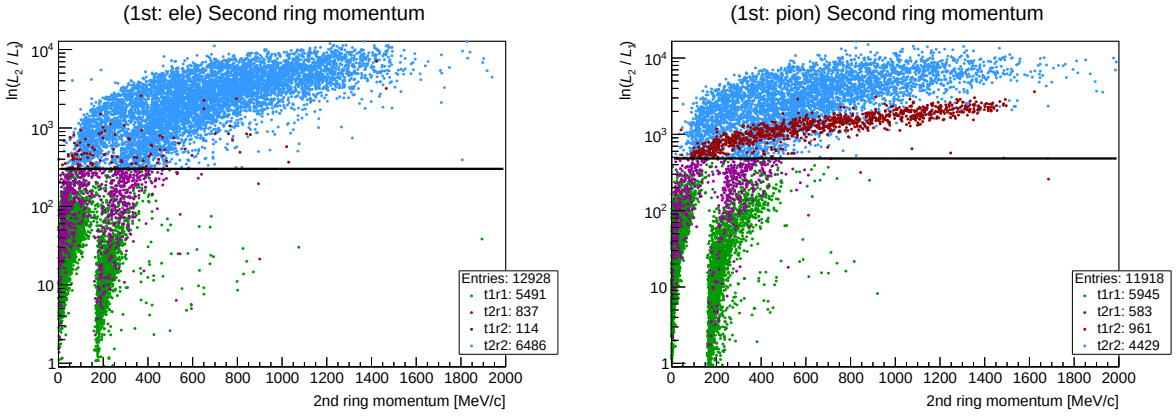


Figure C.4. Scatter plots of the `nll` ratio versus P_2 variables.

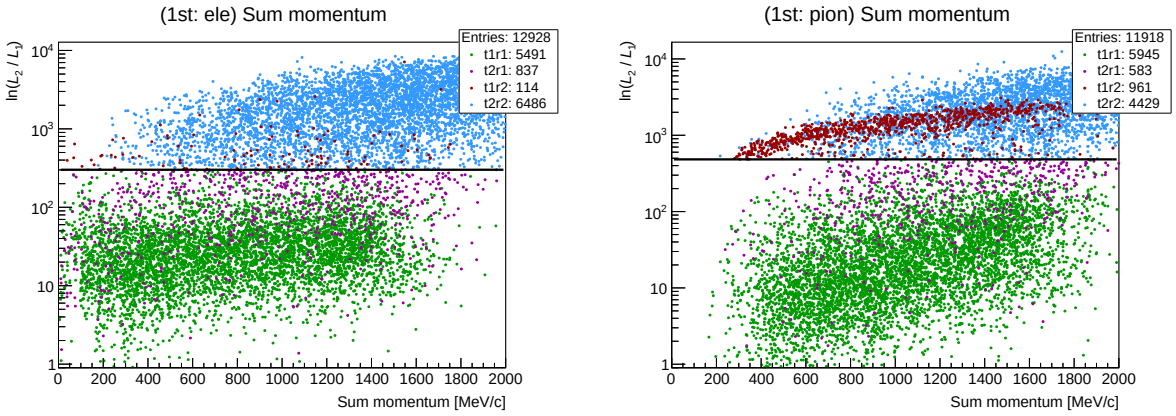


Figure C.5. Scatter plots of the `nll` ratio versus $P_1 + P_2$ variables.

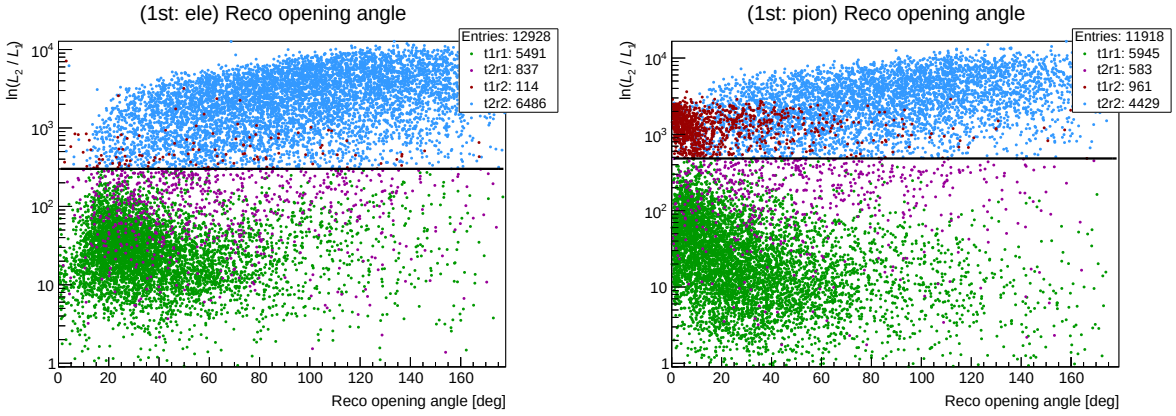


Figure C.6. Scatter plots of the `nll` ratio versus $\Delta\theta$ variables.

C.2 Residuals

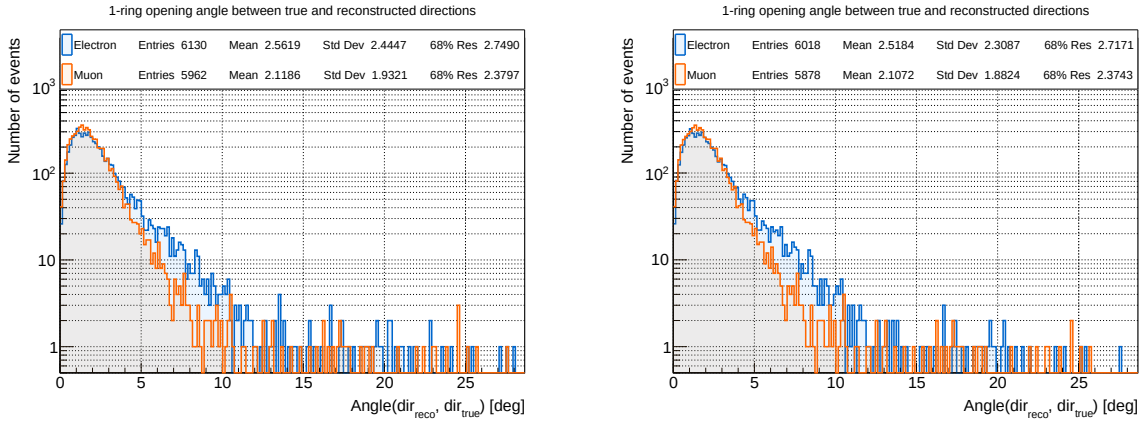


Figure C.7. Residuals distributions of the one-ring data with correctly identified PID for $\Delta\theta$ opening angle variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

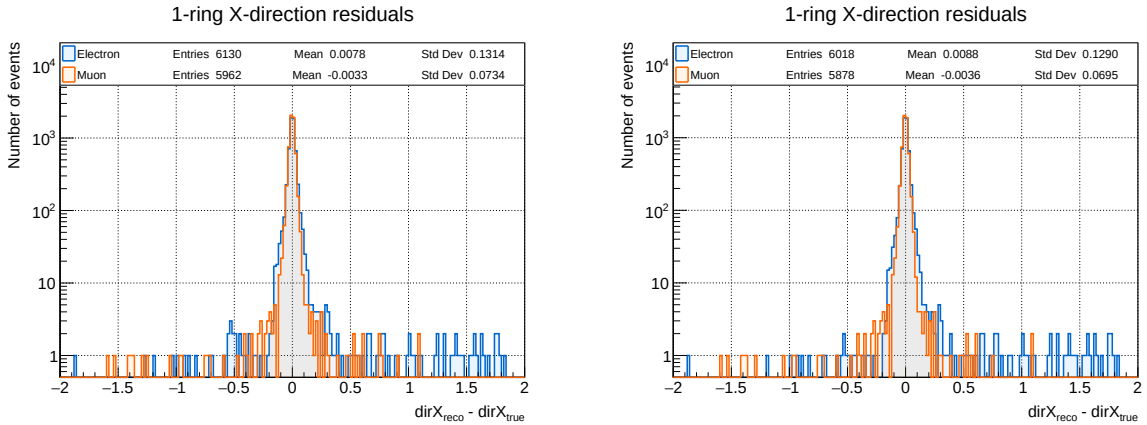


Figure C.8. Residuals distributions of the one-ring data with correctly identified PID for x direction variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

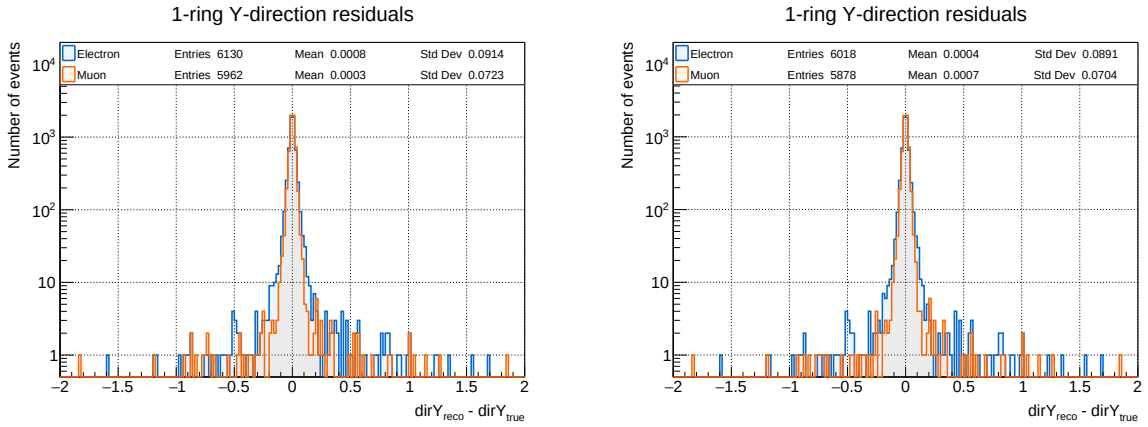


Figure C.9. Residuals distributions of the one-ring data with correctly identified PID for y direction variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

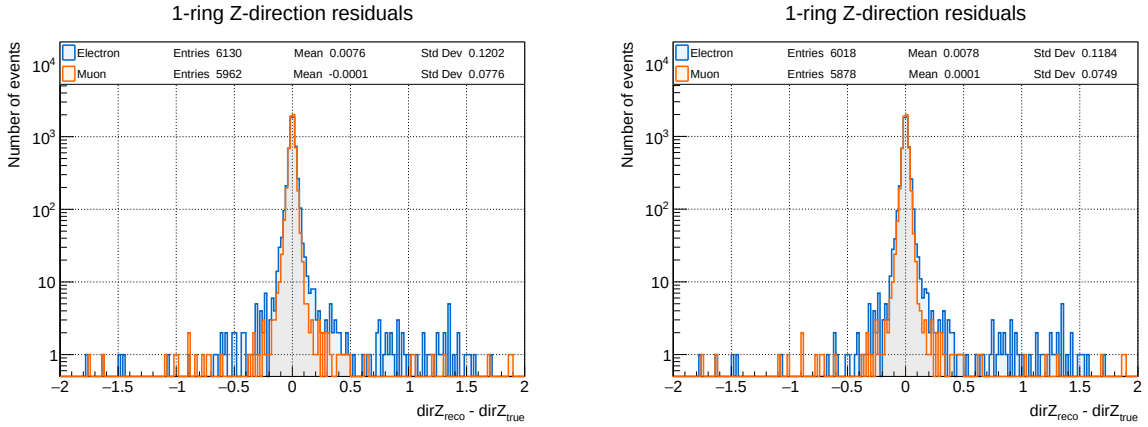


Figure C.10. Residuals distributions of the one-ring data with correctly identified PID for z direction variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

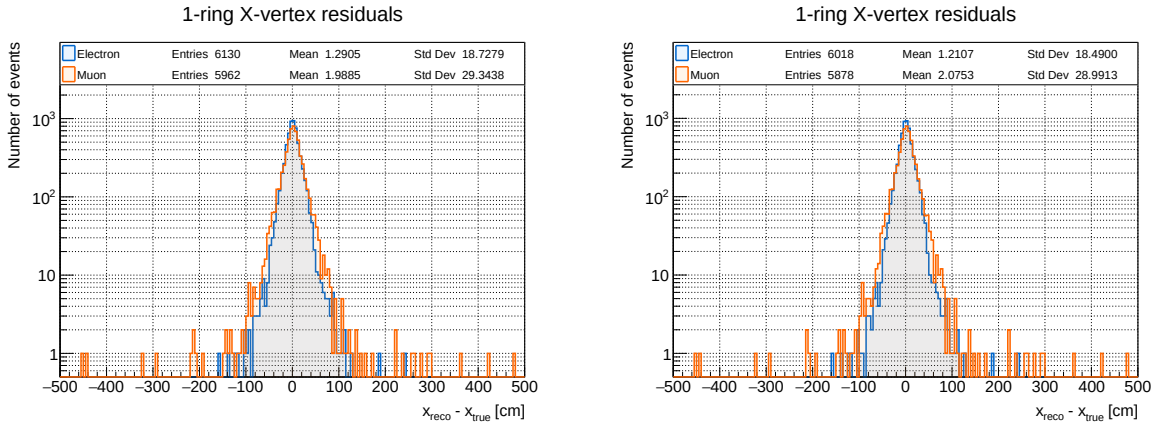


Figure C.11. Residuals distributions of the one-ring data with correctly identified PID for x position variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

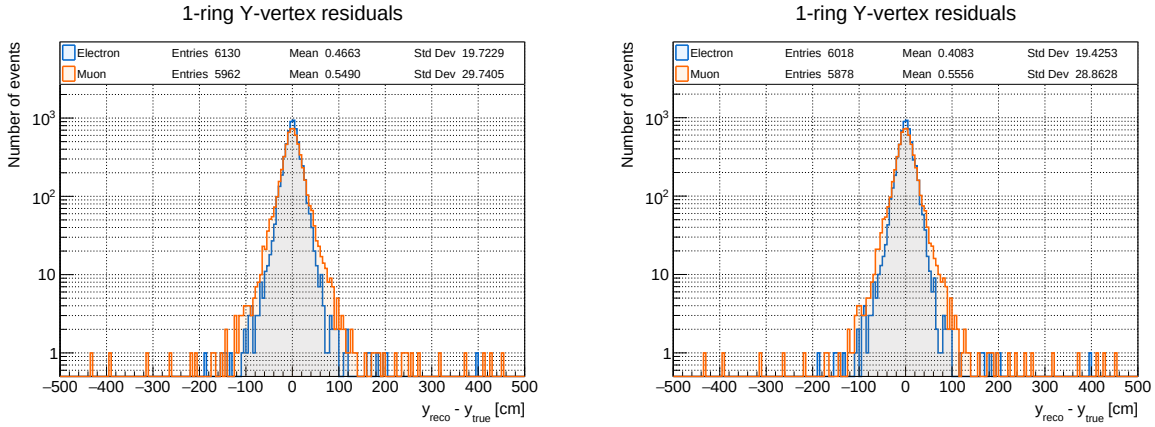


Figure C.12. Residuals distributions of the one-ring data with correctly identified PID for y position variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

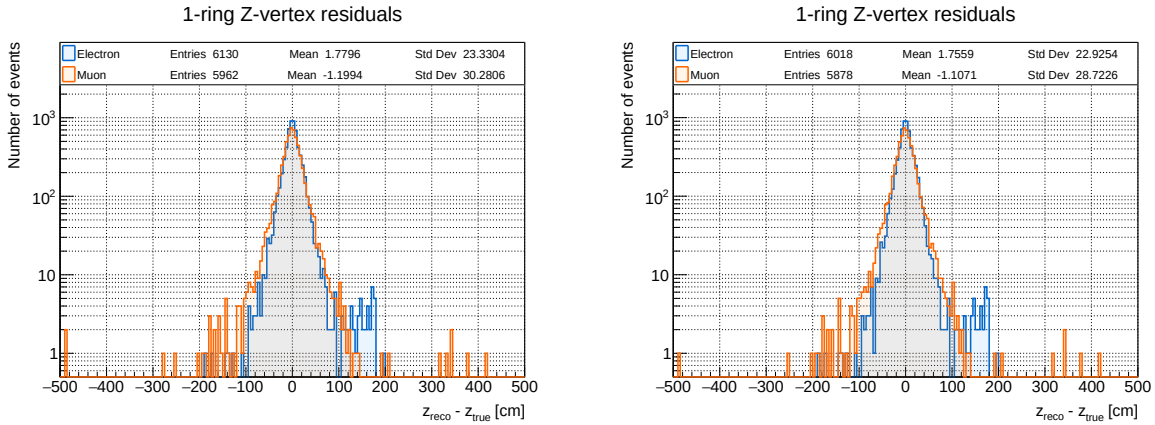


Figure C.13. Residuals distributions of the one-ring data with correctly identified PID for z position variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

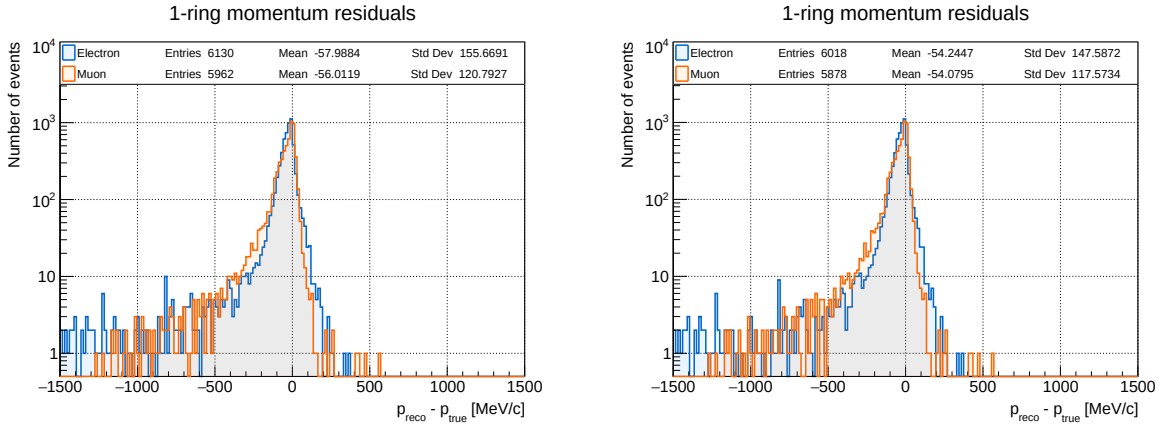


Figure C.14. Residuals distributions of the one-ring data with correctly identified PID for p momentum variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

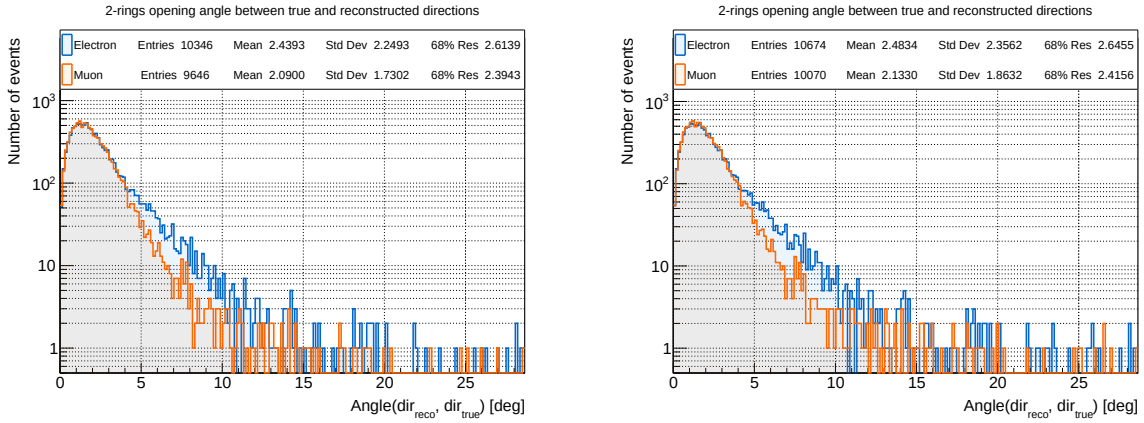


Figure C.15. Residuals distributions of the two-ring data with correctly identified PIDs for $\Delta\theta$ opening angle variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

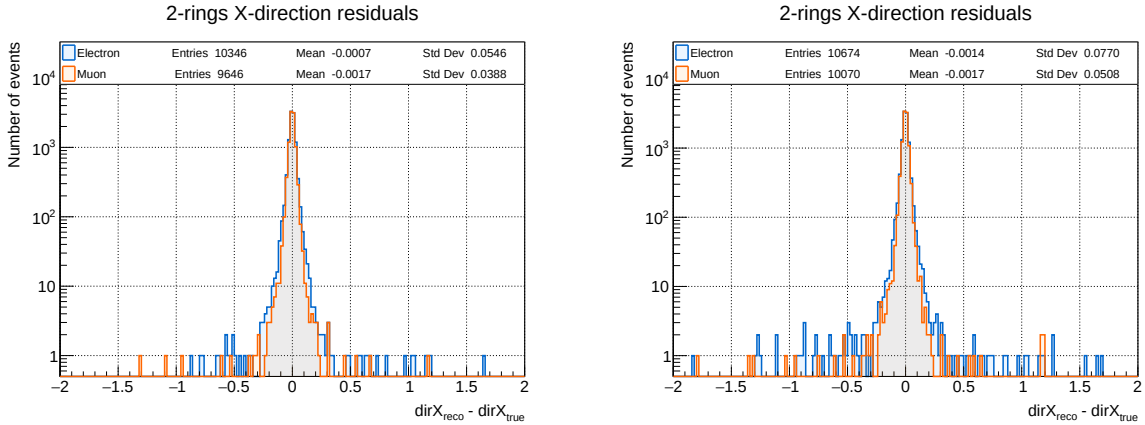


Figure C.16. Residuals distributions of the two-ring data with correctly identified PIDs for x direction variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

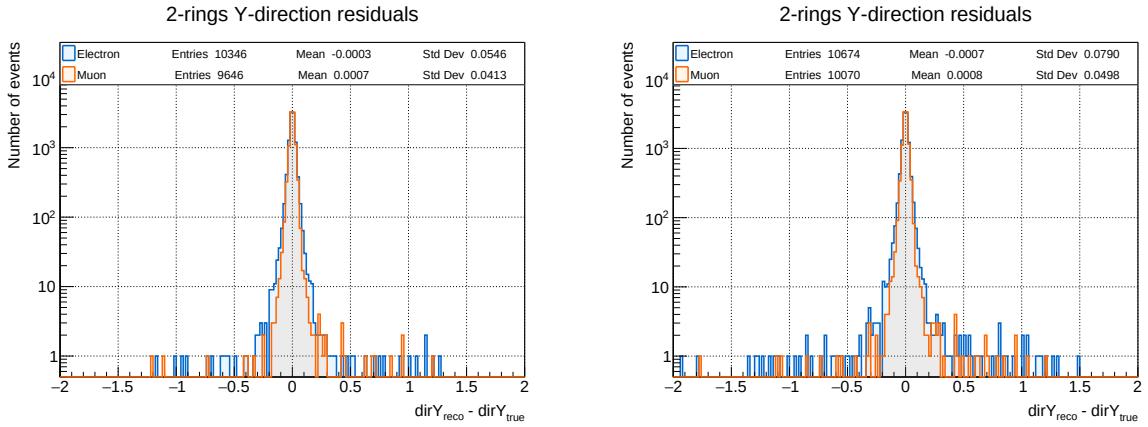


Figure C.17. Residuals distributions of the two-ring data with correctly identified PIDs for y direction variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

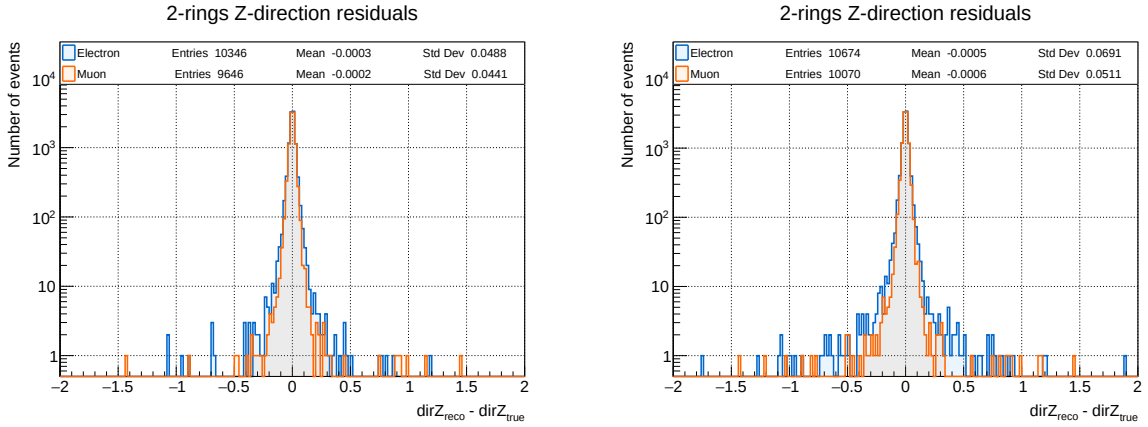


Figure C.18. Residuals distributions of the two-ring data with correctly identified PIDs for z direction variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

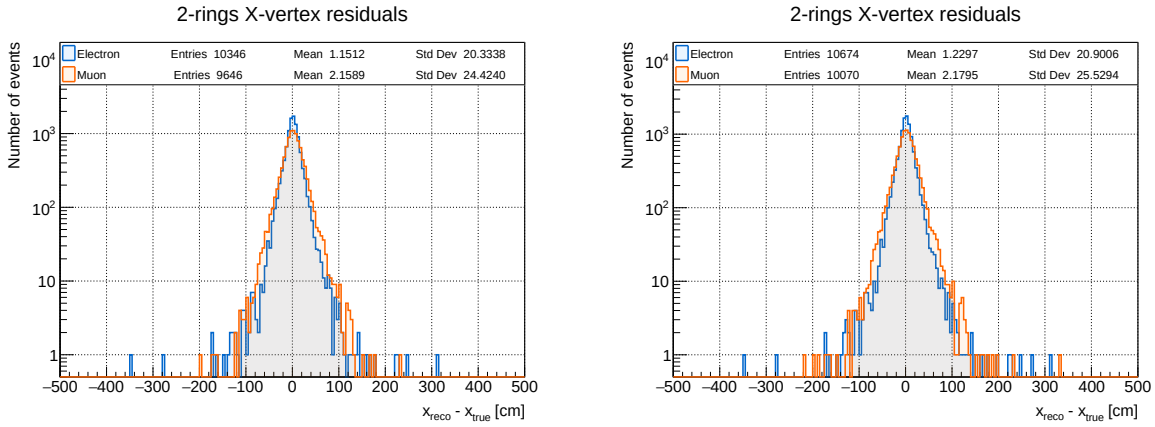


Figure C.19. Residuals distributions of the two-ring data with correctly identified PIDs for x position variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

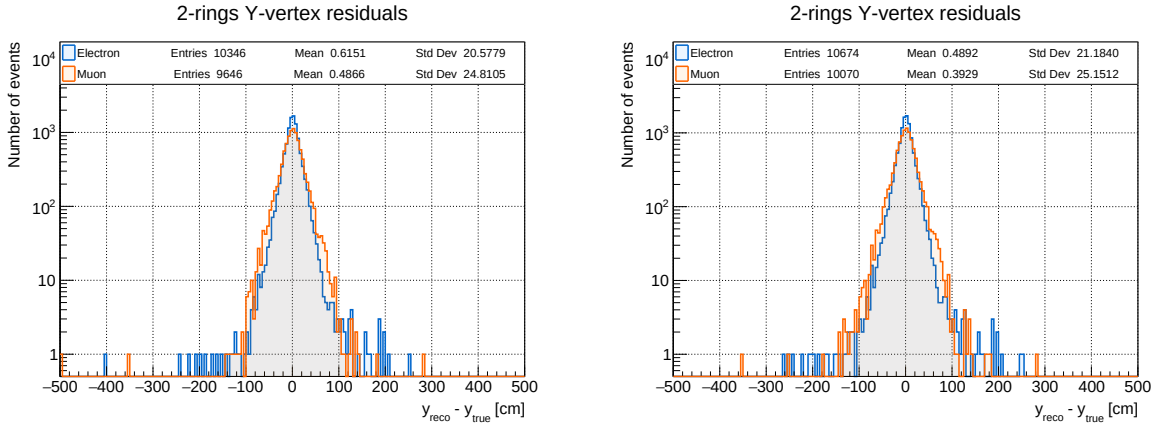


Figure C.20. Residuals distributions of the two-ring data with correctly identified PIDs for y position variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

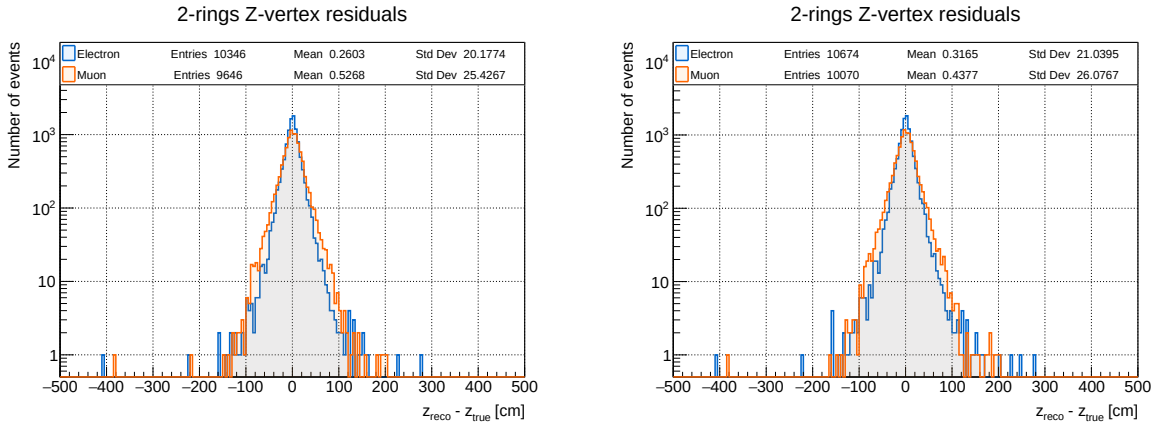


Figure C.21. Residuals distributions of the two-ring data with correctly identified PIDs for z position variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

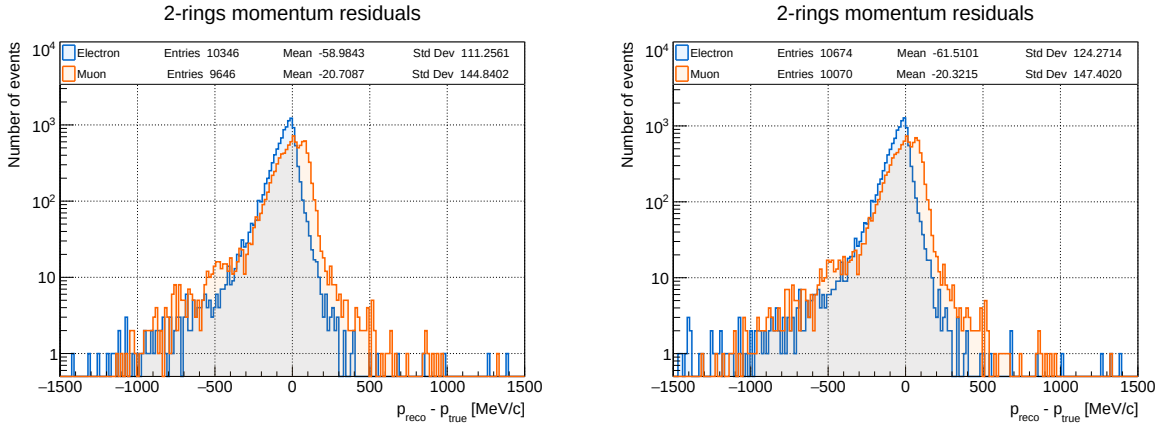


Figure C.22. Residuals distributions of the two-ring data with correctly identified PIDs for p momentum variable. The distributions are shown separately for electron-like and pion-like events. On the left, the results from the standard cuts, while on the right, the results from the new cuts.

Appendix D

List of acronyms

3D Three Dimensional.

ANN Artificial Neural Network.

BDT Boosted Decision Tree.

C.L. Confidence Level.

CC Charged-Current.

CC-DIS Charged-Current Deep Inelastic Scattering.

CC-QE Charged-Current Quasi-Elastic.

CC-RES Charged-Current RESonant.

CCSN Core-Collapse SuperNovae.

CE Collection Efficiency.

CERN European Council for Nuclear Research.

CKM Cabibbo-Kobayashi-Maskawa.

CP Charge-Parity.

CPV Charge-Parity Violation.

DSNB Diffuse Supernova Neutrino Background.

DUNE Deep Underground Neutrino Experiment.

ECAL Electromagnetic CALorimeter.

Fd Far detector.

FGD Fine-Grained Detector.

GUT Grand Unified Theory.

HK Hyper-Kamiokande.

HPC High-Performance Computing.

IC Inner Cylinder.

ID IDentifier.

Id Inner detector.

INGRID Interactive Neutrino GRID.

IWCD Intermediate Water Cherenkov Detector.

J-PARC Japan Proton Accelerator Research Complex.

JUNO Jiangmen Underground Neutrino Observatory.

KamLAND Kamioka Liquid scintillator ANtineutrino Detector.

MC Monte Carlo.

MiniBooNE Mini Booster Neutrino Experiment.

MO Mass Ordering.

mPMT multi-PMT.

NC Neutral-Current.

Nd Near detector.

ND280 Near Detector 280.

ND280-OA ND280 Off-Axis.

NDigits Number of Digitized hits.

NOMAD Neutrino Oscillation MAgnetic Detector.

Od Outer detector.

PDF Probability Distribution Function.

PID Particle IDentification.

PMNS Pontecorvo-Maki-Nakagawa-Sakata.

PMT PhotoMultiplier Tube.

PRISM Precision Reaction Independent Spectrum Measurement.

QBEE QTC-Based Electronics with Ethernet.

QE Quantum Efficiency.

QTC Charge-to-Time Converter.

SK Super-Kamiokande.

SKI SK phase I (April 1996 – July 2001).

SMRD Side Muon Range Detector.

SRN Supernova Relic Neutrinos.

T2K Tokai to Kamioka.

TOF Time of Flight.

TPC Time Projection Chamber.

UA1 Underground Area 1.

Bibliography

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